



Health Insurance

Capstone Project

Project Details:

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1. Introduction

The healthcare industry plays a crucial role in society, as it directly influences individuals' well-being, quality of life and life expectancy. A fundamental aspect of healthcare is health insurance, which serves as a vital safety net, shielding individuals from the potentially crippling costs associated with medical care. Without adequate coverage, people may face financial hardship, especially when faced with serious illnesses or injuries. The high costs of treatments, medications and hospital stays can lead to long-term economic strain, further affecting their quality of life.

However, from the perspective of insurance companies, the challenge lies in balancing the provision of coverage with the management of financial risk. To remain financially viable, insurance providers must optimize their pricing models, which involves accurately assessing the risk associated with insuring an individual.

Traditionally, health insurance premiums are determined based on factors such as age, medical history, and current health status. However, lifestyle choices such as diet, exercise and smoking habits also have a significant impact on an individual's health and the likelihood of requiring costly medical interventions. As a result, insurance companies are increasingly leveraging health and lifestyle data to refine their pricing strategies.

The primary objective of this project is to develop a predictive model that estimates the optimal insurance premium for an individual, considering a variety of health and lifestyle-related factors. By doing so, the project aims to achieve several key outcomes:

- **Personalized and Equitable Pricing:** The development of a predictive model tailored to individual risk profiles will enable insurance companies to provide premiums that reflect the true health risks associated with each person. This approach will lead to more personalized and fair pricing, rather than relying on broad demographic categories such as age or location.
- **Incentivizing Healthier Lifestyles:** By tying insurance premiums to lifestyle choices, the project aims to encourage individuals to adopt healthier behaviours. Offering financial incentives for maintaining a healthy lifestyle such as regular exercise, balanced diets, and avoiding risky habits like smoking can promote a culture of prevention and wellness.
- **Financial Sustainability:** The implementation of more accurate and data-driven pricing models has the potential to reduce overall claims and medical costs, benefiting insurance companies with long-term savings. This creates a more sustainable financial model for both individuals and insurers.
- **Improved Public Health:** By fostering a culture of wellness and preventive care, this approach can lead to better public health outcomes. When individuals are motivated to take charge of their health in ways that directly influence their insurance premiums, society as a whole can benefit from healthier, more informed citizens.

In essence, this project seeks to create a system that accurately predicts and optimizes health insurance costs, ensuring fairness and financial sustainability. By integrating personalized health data into the pricing model, it aims to develop a more equitable, efficient, and health-conscious insurance system that benefits both individuals and the broader healthcare industry.

2. Exploratory Data Analysis

2.1. Data Structure

After importing NumPy and Pandas libraries for tasks like data reading, data computation and data manipulation and loading data, the following observations regarding the data structure were noted.

OBSERVATION	VARIABLES
25000	24

Table 1. Data structure

2.2. Data Types

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 25000 entries, 0 to 24999
Data columns (total 24 columns):
#   Column                                     Non-Null Count  Dtype
---  ---
0   applicant_id                             25000 non-null  int64
1   years_of_insurance_with_us               25000 non-null  int64
2   regular_checkup_last_year               25000 non-null  int64
3   adventure_sports                         25000 non-null  int64
4   Occupation                               25000 non-null  object
5   visited_doctor_last_1_year              25000 non-null  int64
6   cholesterol_level                       25000 non-null  object
7   daily_avg_steps                         25000 non-null  int64
8   age                                      25000 non-null  int64
9   heart_decs_history                      25000 non-null  int64
10  other_major_decs_history                 25000 non-null  int64
11  Gender                                  25000 non-null  object
12  avg_glucose_level                       25000 non-null  int64
13  bmi                                     24010 non-null  float64
14  smoking_status                          25000 non-null  object
15  Year_last_admitted                      13119 non-null  float64
16  Location                                25000 non-null  object
17  weight                                  25000 non-null  int64
18  covered_by_any_other_company            25000 non-null  object
19  Alcohol                                 25000 non-null  object
20  exercise                                25000 non-null  object
21  weight_change_in_last_one_year          25000 non-null  int64
22  fat_percentage                          25000 non-null  int64
23  insurance_cost                          25000 non-null  int64
dtypes: float64(2), int64(14), object(8)
memory usage: 4.6+ MB
```

Table 2. Data types

OBSERVATIONS:

- The data set contains nineteen variables indexed from 0-23 which includes 16 numerical columns, where 14 are of integer data type and 2 are float data type and 8 object columns.
- There are missing values in 2 columns, namely 'bmi' and 'Year_last_admitted' which must be treated.
- The below columns can be treated as categorical columns:
 - A column named 'adventure_sports' which is in integer data type given in terms of 1 and 0, which indicates if a beneficiary is involved with adventure sports or not respectively.
 - A column named 'heart_decs_history' which is in integer data type given in terms of 1 and 0, which indicates if a beneficiary has any past heart diseases or not respectively.
 - A column named 'other_major_decs_history' which is in integer data type given in terms of 1 and 0, which indicates if a beneficiary has any past major diseases apart from heart like any operation or not respectively.
 - A column named 'Year_last_admitted' which gives when the beneficiary has been admitted in the hospital last time.

2.3. Statistical Summary

The following observations were made:

- **years_of_insurance_with_us:** The beneficiaries have been with the health insurance company for 0 to 8 years, with the average tenure being 4 years.
- **regular_checkup_lasy_year:** Last year, beneficiaries had between 0 to 5 regular check-ups, with 75% of them attending only once, and the maximum number of check-ups was 5.
- **Occupation:** Most beneficiaries are students, totalling 10,169.
- **visited_doctor_last_1_year:** In the past year, doctor visits ranged from 0 to 12, with an average of 3 visits.
- **cholesterol_level:** Most beneficiaries have cholesterol levels between 150mg/dL and 175mg/dL, with 8,763 individuals falling into this range.
- **daily_avg_steps:** On average, beneficiaries walk 5,215 steps daily, with a minimum of 2,000 steps and a maximum of 11,255 steps.
- **age:** The age of beneficiaries ranges from 16 to 74 years, with an average age of 45.
- **Gender:** Males make up most of the customer base, with 16,422 male beneficiaries.
- **avg_glucose_level:** The average glucose level is 167mg/dL, ranging from 57mg/dL to 277mg/dL, with a standard deviation of 62.7.
- **bmi:** BMI values vary widely from 12 to 100, but 75% of beneficiaries have a BMI of 35, with a standard deviation of 7.
- **smoking_status:** A large proportion of beneficiaries (9,249) have never smoked.
- **Year_last_admitted:** The year in which the beneficiary was admitted last to the hospital ranges from 1990-2018.
- **Location:** The highest number of hospital location, 1,742, are in Bangalore.
- **weight:** Individual weights range from 52 to 96 kg, with a standard deviation of 9.3 kg.
- **covered_by_any_other_company:** Most beneficiaries do not have insurance from other companies, with only 1,742 holding policies from other providers.
- **Alcohol:** 13,752 beneficiaries out of 25,000 consume alcohol rarely.
- **exercise:** Around 14,638 beneficiaries engage in moderate exercise.
- **weight_change_in_last_one_year:** Weight change in the past year ranges from 0 to 6 kg, with an average change of 2.5 kg.

- **fat_percentage:** The average fat percentage is 28.8%, with the maximum being 42%.
- **insurance_cost:** Insurance premiums range from 2,468 to 67,870, with an average cost of 27,147.

2.4. Univariate analysis

The table below provides a summary of the univariate analysis for all variables.

Variable	Key Observations
years_of_insurance_with_us	The number of years ranges from 0 to 8, with a nearly uniform distribution. The average tenure is around 4 years, with fewer individuals at 2 years.
regular_checkup_last_year	15,000 beneficiaries did not have a regular check-up last year. Around 4,000 had one check-up, and only a few had 4-5 check-ups.
visited_doctor_last_1_year	Most visited the doctor 2 to 4 times. A few visited more than 5-12 times, with outliers in the 8-12 visit range.
daily_avg_steps	Steps range from 2,000 to 11,000. The average is around 5,200, with outliers at both ends (below 3,000 and above 8,000).
age	The age range is from 16 to 75. The distribution is nearly uniform, with a mean and median around 45 years.
avg_glucose_level	Glucose levels range from 57 to 277 mg/dL, with the mean and median around 157 mg/dL. Over half of beneficiaries have glucose >140 mg/dL.
bmi	BMI ranges from 12 to 100, showing a right-skewed distribution. The mean BMI is close to 30, with many classified as obese (BMI >30).
weight	Weight ranges from 52 to 96 kg. The distribution is multimodal with no outliers, and there's a slight right skew.
weight_change_in_last_one_year	Weight change ranges from 0 to 6 kg. Most people had no change or 1-2 kg change, with a few having 5-6 kg change. The distribution is multimodal.
fat_percentage	Fat percentage ranges from 11% to 43%, with many in the 36-37% range. The mean fat percentage is 29%, which is above the ideal threshold.
insurance_cost	Insurance premiums range from 2,500 to 67,000. Most people choose premiums around 30,000, with a slight right skew towards higher premiums.
Year_last_admitted	Admissions span from 1990 to 2018. There was a high frequency of admissions in 2004-2005.
adventure_sports	91.8% of beneficiaries do not participate in adventure sports, suggesting a lower-risk profile for most.
Occupation	40% are students, 40% are business owners, and 19% are salaried. This likely reflects workplace-provided insurance.
cholesterol_level	33.4% have levels between 125-150 mg/dL, and 35.1% between 150-175 mg/dL. 8% have levels >200 mg/dL, indicating higher cardiovascular risk.
Gender	65.7% of beneficiaries are male, and 34.3% are female.

smoking_status	37% have never smoked, 15.5% currently smoke, and 30% have an unknown smoking status.
Location	Locations are distributed fairly evenly, with Bangalore having 7% and Surat at 6.4%.
covered_by_any_other_company	69.7% do not have insurance from other companies, while 30% have multiple policies.
Alcohol	55% consume alcohol rarely, 34% do not consume alcohol, and 10.8% consume it daily.
Exercise	58.6% engage in moderate exercise, 21% in intense exercise, and 20% do not exercise at all.
heart_decs_history	5% have a history of heart disease.
other_major_decs_history	9.8% have a history of other major diseases, such as surgeries, aside from heart-related conditions.

Table 3. Univariate analysis

* Visual plots are provided in the Appendix: Tables and Charts section for reference

2.5. Bivariate Analysis

2.5.1. Relationship between numerical variables

2.5.1.1. Heat Map

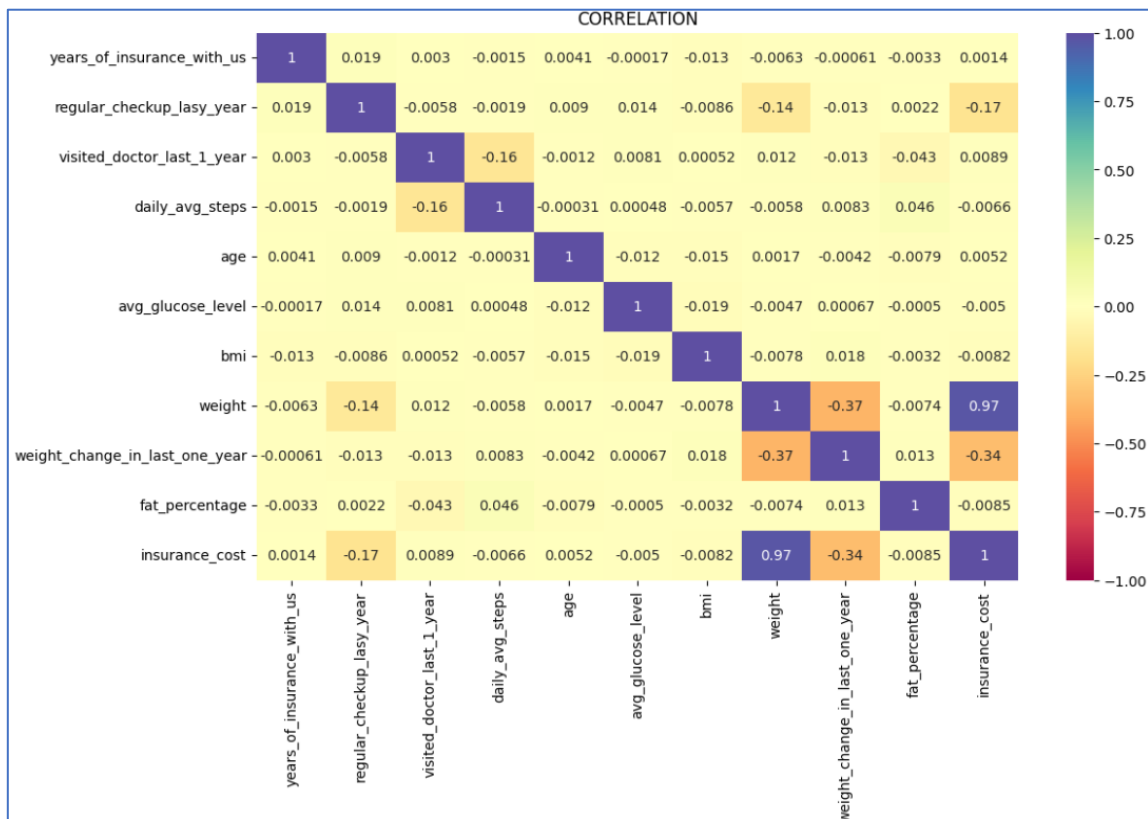


Fig 1. Heat Map

OBSERVATIONS:

- The highest correlation, around 0.97, is observed between weight and insurance cost, showing a strong positive correlation. This suggests that weight is a key factor in determining the insurance cost.
- Weight change over the past year and weight itself have a negative correlation of -0.37, meaning that as weight increases, the change in weight tends to decrease.
- There is also a negative correlation of -0.34 between insurance cost and weight change over the past year, indicating that higher insurance costs are associated with smaller changes in weight.
- The correlation between insurance cost and regular check-ups from the previous year is -0.17, suggesting a weak negative relationship between the two.

2.5.1.2. Pair Plot

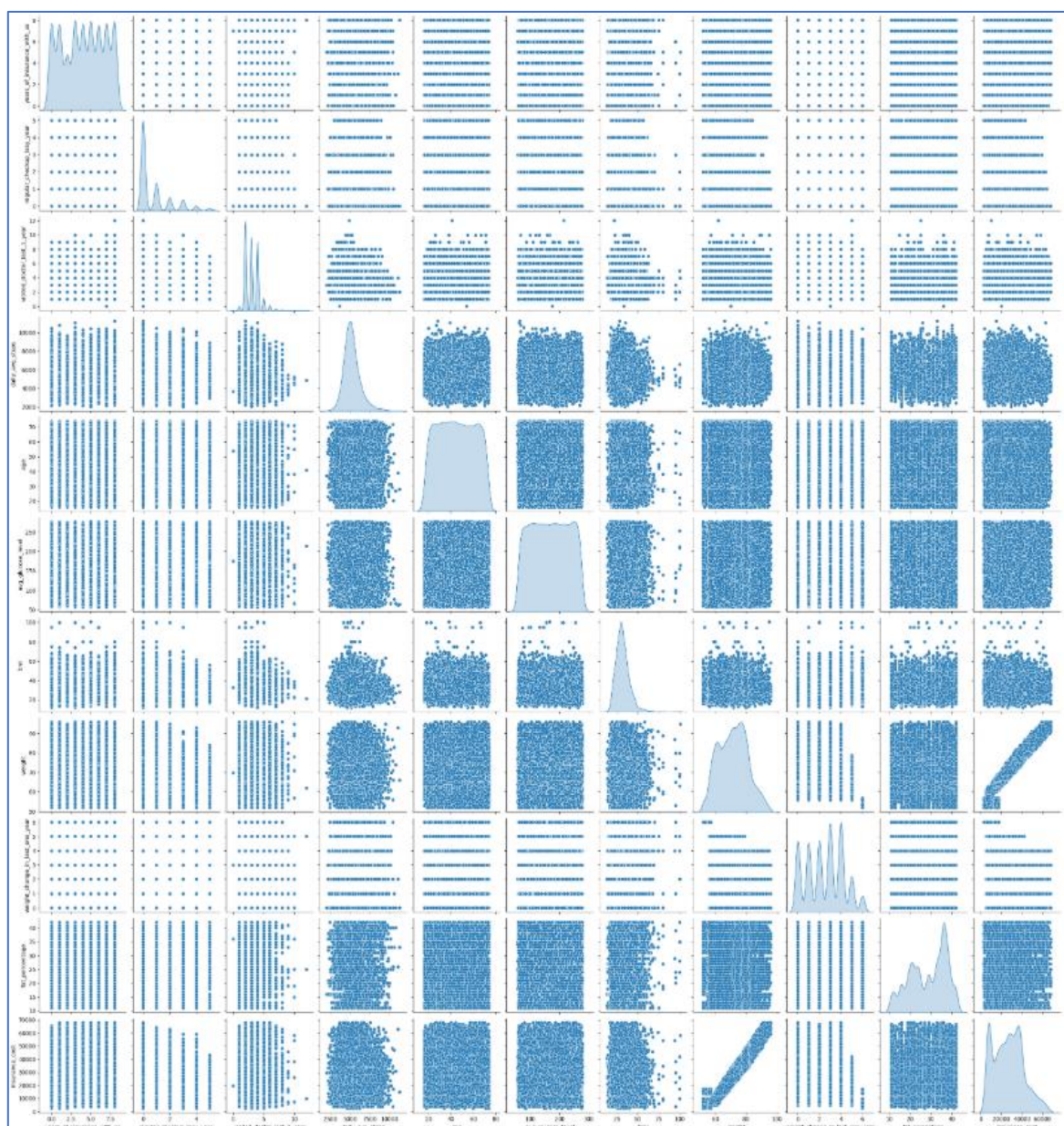


Fig 2. Pair Plot

OBSERVATIONS:

- In the pair plot, there is a strong positive correlation between weight and insurance cost, with both variables increasing together.
- Variables such as years of insurance, age, and average glucose level display a uniform distribution.
- Variables like daily average steps, BMI, and weight follow a normal distribution. BMI decreases as the frequency of regular check-ups and doctor visits in the past year increases.

2.5.2. Relationship between predictor variables and the target variable

The table below illustrates the relationship between the predictor variables and the target variable.

Variable Pair	Key Observations
weight vs insurance_cost	There is a strong positive correlation, indicating that as weight increases, insurance costs tend to rise as well.
weight_change_in_last_one_year vs insurance_cost	Insurance costs are relatively stable for most weight change categories (0-4 kg), but a slight decrease in cost is seen for those with a 5-6 kg weight change.
Occupation vs insurance_cost	Median insurance costs are consistent at 29,000 across all occupation categories (students, salaried, business owners), though salaried individuals show slight variation at the lower end.
Gender vs insurance_cost	Insurance costs are similar across both genders, with the median staying the same at 29,000, but the lower limit for males is slightly lower.
smoking_status vs insurance_cost	Smoking status has minimal impact on insurance costs, as the median cost is nearly identical for all smoking categories (never smoked, smokes, formerly smoked, unknown).
Location vs insurance_cost	Hospital location does not significantly impact insurance costs. The distributions of costs are similar across locations.
covered_by_any_other_company vs insurance_cost	Individuals with coverage from other companies tend to have slightly higher insurance costs, with both lower and upper limits being higher for this group.
Alcohol vs insurance_cost	Frequency of alcohol consumption (daily, rare, or none) does not significantly affect insurance costs. The cost distributions are similar across these groups.
exercise vs insurance_cost	Exercise habits (moderate, intense, none) show no significant impact on insurance cost, as the distributions of costs are similar across all groups.
adventure_sports vs insurance_cost	People who engage in adventure sports tend to have higher insurance costs. The median for this group is above 30,000, and the range of costs is narrower compared to non-participants.

cholesterol_level vs insurance_cost	Insurance costs are slightly higher for individuals with cholesterol levels between 200-225 mg/dL, but the costs are generally consistent across all cholesterol categories.
heart_decs_history vs insurance_cost	A history of heart disease does not significantly impact insurance costs. The distributions and medians of insurance costs are similar for individuals with and without heart disease history.
other_major_decs_history vs insurance_cost	Having a history of other major diseases or surgeries (excluding heart conditions) does not noticeably affect insurance costs, as the distribution remains similar across both groups.
Year_last_admitted vs insurance_cost	Insurance costs decrease over time: individuals admitted between 1990-2000 have higher median costs (~40,000), while costs drop to ~10,000 for admissions in 2013-2018, indicating lower costs for more recent admissions.

Table 4. Bivariate analysis: Predictor vs Target variables

** Visual plots are provided in the Appendix: Tables and Charts section for reference*

Examining the relationship between weight and insurance cost, while considering various other significant factors or categories, is crucial in understanding how these elements influence insurance premiums.

Variable Pair	Key Observations
weight vs insurance_cost (covered_by_any_other_company)	Insurance costs are slightly higher for individuals with coverage from other companies compared to those without.
weight vs insurance_cost (adventure_sports)	Individuals who engage in adventure sports tend to have higher insurance costs compared to those who do not.
weight vs insurance_cost (Year_last_admitted)	Insurance costs decline over time. Individuals admitted in more recent years generally have lower costs, while those admitted earlier have higher costs.

Table 5. Multivariate analysis

** Visual plots are provided in the Appendix: Tables and Charts section for reference*

2.5.3. Relationship between predictor variables

The table below presents the effect of Variables on the History of Heart Disease and Other Major Diseases.

Variable Pair	Key Observations
Gender vs heart_decs_history	Males have a higher prevalence of heart disease history.
age vs heart_decs_history	Individuals in the 26-35 and 46-55 age groups are more likely to have a history of heart disease.
daily_avg_steps vs heart_decs_history	Individuals with heart disease typically walk an average of 4,000 to 6,000 steps daily.

exercise vs heart_decs_history	Individuals with heart disease tend to engage in moderate exercise.
cholesterol_level vs heart_decs_history	Individuals with heart disease often have low cholesterol levels, typically between 125-175 mg/dL.
avg_glucose_level vs heart_decs_history	Individuals with heart disease typically have glucose levels ranging from 150 to 200 mg/dL.
bmi vs heart_decs_history	Individuals with heart disease tend to have a BMI in the range of 30-40.
smoking_status vs heart_decs_history	Individuals who have never smoked or who have smoked in the past are more likely to develop heart disease.
Alcohol vs heart_decs_history	People who consume alcohol occasionally are less likely to have heart disease.
weight vs heart_decs_history	Individuals weighing between 70-80 kg have the highest incidence of heart disease history.
fat_percentage vs heart_decs_history	Individuals with a fat percentage between 35-40 are more likely to have a history of heart disease.
Gender vs other_major_decs_history	Males are more susceptible to major diseases, including surgeries unrelated to heart conditions.
age vs other_major_decs_history	Individuals aged 26-65 are more likely to experience major diseases, including surgeries.
daily_avg_steps vs other_major_decs_history	Individuals with major diseases tend to take 4,000 to 6,000 steps daily.
exercise vs other_major_decs_history	Individuals with major diseases tend to engage in moderate exercise.
cholesterol_level vs other_major_decs_history	Individuals with major diseases apart from heart conditions often have low cholesterol levels, ranging from 125-175 mg/dL.
avg_glucose_level vs other_major_decs_history	Individuals with major diseases typically have glucose levels ranging from 150 to 200 mg/dL.
bmi vs other_major_decs_history	Individuals with major diseases (other than heart disease) tend to have a BMI in the range of 30-40.
smoking_status vs other_major_decs_history	Individuals who have never smoked or who have smoked in the past are more likely to develop major diseases.
Alcohol vs other_major_decs_history	People who consume alcohol occasionally are less likely to have a history of major diseases.
weight vs other_major_decs_history	Individuals weighing between 70-80 kg have the highest incidence of major diseases.
fat_percentage vs other_major_decs_history	Individuals with a fat percentage between 35-40 are more likely to have a history of major diseases.
covered_by_any_other_company heart_decs_history vs	Most individuals without coverage from other insurance companies have a history of heart disease.

covered_by_any_other_company other_major_decs_history	vs	Most individuals without coverage from other insurance companies have a history of major disease.
Alcohol vs exercise		Individuals who rarely consume alcohol or do not consume alcohol tend to exercise more frequently. Conversely, those who drink daily tend to exercise less.
adventure_sports vs heart_decs_history		Individuals not participating in adventure sports have a higher likelihood of having heart disease.
adventure_sports other_major_decs_history	vs	Individuals who do not participate in adventure sports have a higher likelihood of having major diseases.

Table 6. Bivariate analysis: Predictor variables

* *Visual plots are provided in the Appendix: Tables and Charts section for reference*

2.6. Insights based on Exploratory Data Analysis

2.6.1. Based on Univariate Analysis

1. **Insurance and Duration:** Most individuals are insured for 2-8 years, with an average of 4 years, indicating long-term coverage preferences.
2. **Health Check-ups:** A large portion of individuals did not have a regular check-up last year, reflecting gaps in preventive healthcare, which may be due to lack of awareness or access.
3. **Doctor Visits:** Most individuals visit the doctor 2-4 times per year, suggesting moderate healthcare usage. Outliers visiting more often may have chronic health conditions.
4. **Physical Activity:** Daily steps range from 2,000 to 11,000, with an average of 5,200. While most engage in moderate exercise, 20% don't exercise at all, which could indicate health risks.
5. **BMI and Weight:** The average BMI is near 30, signalling a tendency toward overweight or obesity, linked to health issues like heart disease and diabetes.
6. **Cholesterol Levels:** Most have normal cholesterol levels, but 8% with high cholesterol are at increased risk for cardiovascular disease.
7. **Health Conditions:** Chronic conditions like heart disease are not widespread, but those with such conditions may face higher healthcare needs and insurance premiums.
8. **Insurance Costs:** Most individuals opt for premiums around 30,000, indicating a preference for mid-range coverage, while those choosing higher premiums likely seek more extensive coverage.
9. **Alcohol and Smoking:** 55% rarely drink, and 15.5% smoke, which affects health risks, particularly respiratory and cardiovascular diseases.
10. **Demographics:** The population is skewed toward males (65.7%), with most beneficiaries aged around 45, which is a peak age for health insurance.

2.6.2. Based on Bivariate Analysis

1. **Weight & Insurance Cost:** Higher weight correlates with higher insurance costs, as it increases the risk of health issues like heart disease.
2. **Insurance History:** Individuals with coverage from other insurers tend to have higher premiums, reflecting perceived higher risk.
3. **Health-related Factors:** Higher BMI and cholesterol levels correlate with increased premiums, due to the associated health risks. A history of heart disease significantly increases premiums.
4. **Behavioural Factors:** Smoking and alcohol have a minimal direct impact, but both contribute to indirect health risks. Regular exercisers tend to have lower premiums due to better health.
5. **Age & Gender:** Middle-aged groups (26-35, 46-55) face higher premiums due to a higher prevalence of heart disease. Gender has a minimal effect, though men are at higher risk.
6. **Health Conditions & Medical History:** Older individuals, those with a higher fat percentage, and those who don't exercise are more likely to face higher healthcare costs and premiums due to increased health risks.

2.6 Business Impact from Exploratory Data Analysis

1. **Improved Customer Acquisition Strategy:** By understanding the patterns and behaviours of potential customers, the company can efficiently allocate its marketing budget, leading to increased acquisition rates.
2. **Claims Management and Fraud Detection:** With better predictions of claims and a deeper understanding of customer behaviours, the company can optimize insurance payouts, reduce fraudulent claims, and effectively manage costs.
3. **Operational Efficiency and Cost Optimization:** Insights from the data allow the company to streamline operations, reduce unnecessary costs, and improve service delivery by better managing resources.
4. **Enhanced Customer Experience and Service Delivery:** The analysis provides a way to personalize customer interactions, improving customer satisfaction and engagement, while also streamlining the claims process for a smoother experience.
5. **Regulatory Compliance and Risk Mitigation:** By identifying patterns that may indicate non-compliance or risk, the company can reduce legal and financial risks, ensuring it adheres to regulatory standards and building customer trust with fair practices.
6. **Trend Analysis and Future Growth Opportunities:** By understanding emerging trends, the company can seize new opportunities, anticipate customer shifts, and prepare for future growth with innovative products.
7. **Customer Education and Engagement:** Through data-driven insights, the company can run targeted educational campaigns that improve customer health behaviours and engagement, fostering long-term loyalty by proactively addressing customer needs.

3. Data Preprocessing

The columns 'applicant_id' and 'Location' have been removed because they do not provide significant value to the data. The 'applicant_id' consists of unique entries, and the 'Location' column does not appear to contribute meaningfully in the exploratory data analysis.

3.1. Duplicate value check

- No duplicate values are found in the data set.

3.2. Missing value treatment

3.2.1. Missing value check

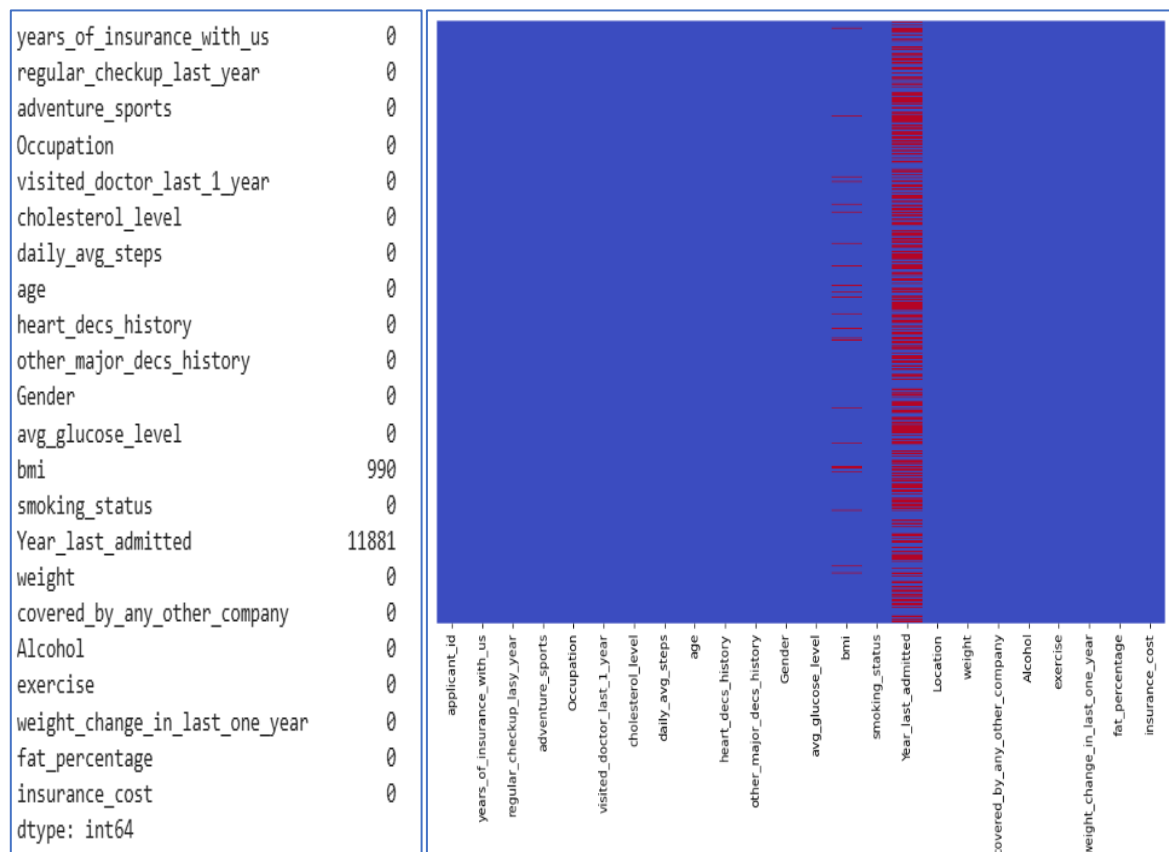


Table 7. Missing values in the data set

OBSERVATIONS:

- There are missing values in 2 columns, namely 'bmi' and 'Year_last_admitted' which represent 3.96% and 47.5% of the data set respectively.

3.2.2. Imputing null values in the bmi column

The missing values in the 'bmi' column are scaled using **StandardScaler** and further imputed using **KNN Imputer (number of nearest neighbours as 5)** which works by filling missing values based on the values of the nearest neighbours in the dataset.

years_of_insurance_with_us	0
regular_checkup_last_year	0
adventure_sports	0
Occupation	0
visited_doctor_last_1_year	0
cholesterol_level	0
daily_avg_steps	0
age	0
heart_decs_history	0
other_major_decs_history	0
Gender	0
avg_glucose_level	0
bmi	0
smoking_status	0
Year_last_admitted	11881
weight	0
covered_by_any_other_company	0
Alcohol	0
exercise	0
weight_change_in_last_one_year	0
fat_percentage	0
insurance_cost	0
dtype: int64	

Table 8. Missing values after imputing bmi column

3.2.3. Imputing null values in the Year_last_admitted column

The column 'Year_last_admitted' is used to create a new column, 'Year_since_last_admitted', which is calculated by subtracting the value from the current year. Missing values in this new column are imputed with -1, indicating that these values are excluded from calculations and will not skew or bias the dataset.

years_of_insurance_with_us	0
regular_checkup_last_year	0
adventure_sports	0
Occupation	0
visited_doctor_last_1_year	0
cholesterol_level	0
daily_avg_steps	0
age	0
heart_decs_history	0
other_major_decs_history	0
Gender	0
avg_glucose_level	0
bmi	0
smoking_status	0
weight	0
covered_by_any_other_company	0
Alcohol	0
exercise	0
weight_change_in_last_one_year	0
fat_percentage	0
insurance_cost	0
Years_since_last_admitted	0
dtype: int64	

Table 9. Missing values after imputing bmi and Year_last_admitted column

3.3. Outlier Detection and Treatment

3.3.1. Outlier Detection

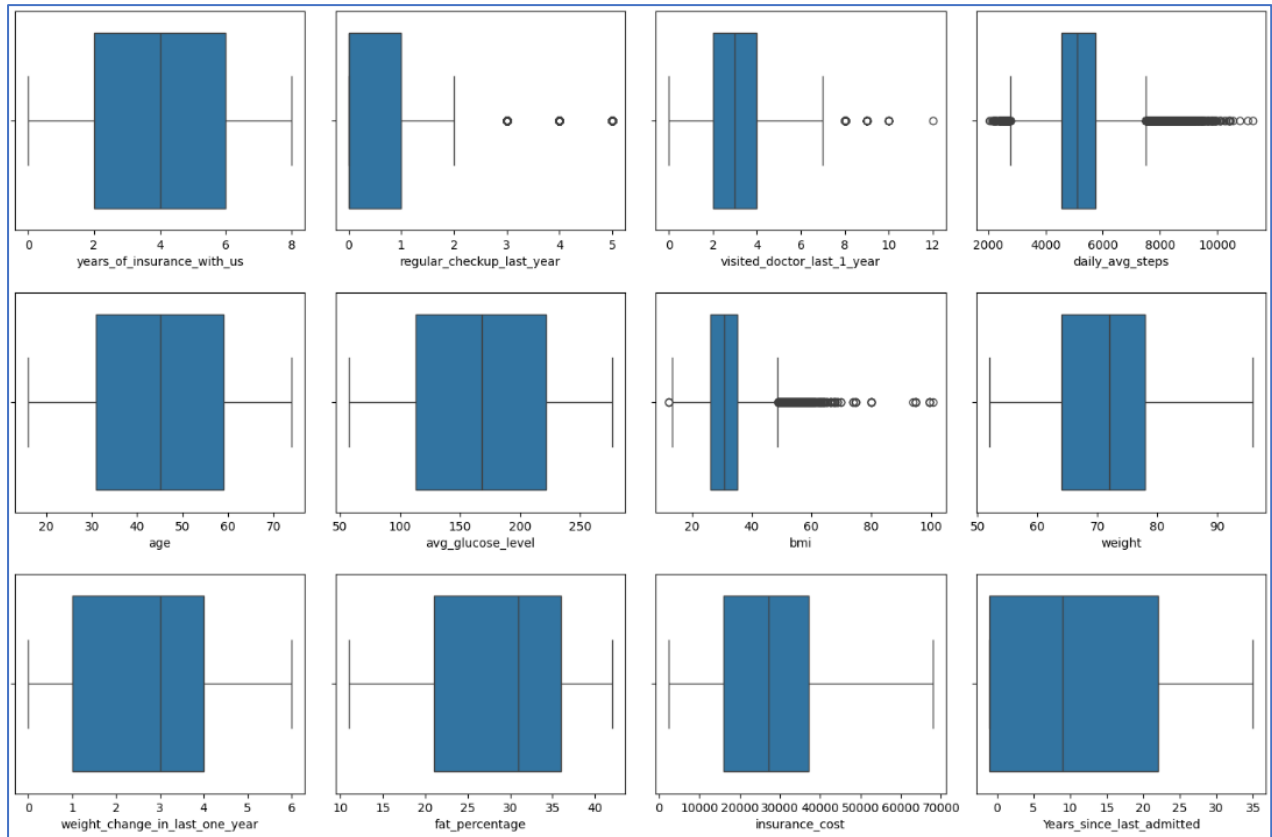


Fig 3. Outlier detection

OBSERVATIONS:

- Outliers are observed in four variables of the dataset: regular_checkup_last_year, visited_doctor_last_1_year, daily_avg_steps and bmi.

3.3.2. Outlier Treatment

3.3.2.1. regular_checkup_last_year and visited_doctor_last_1_year

- The maximum number of regular check-ups is 5, and the maximum number of doctor visits is 12.
- These values are deemed acceptable as individuals with health conditions may have had a higher number of doctor visits.
- Given that these elevated figures could reflect significant health issues and represent valid cases, it may be appropriate to retain these outliers.

3.3.2.2. daily_avg_steps

- The daily_avg_steps is further analysed to explore the distribution of individuals at both the upper and lower limits, to understand its relationship with BMI, fat percentage, and weight.
- For individuals in the upper limit, the average weight is 71.2, average BMI is 31, and average fat percentage ranges between 28-30. While the correlation between these parameters appears weak, it's plausible that people with higher BMI and fat percentage may walk more steps. Alternatively, the higher step count could result from daily activities that aren't necessarily health related. Therefore, the upper limit is considered valid data.
- A similar analysis is done for the lower limit, where the average weight is within similar ranges. Most individuals in this group engage in moderate or extreme exercise activities.
- Given this, the daily_avg_steps appears to represent valid cases, and retaining these outliers is deemed appropriate.

3.3.2.3. Bmi

- A BMI value exceeding 50 or 60 is unusually high and likely represents an outlier.
- This value is far beyond the typical BMI range, which generally peaks around 40-50 for individuals classified as morbidly obese. Such a high BMI could indicate significant data errors, or it may represent extreme cases of obesity.
- Interestingly, the weight and fat percentage of individuals with these extremely high BMI values appear normal. This suggests that these values are likely data errors, which could negatively impact the model. There are approximately 550 observations with such high BMI values.
- Therefore, these outliers are capped using Interquartile Range (IQR) method, where the limits are calculated based on the Q1 (25th percentile) and Q3 (75th percentile) values, and the outliers are defined as values outside the range of $Q1 - 1.5 * IQR$ and $Q3 + 1.5 * IQR$.
- The values below the lower limit are replaced with the lower limit (12.8) and the values above the upper limit are replaced with the upper limit (48.5).

4. Model Building

Data preparation is a key step in model building as it directly impacts model performance by ensuring the data is clean, structured, and relevant.

Key steps include:

- **Dropping the Target Variable:** Removing the target variable (insurance cost) before encoding prevents data leakage and ensures the model learns only from the predictors.
- **Encoding Categorical Features:** Converting categorical features into numerical values enables the model to process them effectively.
- **Including an Intercept:** Adding an intercept allows the model to capture the baseline factors influencing the target variable.
- **Creating Dummy Variables:** Dummy variables convert categorical variables into binary format, ensuring the model evaluates each category's effect on the target without assuming any order.
- **Splitting the Data into Train and Test:** The data is divided into 70% for training and 30% for testing, ensuring the model is evaluated on unseen data for better performance assessment and generalization.

```
Number of rows in train data = 17500
Number of rows in test data = 7500
```

The dataset is a regression problem aimed at predicting personalized insurance premiums based on an individual's health and lifestyle factors.

4.1. Model 1: LINEAR REGRESSION

Linear regression is a great baseline model as it efficiently handles multiple predictors, showing their collective impact on insurance costs, while being computationally simple and ideal for large datasets.

4.1.1. Justification for Model Selection

- Linear models are often a good starting point. The dataset includes both dependent and independent variables, making linear regression an ideal choice for modeling the relationship between the dependent and predictor variables.
- The exploratory data analysis revealed a strong positive correlation between the predictor variable 'weight' and the target variable 'insurance_cost' which indicates that as weight increases, insurance cost also increases, implying a direct, proportional relationship between the two variables.
- This model was initially tested for understanding basic relationships between predictors and insurance cost.
- Feature selection is essential, and linear regression is justified when predictors are carefully selected, either based on domain expertise or by using the VIF test to remove variables with high VIF values and p-values greater than 0.05.
- Linear regression is often favored for its simplicity and interpretability, as the model's coefficients directly highlight the impact of each predictor variable on the dependent variable.

4.1.2. Building Linear Regression Model using the Train Data

The linear regression model uses the Ordinary Least Squares (OLS) method to estimate parameters by minimizing the sum of squared differences between observed and predicted values.

4.1.3. Linear Regression Results

- Adjusted. R-squared is 0.945 which is a higher value and indicates a better fit, assuming certain conditions are met.
- The Y-intercept of -78,590 represents the predicted insurance cost when all predictors are zero. This result might indicate that the model is not capturing certain real-world constraints or that the relationship between the predictors and target variable needs to be adjusted, where the linear regression assumes linear relationship between variables.

OLS Regression Results						
=====						
Dep. Variable:	insurance_cost	R-squared:	0.945			
Model:	OLS	Adj. R-squared:	0.945			
Method:	Least Squares	F-statistic:	1.039e+04			
Date:	Fri, 14 Mar 2025	Prob (F-statistic):	0.00			
Time:	13:24:03	Log-Likelihood:	-1.6695e+05			
No. Observations:	17500	AIC:	3.340e+05			
DF Residuals:	17470	BIC:	3.342e+05			
DF Model:	29					
Covariance Type:	nonrobust					
=====						
	coef	std err	t	P> t	[0.025	0.975]
const	-7.859e+04	380.982	-206.270	0.000	-7.93e+04	-7.78e+04
years_of_insurance_with_us	-103.5772	11.987	-8.641	0.000	-127.074	-80.081
regular_checkup_last_year	-366.2630	22.333	-16.400	0.000	-410.037	-322.489
visited_doctor_last_1_year	-35.5396	22.897	-1.552	0.121	-80.420	9.341
daily_avg_steps	-0.0300	0.025	-1.203	0.229	-0.079	0.019
age	2.7522	1.581	1.741	0.082	-0.346	5.851
avg_glucose_level	0.3316	0.406	0.817	0.414	-0.464	1.127
bmi	-0.7016	3.929	-0.179	0.858	-8.403	7.000
weight	1470.6918	3.288	447.272	0.000	1464.247	1477.137
weight_change_in_last_one_year	171.5432	16.269	10.544	0.000	139.655	203.432
fat_percentage	-0.9141	3.301	-0.277	0.782	-7.384	5.555
Years_since_last_admitted	39.6287	2.804	14.132	0.000	34.132	45.125
adventure_sports_Yes	117.3540	93.586	1.254	0.210	-66.085	300.793
Occupation_Salaried	32.7082	138.567	0.236	0.813	-238.896	304.313
Occupation_Student	55.2799	88.230	0.627	0.531	-117.660	228.220
cholesterol_level_150 to 175	-46.0294	81.836	-0.562	0.574	-206.436	114.377
cholesterol_level_175 to 200	11.5155	129.466	0.089	0.929	-242.252	265.283
cholesterol_level_200 to 225	89.5378	129.311	0.692	0.489	-163.925	343.001
cholesterol_level_225 to 250	146.9884	122.618	1.199	0.231	-93.355	387.331
heart_decs_history_Yes	123.3886	114.666	1.076	0.282	-101.368	348.145
other_major_decs_history_Yes	65.6458	88.765	0.740	0.460	-108.343	239.635
Gender_Male	46.8420	60.143	0.779	0.436	-71.044	164.728
smoking_status_formerly smoked	-61.1212	81.503	-0.750	0.453	-220.875	98.632
smoking_status_never smoked	19.5694	65.977	0.297	0.767	-109.752	148.891
smoking_status_smokes	-33.9780	83.408	-0.407	0.684	-197.467	129.511
covered_by_any_other_company_Yes	1140.0480	57.681	19.765	0.000	1026.988	1253.108
Alcohol_No	-32.3856	90.255	-0.359	0.720	-209.294	144.523
Alcohol_Rare	-10.2360	85.056	-0.120	0.904	-176.954	156.482
exercise_Moderate	13.9480	65.111	0.214	0.830	-113.675	141.571
exercise_No	11.5342	79.709	0.145	0.885	-144.704	167.772
=====						
Omnibus:	558.091	Durbin-Watson:	1.981			
Prob(Omnibus):	0.000	Jarque-Bera (JB):	669.168			
Skew:	0.393	Prob(JB):	4.92e-146			
Kurtosis:	3.546	Cond. No.	8.19e+04			
=====						
Notes:						
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.						
[2] The condition number is large, 8.19e+04. This might indicate that there are strong multicollinearity or other numerical problems.						

Table 10. Linear Regression Result-1

4.1.4. Model Performance Evaluation

To evaluate the performance of a regression model, several performance metrics helps measure how well the model fits the data and how accurate its predictions are.

- **Root Mean Squared Error (RMSE):** Measures average prediction error, with larger errors weighted more. Lower RMSE indicates better model fit.
- **Mean Absolute Error (MAE):** Averages absolute prediction errors. Lower MAE means smaller average error.
- **R-squared:** Shows the proportion of variance explained by the model. Higher values indicate better performance, but can be misleading in case of overfitting.
- **Adjusted R-squared:** Adjusts R-squared for the number of predictors, penalizing irrelevant ones. Higher values indicate a better model fit.
- **Mean Absolute Percentage Error (MAPE):** Measures prediction accuracy in percentage terms. Lower MAPE values indicate better performance.

Checking model performance on Train Data

Training Performance					
	RMSE	MAE	R-squared	Adj. R-squared	MAPE
0	3364.244496	2707.519836	0.945185	0.94509	15.224003

Checking model performance on Test Data

Test Performance					
	RMSE	MAE	R-squared	Adj. R-squared	MAPE
0	3322.370014	2693.729313	0.945377	0.945158	14.98055

Observations:

- The model's adjusted R-squared of 0.945 indicates it is not underfitting.
- MAE of 2693.72 shows the model predicts insurance cost with a mean error of around 2693.72 on the test data.
- MAPE of 14.98% suggests predictions are within 14.98% of the actual insurance cost.
- RMSE is slightly lower for the test set (3322.37) than the training set (3364.24), suggesting good generalization to new data.

4.1.5. Checking Linear Regression Assumptions

Linear regression models rely on certain assumptions to ensure valid results. Violations of these assumptions can lead to biased or inaccurate predictions.

4.1.5.1. Test for Multicollinearity

- **VIF result:** Most predictors have low VIFs (below 5), indicating minimal multicollinearity, though some variables have moderate VIFs (over 2), suggesting slight correlations.
- **High p-value variables:** Variables with p-values > 0.05 were dropped iteratively. The final model had no variables with p-values > 0.05, with an adjusted R-squared of 0.945, indicating that 94.5% of variance is explained.

- **Model performance on train and test data:** The adjusted R-squared values for both training and test sets were 0.945, showing no overfitting and confirming the assumption of multicollinearity is satisfied.

4.1.5.2. Test for Linearity and Independence

- No patterns were observed in the residuals vs. fitted values plot, suggesting linearity and independence are satisfied.

4.1.5.3. Test for Normality

- The histogram of residuals is slightly right-skewed.
- The Q-Q plot shows that residuals mostly follow a straight line, except for the tails.
- Shapiro-Wilk test indicated non-normal residuals (p-value < 0.05), suggesting a deviation from normality. Thus, the assumption of normality is not satisfied.

4.1.5.4. Test for Homoscedasticity

- The Goldfeld-Quandt test p-value was higher than 0.05, confirming that residuals are homoscedastic.
- Thus, the assumption of homoscedasticity is satisfied.

4.1.6. Final Model

OLS Regression Results						
=====						
Dep. Variable:	insurance_cost	R-squared:	0.945			
Model:	OLS	Adj. R-squared:	0.945			
Method:	Least Squares	F-statistic:	5.022e+04			
Date:	Sat, 15 Mar 2025	Prob (F-statistic):	0.00			
Time:	08:43:24	Log-Likelihood:	-1.6696e+05			
No. Observations:	17500	AIC:	3.339e+05			
Df Residuals:	17493	BIC:	3.340e+05			
Df Model:	6					
Covariance Type:	nonrobust					
=====						
	coef	std err	t	P> t	[0.025	0.975]

const	-7.865e+04	256.920	-306.135	0.000	-7.92e+04	-7.81e+04
years_of_insurance_with_us	-104.1256	11.979	-8.692	0.000	-127.607	-80.645
regular_checkup_last_year	-365.3913	22.312	-16.376	0.000	-409.126	-321.657
weight	1470.7921	3.281	448.294	0.000	1464.361	1477.223
weight_change_in_last_one_year	171.8021	16.252	10.571	0.000	139.947	203.657
Years_since_last_admitted	39.6480	2.802	14.151	0.000	34.156	45.140
covered_by_any_other_company_Yes	1141.1216	57.626	19.802	0.000	1028.169	1254.074
=====						
Omnibus:	560.169	Durbin-Watson:	1.980			
Prob(Omnibus):	0.000	Jarque-Bera (JB):	672.015			
Skew:	0.394	Prob(JB):	1.19e-146			
Kurtosis:	3.548	Cond. No.	740.			
=====						
Notes:						
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.						

Table 11. Linear Regression Result-Final

Checking model performance on Train Data

Training Performance					
	RMSE	MAE	R-squared	Adj. R-squared	MAPE
0	3365.987404	2709.141195	0.945128	0.945106	15.23184

Checking model performance on Test Data

Test Performance					
	RMSE	MAE	R-squared	Adj. R-squared	MAPE
0	3323.951853	2695.567318	0.945325	0.945274	14.988497

Observations:

- The model is able to explain approximately 94.5% of the variation in the data.
- The train and test RMSE and MAE are comparable. So, the model is not suffering from overfitting. The MAPE on the test set suggests we can predict within 14.98% of the insurance cost.

Equation of linear regression

The equation of linear regression obtained from the data set is calculated as follows:

$$\text{insurance_cost} = -78652.26390920352 + -104.12560991566231 * (\text{years_of_insurance_with_us}) + -365.39134347650344 * (\text{regular_checkup_last_year}) + 1470.7921358383692 * (\text{weight}) + 171.8021349080226 * (\text{weight_change_in_last_one_year}) + 39.64803991996878 * (\text{Years_since_last_admitted}) + 1141.1215941987866 * (\text{covered_by_any_other_company_Yes})$$

OBSERVATIONS:

- The linear regression model identifies key predictors for insurance cost, including years of insurance, regular checkups, weight, weight change, years since last admission, and whether the individual has additional insurance coverage.
- However, the negative intercept suggests potential issues with the model, indicating that the linear regression approach may not effectively capture the relationships between variables.
- This calls for exploring alternative modelling techniques to improve the model's fit and address the data's characteristics.

4.2. ENSEMBLE METHODS

Ensemble Methods are techniques that combine the predictions of multiple models to improve the overall performance and accuracy compared to individual models. These methods work on the principle that a group of weak learners (models) can produce a stronger learner, leading to better results.

4.2.1. Justification for Model Selection

- These models were chosen because they can handle the complex interactions and non-linear relationships between features.
- They are also more robust to outliers and noise. Considering the potential presence of outliers in the dataset, particularly for variables like bmi and daily_avg_steps, ensemble methods help mitigate the impact of these outliers and provides more stable and accurate predictions by averaging the outputs of several models.

- By combining predictions from multiple models instead of depending on a single regression model, these methods are likely to yield more accurate insurance cost predictions for a broad spectrum of individuals, particularly those with varying health histories and lifestyles.
- The exploratory analysis revealed important insights, such as the impact of covered_by_any_other_company and weight on insurance_cost. Specifically, individuals with the same weight who have insurance from another company tend to pay higher premiums. The interaction between covered_by_any_other_company and weight might influence the insurance premium in ways that a single model might overlook.
- The ensemble methods offer valuable insights into which features are the most important for predicting the target variable. It does this by calculating feature importance, which ranks the predictors based on their contribution to reducing prediction error.
- They also minimize the risk of overfitting compared to individual decision trees by averaging the predictions from multiple trees, which helps in creating a more stable and accurate model.
- The current data set comprises a mix of numerical and categorical data, which can be handled by the ensemble methods more effectively. Random Forests specifically can handle categorical variables by splitting them into multiple branches based on their values, while simultaneously capturing complex relationships between numerical features.
- Hyperparameter tuning offers the advantage of improving model performance by finding the right balance between underfitting and overfitting.

4.2.2. MODEL 2: BAGGING REGRESSOR

The Bagging Regressor improves prediction stability by training multiple models on random subsets of data and combining their outputs. It reduces variance, providing more reliable insurance cost predictions.

Data preparation is similar to linear regression, with the only difference being the inclusion of an intercept for linear regression.

4.2.2.1. Fitting the Model on Train Data

The Bagging Regressor is initialized and trained on the provided training data. The random_state parameter ensures that the bootstrapping process is consistent, and the model learns from multiple subsets of the data to improve accuracy and reduce overfitting.

```
BaggingRegressor
BaggingRegressor(random_state=1)
```

4.2.2.2. Model Performance Evaluation

The evaluation metrics used are RMSE, MAE, R-squared, Adjusted R-squared, and MAPE, as described above.

Checking model performance on Train Data

	RMSE	MAE	R-squared	Adj. R-squared	MAPE
0	1338.711021	987.4468	0.99132	0.991306	4.723092

Checking model performance on Test Data

	RMSE	MAE	R-squared	Adj. R-squared	MAPE
0	3193.89683	2520.107707	0.94952	0.949324	12.054849

Observations:

- The model shows overfitting, as evidenced by a high R-squared (0.99) on training data and a lower R-squared (0.94) on test data.
- The significant drop in performance on test data and higher MAPE suggest that the model is too tailored to the training data and doesn't generalize well to unseen data.

4.2.3. Model 3: RANDOM FOREST

Random Forest is an ensemble method that combines multiple decision trees to enhance accuracy and reduce overfitting. It uses bootstrapped samples of the training data, making it a powerful and robust algorithm for complex data.

The data preparation remains the same as with linear regression, with the addition of an intercept for linear models.

4.2.3.1 Fitting the Model on Train Data

The RandomForestRegressor is initialized and trained on the training data. The random_state=1 ensures that the process is reproducible.

```
RandomForestRegressor
RandomForestRegressor(random_state=1)
```

4.2.3.2. Model Performance Evaluation

The evaluation metrics used are RMSE, MAE, R-squared, Adjusted R-squared, and MAPE, as described above.

Checking model performance on Train Data

	RMSE	MAE	R-squared	Adj. R-squared	MAPE
0	1138.41311	896.518629	0.993723	0.993713	4.308465

Checking model performance on Test Data

	RMSE	MAE	R-squared	Adj. R-squared	MAPE
0	3055.870644	2421.99648	0.953789	0.953609	11.562507

Observations:

- The model shows overfitting, with high performance on the training data (R-squared: 0.9937) but poor performance on the test data (R-squared: 0.9538).
- The low errors on training data are misleading, as the model struggles with larger errors on the test data, indicating a need for better generalization.

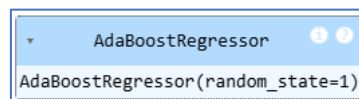
4.2.4. Model 4: ADABOOST REGRESSOR

AdaBoost Regressor improves accuracy by focusing on misclassified data points and handling complex relationships. It is less prone to overfitting, making it a strong choice for the dataset.

Data preparation remains the same, with the exception of including an intercept for linear regression.

4.2.4.1. Fitting the Model on Train Data

AdaBoostRegressor is the model being initialized which is tailored for regression tasks. The `random_state=1` ensures that the model's training process is reproducible.



```
AdaBoostRegressor(random_state=1)
```

4.2.4.2. Model Performance Evaluation

The evaluation metrics used are RMSE, MAE, R-squared, Adjusted R-squared, and MAPE, as described above.

Checking model performance on Train Data

	RMSE	MAE	R-squared	Adj. R-squared	MAPE
0	3249.676128	2672.964057	0.948854	0.948769	15.018419

Checking model performance on Test Data

	RMSE	MAE	R-squared	Adj. R-squared	MAPE
0	3247.414907	2671.945571	0.947814	0.947611	14.985721

Observations:

- The model shows good generalization, with similar performance on both train and test data, indicating it is not overfitting. High R-squared values (94.88% for train, 94.78% for test) suggest a strong fit, though RMSE and MAE are relatively high, indicating room for improvement in prediction accuracy.
- MAPE is around 15%, reflecting moderate error.

4.2.5. Model 5: GRADIENT BOOSTING REGRESSOR

Gradient Boosting is an ensemble technique that builds models sequentially, correcting errors from previous models. It handles complex, non-linear relationships well, making it ideal for datasets like customer health and insurance data.

Data preparation remains the same, with the exception of including an intercept for linear regression.

4.2.5.1. Fitting the Model on Train Data

GradientBoostRegressor is the model being initialized which is tailored for regression tasks. The `random_state=1` ensures that the model's training process is reproducible.

```
GradientBoostRegressor
GradientBoostRegressor(random_state=1)
```

4.2.5.2. Model Performance Evaluation

The evaluation metrics used are RMSE, MAE, R-squared, Adjusted R-squared, and MAPE, as described above.

Checking model performance on Train Data

	RMSE	MAE	R-squared	Adj. R-squared	MAPE
0	2927.596536	2354.122354	0.95849	0.958421	11.487394

Checking model performance on Test Data

	RMSE	MAE	R-squared	Adj. R-squared	MAPE
0	2970.611098	2392.078909	0.956331	0.956162	11.612841

Observations:

- The model shows similar metrics for both training and test data, indicating good generalization and no overfitting.
- High R-squared values (0.95849 for training, 0.95633 for test) suggest the model explains 96% of the variance.
- RMSE and MAE values suggest some prediction errors, while MAPE of 11.5% indicates moderate prediction accuracy with room for improvement.

4.2.6. Model 6: XGBOOST REGRESSOR

XGBoost (eXtreme Gradient Boosting) is an optimized version of Gradient Boosting, known for its speed, efficiency, and scalability. It performs well in regression tasks, especially with structured data, and provides valuable feature importance insights.

The data preparation remains the same, with the only difference being the inclusion of an intercept in linear regression.

4.2.6.1. Fitting the Model on Train Data

XGBRegressor is the model being initialized which is tailored for regression tasks. The `random_state=1` ensures that the model's training process is reproducible.

```
XGBRegressor
XGBRegressor(base_score=None, booster=None, callbacks=None,
              colsample_bylevel=None, colsample_bynode=None,
              colsample_bytree=None, device=None, early_stopping_rounds=None,
              enable_categorical=False, eval_metric=None, feature_types=None,
              gamma=None, grow_policy=None, importance_type=None,
              interaction_constraints=None, learning_rate=None, max_bin=None,
              max_cat_threshold=None, max_cat_to_onehot=None,
              max_delta_step=None, max_depth=None, max_leaves=None,
              min_child_weight=None, missing=nan, monotone_constraints=None,
              multi_strategy=None, n_estimators=None, n_jobs=None,
              num_parallel_tree=None, random_state=1, ...)
```

4.2.6.2. Model Performance Evaluation

The evaluation metrics used are RMSE, MAE, R-squared, Adjusted R-squared, and MAPE, as described above.

Checking model performance on Train Data

	RMSE	MAE	R-squared	Adj. R-squared	MAPE
0	2112.222275	1658.957079	0.978392	0.978356	8.220348

Checking model performance on Test Data

	RMSE	MAE	R-squared	Adj. R-squared	MAPE
0	3088.892844	2459.27551	0.952784	0.952601	11.835337

Observations:

- The model performs well with an R-squared of 97.84% on the training set, capturing most of the variance.
- However, test metrics show slightly worse performance with increased RMSE, MAE, and a drop in R-squared.
- The test MAPE (11.84%) is higher than the training MAPE (8.22%), indicating room for improvement.
- Despite this, the model shows a good balance of fit and generalization, with robust performance overall.

4.3. Model 7: DECISION TREE

A Decision Tree is a supervised learning algorithm that splits data based on features to minimize prediction error. It's ideal for handling both numerical and categorical data, capturing non-linear relationships, and offering interpretability.

4.3.1. Justification For Model Selection

- This model was selected for its interpretability and ability to capture non-linear relationships and feature interactions.
- It is easy to understand and can directly show how different feature thresholds impact insurance cost predictions.
- The model is simple, efficient, and can be pruned to avoid overfitting, ensuring good generalization.
- Decision trees can be trained quickly, making them suitable for large-scale prediction tasks.

4.3.2. Fitting the Model on Train Data

DecisionTreeRegressor is the model being initialized which is tailored for regression tasks. The `random_state=1` ensures that the model's training process is reproducible.

```
DecisionTreeRegressor
DecisionTreeRegressor(random_state=1)
```

4.3.3. Model Performance Evaluation

The evaluation metrics used are RMSE, MAE, R-squared, Adjusted R-squared, and MAPE, as described above.

Checking model performance on Train Data

	RMSE	MAE	R-squared	Adj. R-squared	MAPE
0	0.0	0.0	1.0	1.0	0.0

Checking model performance on Test Data

	RMSE	MAE	R-squared	Adj. R-squared	MAPE
0	4236.557614	3284.249867	0.911181	0.910836	15.37568

Observations:

- The model performs perfectly on the training set (RMSE=0, MAE=0), but struggles on the test set with high RMSE, MAE, and a significant drop in R-squared.
- This indicates overfitting, where the model has learned noise and outliers in the training data, likely due to an overly deep tree or insufficient pruning.

5. Model Performance Improvement

To improve model performance:

- **Hyperparameter Tuning:** Grid Search systematically tests all hyperparameter combinations using cross-validation to find the optimal values.
- **Pruning Decision Trees:** Limiting depth and splitting constraints helps reduce tree complexity, improving generalization and preventing overfitting.

5.3. BAGGING REGRESSOR

5.3.2. Model Tuning

Grid search is used for hyperparameter tuning of the model. The parameters include:

- **max_samples:** Specifies the fraction of the dataset used for training each model in the ensemble, trying values of 0.7, 0.8, and 0.9.
- **max_features:** Defines the fraction of features used for training each model, with values of 0.7, 0.8, and 0.9.
- **n_estimators:** The number of base learners in the ensemble, tested from 90 to 110 in steps of 10.

A 5-fold cross-validation (cv=5) splits the data into 5 parts, training the model on 4 and validating on the remaining part. The best model with optimized hyperparameters is then trained on the full dataset.

```
BaggingRegressor
BaggingRegressor(max_features=0.9, max_samples=0.7, n_estimators=100,
random_state=1)
```

5.3.3. Model Performance Evaluation

The evaluation metrics used are RMSE, MAE, R-squared, Adjusted R-squared, and MAPE, as described above.

Checking model performance on Train Data

	RMSE	MAE	R-squared	Adj. R-squared	MAPE
0	1627.857136	1292.683399	0.987166	0.987145	6.646224

Checking model performance on Test Data

	RMSE	MAE	R-squared	Adj. R-squared	MAPE
0	3220.033696	2558.292603	0.94869	0.948491	13.051561

Observations:

- The model is performing well on the training set, with high R-squared (98.72) and low errors (RMSE: 1627.86, MAE: 1292.68), indicating it fits the training data well.
- However, it shows signs of overfitting, as its performance on the test set drops (R-squared: 94.87, RMSE: 3220.03, MAE: 2558.29).
- The test set errors are significantly higher, suggesting the model struggles to generalize to new data. Improvements in model generalization are needed for better performance on unseen data.
- The model emphasizes the key factors that influence insurance costs, helping us identify which variables have the most significant impact.
- Among these, heart_decs_history, visited_doctor_last_1_year, and covered_by_any_other_company are the top three features contributing most to the prediction.

5.4. RANDOM FOREST

5.4.2. Model Tuning

Grid search is used for hyperparameter tuning of the model.

The parameters include:

- **max_depth:** This defines the maximum depth of each tree. It can be an integer value which means there's no maximum depth, and trees will expand until a stopping criterion is met.
- **max_features:** This specifies the number of features to consider when splitting a node. It can be sqrt, log2 or None, where the sqrt uses the square root of the total number of features, log2 uses the base-2 logarithm of the total features and None uses all features.
- **n_estimators:** This defines the number of trees in the forest. More trees usually improve model performance but can also lead to higher computational costs.

This parameter grid will be used to try different combinations of hyperparameters to find the best configuration for the model.

A 5-fold cross-validation (cv=5) is applied, which means the data will be split into 5 parts, and the model will be trained on 4 parts and validated on the remaining part, repeated 5 times.

The best model with the optimized hyperparameters to the full training dataset is initialized.

```
RandomForestRegressor(max_depth=8, max_features=None, n_estimators=110,
random_state=1)
```

5.4.3. Model Performance Evaluation

The evaluation metrics used are RMSE, MAE, R-squared, Adjusted R-squared, and MAPE, as described above.

Checking model performance on Train Data

	RMSE	MAE	R-squared	Adj. R-squared	MAPE
0	2766.609652	2211.52471	0.96293	0.962868	10.632904

Checking model performance on Test Data

	RMSE	MAE	R-squared	Adj. R-squared	MAPE
0	2998.60661	2383.236471	0.955504	0.955331	11.344109

OBSERVATIONS:

- The model performs well on the training set, with a strong R-squared of 96.29 and low error metrics (RMSE: 2766.61, MAE: 2211.52), indicating it fits the training data well.
- However, its performance on the test set drops slightly (R-squared: 95.55, RMSE: 2998.61, MAE: 2383.24).
- The test set errors are higher, suggesting the model faces some difficulty in generalizing to unseen data. While the model is effective, improvements in generalization could enhance its performance on new data.

5.5. ADABOOST REGRESSOR

5.5.2. Model Tuning

Grid search is used for hyperparameter tuning of the model. The parameters include:

- **n_estimators**: This gives the number of weak learners (decision trees) to use in the boosting process. The grid searches values between 10 and 100 (inclusive), in steps of 10.
- **learning_rate**: This controls the contribution of each weak learner to the final prediction. A lower learning rate means the model will learn slower, while a higher learning rate speeds up learning but may overfit. The grid explores values [1, 0.1, 0.5, 0.01].

A 5-fold cross-validation (cv=5) splits the data into 5 parts, training the model on 4 and validating on the remaining part. The best model with optimized hyperparameters is then trained on the full dataset.

```
AdaBoostRegressor(learning_rate=0.1, n_estimators=90, random_state=1)
```

5.5.3. Model Performance Evaluation

The evaluation metrics used are RMSE, MAE, R-squared, Adjusted R-squared, and MAPE, as described above.

Checking model performance on Train Data

	RMSE	MAE	R-squared	Adj. R-squared	MAPE
0	3243.498879	2641.603294	0.949049	0.948964	14.479256

Checking model performance on Test Data

	RMSE	MAE	R-squared	Adj. R-squared	MAPE
0	3257.688577	2647.570537	0.947483	0.947279	14.512683

OBSERVATIONS:

- The model performs well on both the training and test sets, with an R-squared of 94.90% on the training set and 94.75% on the test set, indicating a solid fit to the data.
- The error metrics (RMSE: 3243.5 on the training set, 3257.7 on the test set; MAE: 2641.6 on the training set, 2647.6 on the test set) are relatively close, suggesting that the model generalizes well and is not overfitting.

The model emphasizes the key factors that influence insurance costs, helping us identify which variables have the most significant impact. Among these, weight is only feature contributing most to the prediction.

5.6. GRADIENT BOOSTING REGRESSOR

5.6.2. Model Tuning

Grid search is used for hyperparameter tuning of the model. The parameters include:

- **n_estimators:** This gives the number of boosting stages to run. The grid searches for values between 100 and 300, with increments of 50.
- **subsample:** This gives the fraction of samples used for fitting each base learner. The grid tests values [0.8, 0.9, 1], where lower values reduce variance but can increase bias.
- **max_features:** This gives the number of features to consider when looking for the best split. The grid searches values [0.8, 0.9, 1].
- **max_depth:** This gives the maximum depth of the individual trees in the boosting process. The grid checks for depths of 5, 7, and 10, where deeper trees can capture more complex relationships but may risk overfitting.

A 5-fold cross-validation (cv=5) splits the data into 5 parts, training the model on 4 and validating on the remaining part. The best model with optimized hyperparameters is then trained on the full dataset.

```
GradientBoostingRegressor
GradientBoostingRegressor(max_depth=5, max_features=0.9, random_state=1,
                           subsample=0.9)
```

5.6.3. Model Performance Evaluation

The evaluation metrics used are RMSE, MAE, R-squared, Adjusted R-squared, and MAPE, as described above.

Checking model performance on Train Data

	RMSE	MAE	R-squared	Adj. R-squared	MAPE
0	2744.25139	2199.190921	0.963527	0.963466	10.720271

Checking model performance on Test Data

	RMSE	MAE	R-squared	Adj. R-squared	MAPE
0	2959.984817	2369.445986	0.956643	0.956475	11.417282

OBSERVATIONS:

- The model shows strong performance with an R-squared of 96.35% on the training set and 95.66% on the test set, indicating a good fit.
- Error metrics (RMSE and MAE) are close between training and test sets, suggesting good generalization.

5.7. XGBOOST REGRESSOR

5.7.2. Model Tuning

Grid search is used for hyperparameter tuning of the model. The parameters include:

- **n_estimators**: This defines the number of boosting rounds (i.e., the number of trees in the ensemble). This is set to [100, 125, 150].
- **subsample**: This gives the fraction of training data to use for fitting each individual tree. This helps control overfitting. The grid checks values [0.8, 0.9, 1].
- **gamma**: This gives the minimum loss reduction required to make a further partition. It helps control model complexity and overfitting. The grid includes [0, 1, 3].
- **colsample_bytree**: This defines the fraction of features to be used for each individual tree. It helps control overfitting. The grid checks values [0.8, 0.9, 1].
- **colsample_bylevel**: This gives the fraction of features to use for each level (split) of the tree. The grid tests values [0.8, 0.9, 1].

A 5-fold cross-validation (cv=5) is applied, which means the data will be split into 5 parts, and the model will be trained on 4 parts and validated on the remaining part, repeated 5 times.

```
XGBRegressor(base_score=None, booster=None, callbacks=None, colsample_bylevel=1,
colsample_bynode=None, colsample_bytree=1, device=None,
early_stopping_rounds=None, enable_categorical=False,
eval_metric=None, feature_types=None, gamma=0, grow_policy=None,
importance_type=None, interaction_constraints=None,
learning_rate=None, max_bin=None, max_cat_threshold=None,
max_cat_to_onehot=None, max_delta_step=None, max_depth=None,
max_leaves=None, min_child_weight=None, missing=nan,
monotone_constraints=None, multi_strategy=None, n_estimators=100,
n_jobs=None, num_parallel_tree=None, random_state=1, ...)
```

5.7.3. Model Performance Evaluation

The evaluation metrics used are RMSE, MAE, R-squared, Adjusted R-squared, and MAPE, as described above.

Checking model performance on Train Data

	RMSE	MAE	R-squared	Adj. R-squared	MAPE
0	2112.222275	1658.957079	0.978392	0.978356	8.22034

Checking model performance on Test Data

	RMSE	MAE	R-squared	Adj. R-squared	MAPE
0	3088.892844	2459.27551	0.952784	0.952601	11.835337

OBSERVATIONS:

- The model shows strong performance on the training set, with a high R-squared of 97.84% and low error metrics (RMSE: 2112.22, MAE: 1658.96), indicating a good fit to the training data.
- However, there is a noticeable drop in performance on the test set, with R-squared dropping to 95.28% and higher errors (RMSE: 3088.89, MAE: 2459.28).
- This suggests some overfitting, where the model is performing well on the training data but struggles to generalize to unseen data.

The model emphasizes the key factors that influence insurance costs, helping us identify which variables have the most significant impact. Among these, weight, Years_since_last_admitted and covered_by_any_other_company are the three important features contributing most to the prediction.

5.8. DECISION TREE

5.8.2. Model Tuning

Grid search is used for hyperparameter tuning of the model. The parameters include:

- **max_depth**: Specifies the maximum depth of the tree; larger values create more complex trees, while None allows expansion until leaves are pure.
- **min_samples_leaf**: Sets the minimum number of samples required in each leaf node, preventing overfitting by ensuring more data in each leaf.
- **max_leaf_nodes**: Limits the number of leaf nodes, restricting the tree's size.
- **min_impurity_decrease**: Controls the minimum reduction in impurity required to split a node, helping to prevent unnecessary splits and overfitting.

A 5-fold cross-validation (cv=5) is applied, which means the data will be split into 5 parts, and the model will be trained on 4 parts and validated on the remaining part, repeated 5 times.

The best model with the optimized hyperparameters to the full training dataset is initialized.

```
DecisionTreeRegressor
DecisionTreeRegressor(max_depth=7, min_impurity_decrease=0.001,
min_samples_leaf=10, random_state=1)
```

5.8.3. Model Performance Evaluation

The evaluation metrics used are RMSE, MAE, R-squared, Adjusted R-squared, and MAPE, as described above.

Checking model performance on Train Data

	RMSE	MAE	R-squared	Adj. R-squared	MAPE
0	2938.92424	2342.91773	0.958168	0.958099	11.234701

Checking model performance on Test Data

	RMSE	MAE	R-squared	Adj. R-squared	MAPE
0	3050.575606	2420.403523	0.953949	0.95377	11.509848

OBSERVATIONS:

- The model performs well on the training set with an R-squared of 95.82% and low errors (RMSE: 2938.92, MAE: 2342.92).
- Performance on the test set declines slightly, with R-squared dropping to 95.39% and higher errors (RMSE: 3050.58, MAE: 2420.40).
- This indicates a minor overfitting issue, where the model is not generalizing as effectively to new data.

The model emphasizes the key factors that influence insurance costs, helping us identify which variables have the most significant impact. Among these, weight, Years_since_last_admitted and covered_by_any_other_company are the three important features contributing most to the prediction.

6. Model Validation

6.1. Model Performance Evaluation for Train Data

Training performance comparison:				
	Decision Tree	Decision Tree Tuned	Bagging Regressor \	
RMSE	0.0	2938.924240	1338.711021	
MAE	0.0	2342.917730	987.446800	
R-squared	1.0	0.958168	0.991320	
Adj. R-squared	1.0	0.958099	0.991306	
MAPE	0.0	11.234701	4.723092	
	Bagging Regressor Tuned	Random Forest Estimator \		
RMSE	1627.857136	1138.413110		
MAE	1292.683399	896.518629		
R-squared	0.987166	0.993723		
Adj. R-squared	0.987145	0.993713		
MAPE	6.646224	4.308465		
	Random Forest Tuned	Adaboost Regressor	Adaboost Tuned \	
RMSE	2766.609652	3249.676128	3243.498879	
MAE	2211.524710	2672.964057	2641.603294	
R-squared	0.962930	0.948854	0.949049	
Adj. R-squared	0.962868	0.948769	0.948964	
MAPE	10.632904	15.018419	14.479256	
	Gradient Boost Estimator	Gradient Boost Tuned	XGB	
RMSE	2927.596536	2744.251390	2112.222275	
MAE	2354.122354	2199.190921	1658.957079	
R-squared	0.958490	0.963527	0.978392	
Adj. R-squared	0.958421	0.963466	0.978356	
MAPE	11.487394	10.720271	8.220348	
	XGB Tuned	Linear Regression	Linear Regression Final	
RMSE	2112.222275	3364.244496	3365.987404	
MAE	1658.957079	2707.519836	2709.141195	
R-squared	0.978392	0.945185	0.945128	
Adj. R-squared	0.978356	0.945090	0.945106	
MAPE	8.220348	15.224003	15.231840	

6.2. Model Performance Evaluation for Test Data

Test performance comparison:				
	Decision Tree	Decision Tree Tuned	Bagging Regressor \	
RMSE	4236.557614	3050.575606	3193.896830	
MAE	3284.249867	2420.403523	2520.107707	
R-squared	0.911181	0.953949	0.949520	
Adj. R-squared	0.910836	0.953770	0.949324	
MAPE	15.375680	11.509848	12.054849	
	Bagging Regressor Tuned	Random Forest Estimator \		
RMSE	3220.033696	3055.870644		
MAE	2558.292603	2421.996480		
R-squared	0.948690	0.953789		
Adj. R-squared	0.948491	0.953609		
MAPE	13.051561	11.562507		
	Random Forest Tuned	Adaboost Regressor	Adaboost Tuned \	
RMSE	2998.606610	3247.414907	3257.688577	
MAE	2383.236471	2671.945571	2647.570537	
R-squared	0.955504	0.947814	0.947483	
Adj. R-squared	0.955331	0.947611	0.947279	
MAPE	11.344109	14.985721	14.512683	
	Gradient Boost Estimator	Gradient Boost Tuned	XGB \	
RMSE	2970.611098	2959.984817	3088.892844	
MAE	2392.078909	2369.445986	2459.275510	
R-squared	0.956331	0.956643	0.952784	
Adj. R-squared	0.956162	0.956475	0.952601	
MAPE	11.612841	11.417282	11.835337	
	XGB Tuned	Linear Regression	Linear Regression Final	
RMSE	3088.892844	3322.370014	3323.951853	
MAE	2459.275510	2693.729313	2695.567318	
R-squared	0.952784	0.945377	0.945325	
Adj. R-squared	0.952601	0.945158	0.945274	
MAPE	11.835337	14.980550	14.988497	

6.3. Observations from Final Model Train and Test Data

Gradient Boost Tuned

Training Performance: Lower RMSE (2744.25), MAE (2199.19), and MAPE (10.72), higher R-squared (0.9635).

Test Performance: RMSE (2959.98), MAE (2369.45), R-squared (0.9566), and MAPE (11.42).

Gradient Boost Tuned performs consistently well on both training and test data, with the best balance between accuracy (low RMSE and MAE) and generalization (low MAPE and good R-squared).

Random Forest Tuned

Training Performance: RMSE (2766.61), MAE (2211.52), MAPE (10.63), and R-squared (0.9629).

Test Performance: RMSE (2998.61), MAE (2383.24), MAPE (11.34), and R-squared (0.9555).

Random Forest Tuned shows good performance with low error metrics and high R-squared, but it has slightly higher test RMSE and MAE compared to Gradient Boost Tuned.

Gradient Boost Estimator

Training Performance: RMSE (2927.60), MAE (2354.12), MAPE (11.49), and R-squared (0.9585).

Test Performance: RMSE (2970.61), MAE (2392.08), MAPE (11.61), and R-squared (0.9563).

Gradient Boost Estimator has strong training and test performance with good generalization, but it is slightly behind in terms of MAPE compared to Gradient Boost Tuned.

6.4. Feature Importance

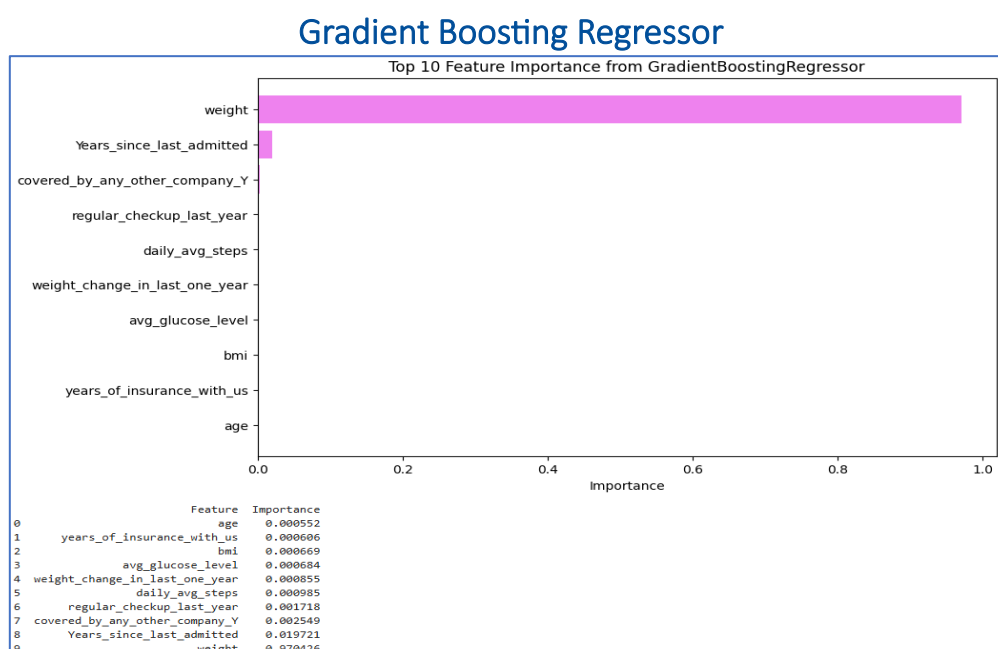


Fig 4. Feature Importance – Gradient boosting regressor

Random Forest Regressor

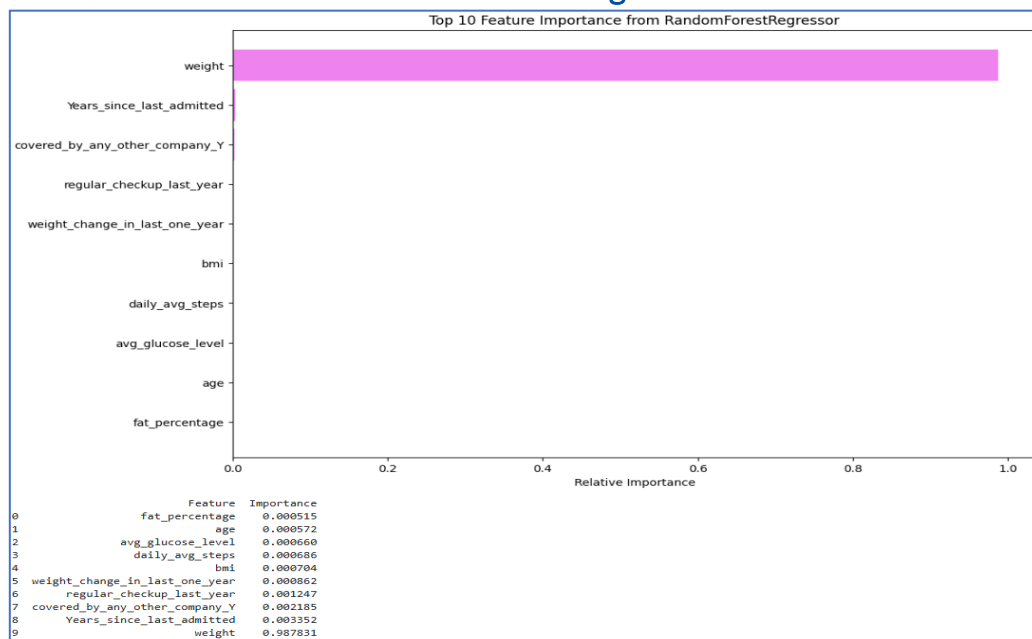


Fig 5. Feature Importance – Random Forest regressor

OBSERVATIONS:

- The model emphasizes the key factors that influence insurance costs, helping us identify which variables have the most significant impact.
- **weight:** One of the most significant contributors to insurance cost. As weight increases, the insurance cost tends to rise. This suggests that individuals with higher body weight may have higher health-related risks, which in turn affects their premiums.
- **Years_since_last_admitted:** The number of years since the last hospital admission also significantly influences insurance costs. As this number increases, it appears that individuals may have higher insurance premiums, possibly due to accumulated health risks over time.
- **covered_by_any_other_company:** Individuals with insurance from another company tend to have higher insurance costs with the current provider. This could reflect the higher risk associated with individuals who already have coverage, indicating a potentially higher likelihood of claims or more comprehensive coverage needs.

6.5. CONCLUSION

- When selecting the optimal model, the goal is to achieve the best balance between accuracy (minimizing prediction errors) and generalization (ensuring that the model performs well on unseen data).
- The top three models, Gradient Boost Tuned, Random Forest Tuned and Gradient Boost Estimator stand out due to their consistent performance across both training and test datasets.
- These models achieve a strong balance between minimizing error metrics such as RMSE, MAE, and MAPE, while maintaining high R-squared values that reflect their ability to generalize well to unseen data.
- Gradient Boost Tuned performs slightly better overall, with the lowest RMSE and MAPE, but Random Forest Tuned and Gradient Boost Estimator are also strong contenders, offering competitive accuracy and solid generalization. Their overall reliability and balanced error reduction make them the top choices for predicting insurance costs.

After building the final optimum model, clustering is performed to analyse how the insurance cost is influenced by other predictor variables and to segment customers based on these factors.

7. K-means Clustering

- **Elbow Method:** The distortion score against different values of k were plotted to identify the "elbow" point, where the rate of improvement slows down.
- **Silhouette Score:** Using the silhouette_score algorithm from sklearn.metrics, how similar each data point was to its own cluster compared to other clusters is assessed. Higher silhouette scores indicate better-defined clusters.

Both methods were applied, and the best number of clusters was determined to be 3.

- Based on the silhouette plots, k=3 showed a good silhouette score with relatively low computation time.
- Choosing too few clusters can oversimplify the data, masking important patterns and variations, which could result in the loss of valuable insights during analysis.
- While a lower number of clusters can simplify the analysis, it often sacrifices depth and accuracy, making it more difficult to draw meaningful conclusions from the data.

Thus, k=3 was chosen for further analysis. Afterward, cluster profiling was performed, yielding three distinct clusters with the following observations.

In order to identify distinct customer segments based on their spending habits, the insurance_cost variable is used for grouping.

7.1 Clustering Profiling using Mean

- **Cluster 0** represents individuals with a long history of insurance coverage, moderate health engagement, and a cautious lifestyle with balanced activity levels. They tend to be middle-aged, with a moderate health risk profile but high insurance costs.
- **Cluster 1** consists of individuals with slightly longer insurance durations and a more active lifestyle. They have moderate health management habits and are more inclined towards higher insurance coverage options. They show slightly more engagement in physical activities and have a balanced gender distribution.
- **Cluster 2** features individuals with a moderate insurance duration but a higher health risk profile. They tend to have higher BMI and fat percentage, indicating potential health risks. Despite this, they are somewhat active and tend to have more moderate insurance coverage preferences. They also have a higher incidence of heart disease.

8. Business Recommendations

Based on the Final Optimum Model

- **Weight Management Programs:** Encourage weight management to reduce premiums, offering discounts or incentives for those who engage in fitness programs.
- **Competitive Plans for Dual Coverage Holders:** Offer targeted insurance options for individuals with existing coverage from other companies, with value-added services to reduce risks and avoid policy overlaps.
- **Regular Checkups:** Promote regular health checkups to detect early health issues, reducing long-term risks and premiums. Offer discounts for policyholders who complete annual checkups.
- **Targeted Health Plans for High-Risk Individuals:** Offer personalized plans for those with a history of medical conditions or recent hospital admissions, focusing on prevention and regular health monitoring.
- **Specialized Coverage for Adventure Sports:** Create tailored policies for individuals involved in high-risk activities, with safety training or guidelines to reduce risks and premiums.
- **Health Monitoring Incentives:** Reward active individuals by integrating wearable trackers and offering premium discounts for meeting physical activity targets.
- **Risk-Specific Policies:** Develop policies for older customers or those with pre-existing conditions, offering specialized coverage like health screenings or chronic disease management.

Based on Clustering

Cluster 0 (Moderate Insurance Cost)

- **Offer long-term, moderate coverage:** Focus on plans that cater to moderate health risks with a focus on preventive care.
- **Loyalty Discounts:** Reward customers for maintaining long-term coverage with premium discounts or bonuses.
- **Health Monitoring Tools:** Provide tools for health risk assessments to help manage moderate health risks.

Cluster 1 (Highest Insurance Cost)

- **Premium Active Lifestyle Plans:** Offer comprehensive insurance with coverage for fitness-related needs (sports injuries, therapy).
- **Activity-based Discounts:** Provide discounts or rewards for verified physical activity (gym memberships, fitness trackers).
- **Higher Coverage Options:** Promote plans with extensive coverage, targeting their higher insurance coverage preference.

Cluster 2 (Moderate to High Insurance Cost)

- **Chronic Condition Coverage:** Focus on insurance plans that cover heart disease, obesity-related issues, and preventive care.
- **Health Improvement Incentives:** Offer discounts for engaging in health programs (weight loss, heart disease prevention).
- **Targeted Communication:** Educate on managing health risks, with a focus on heart disease and weight management.

9. Appendix: Tables and Charts

9.1 Univariate Analysis

9.1.1 years_of_insurance_with_us

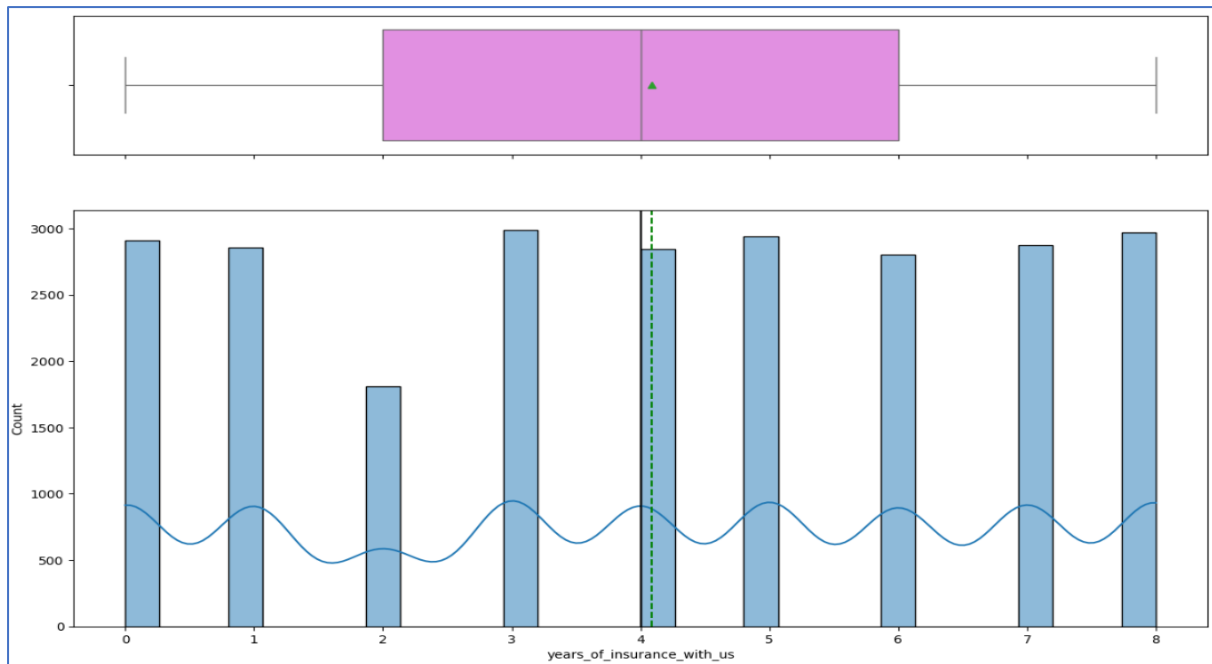


Fig 1. years_of_insurance_with_us

9.1.2 regular_checkup_lasy_year

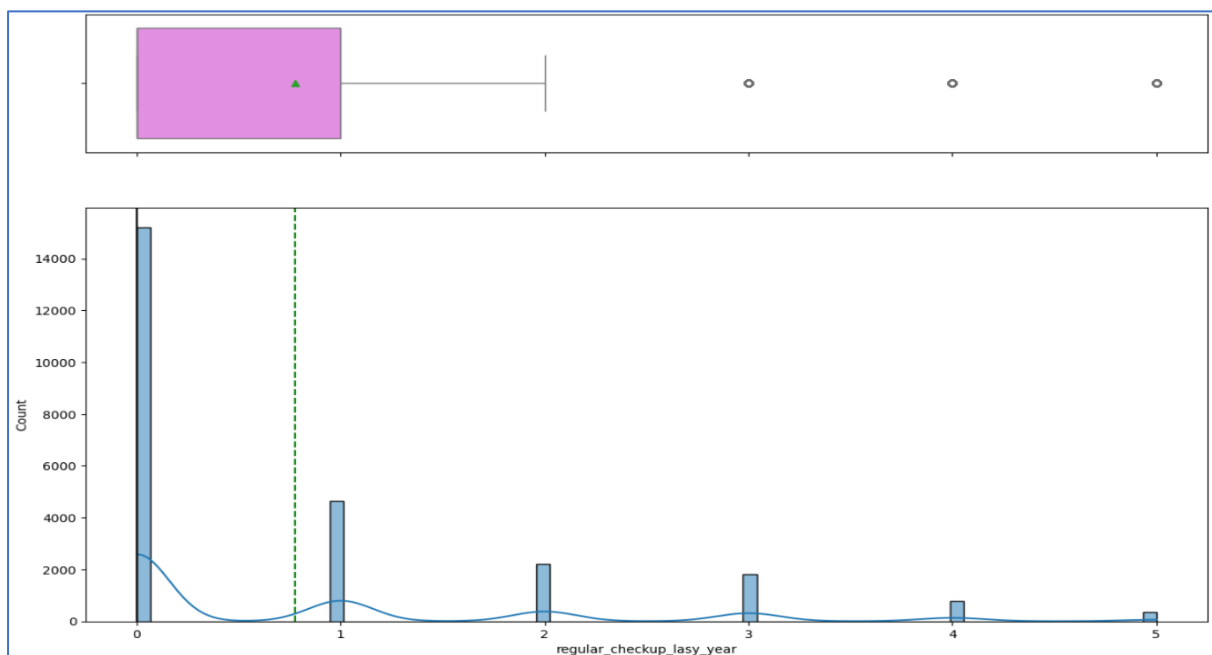


Fig 2. regular_checkup_lasy_year



9.1.3 visited_doctor_last_1_year

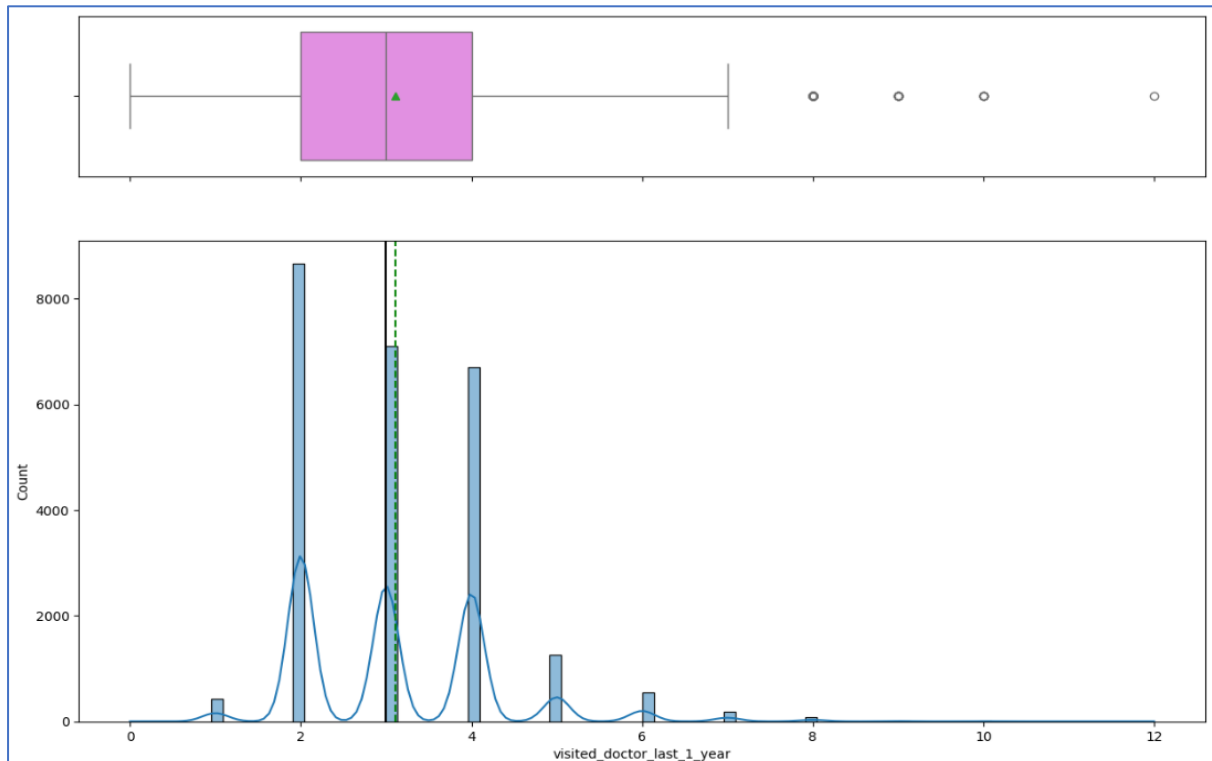


Fig 3. visited_doctor_last_1_year

9.1.4 daily_avg_steps

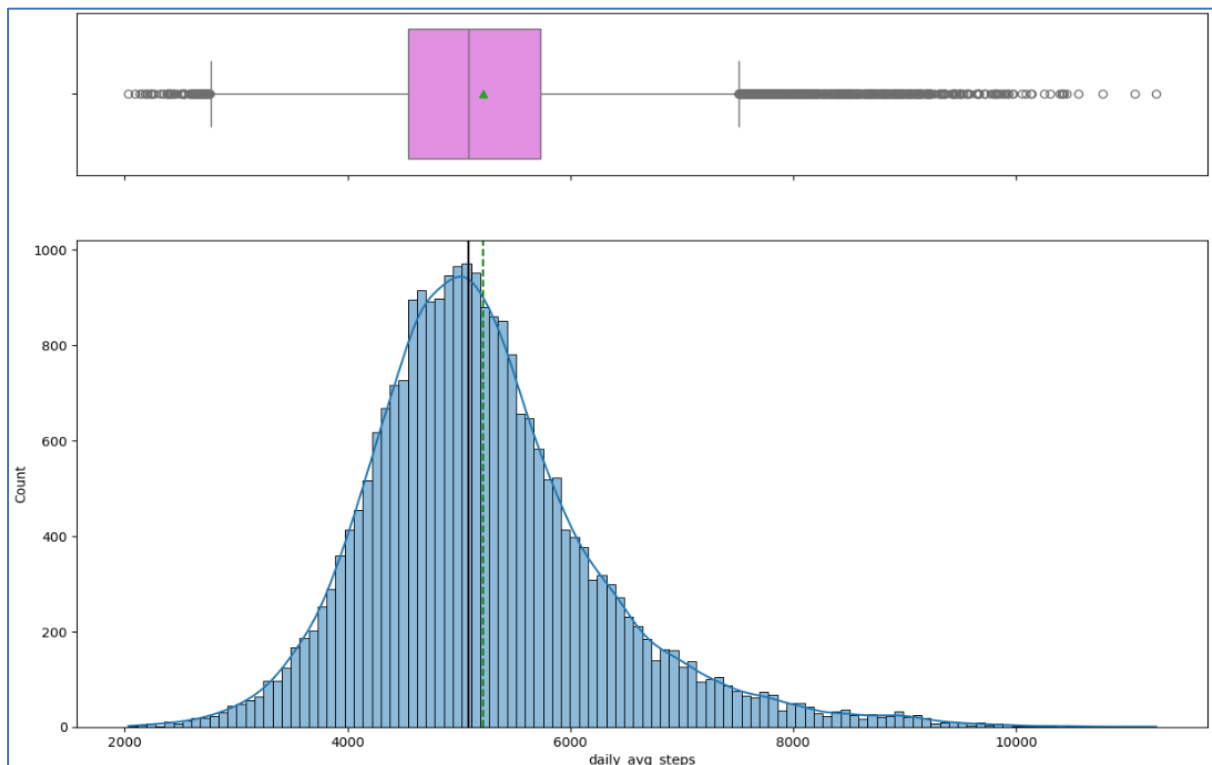


Fig 4. daily_avg_steps



9.1.5 age

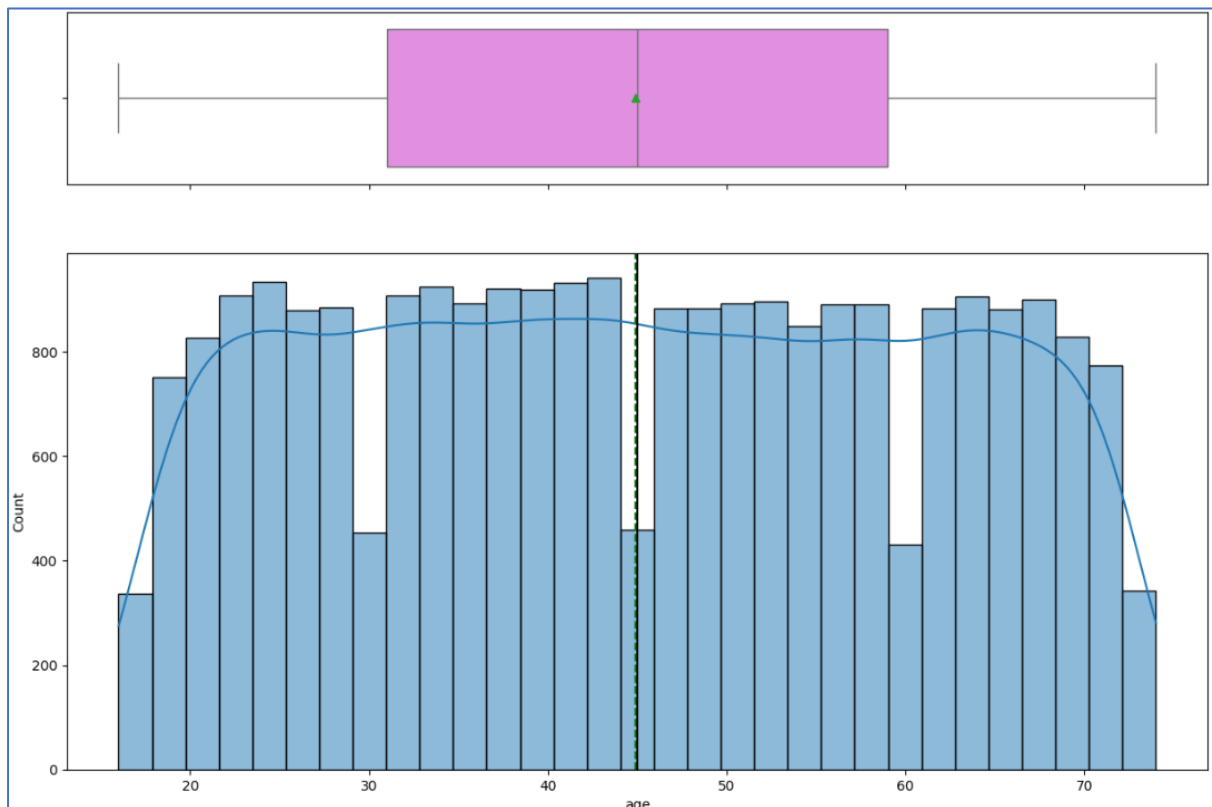


Fig 5. age

9.1.6 avg_glucose_level

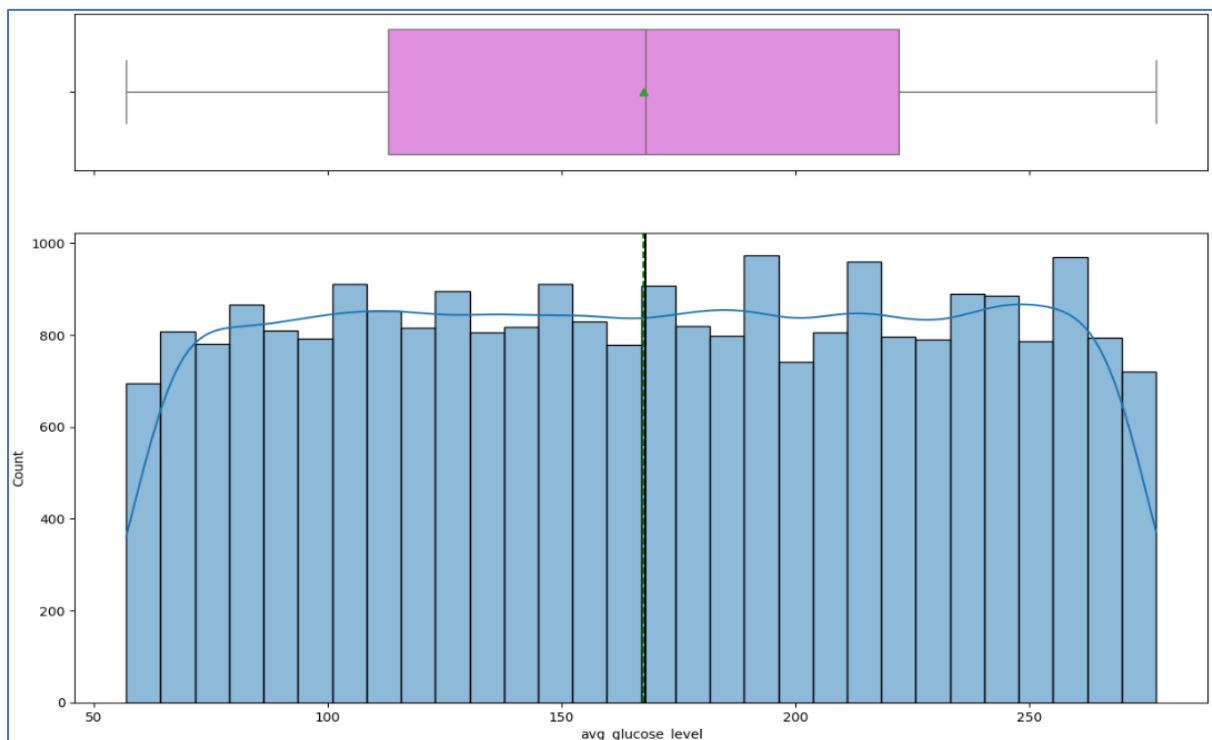


Fig 6. avg_glucose_level

9.1.7 bmi

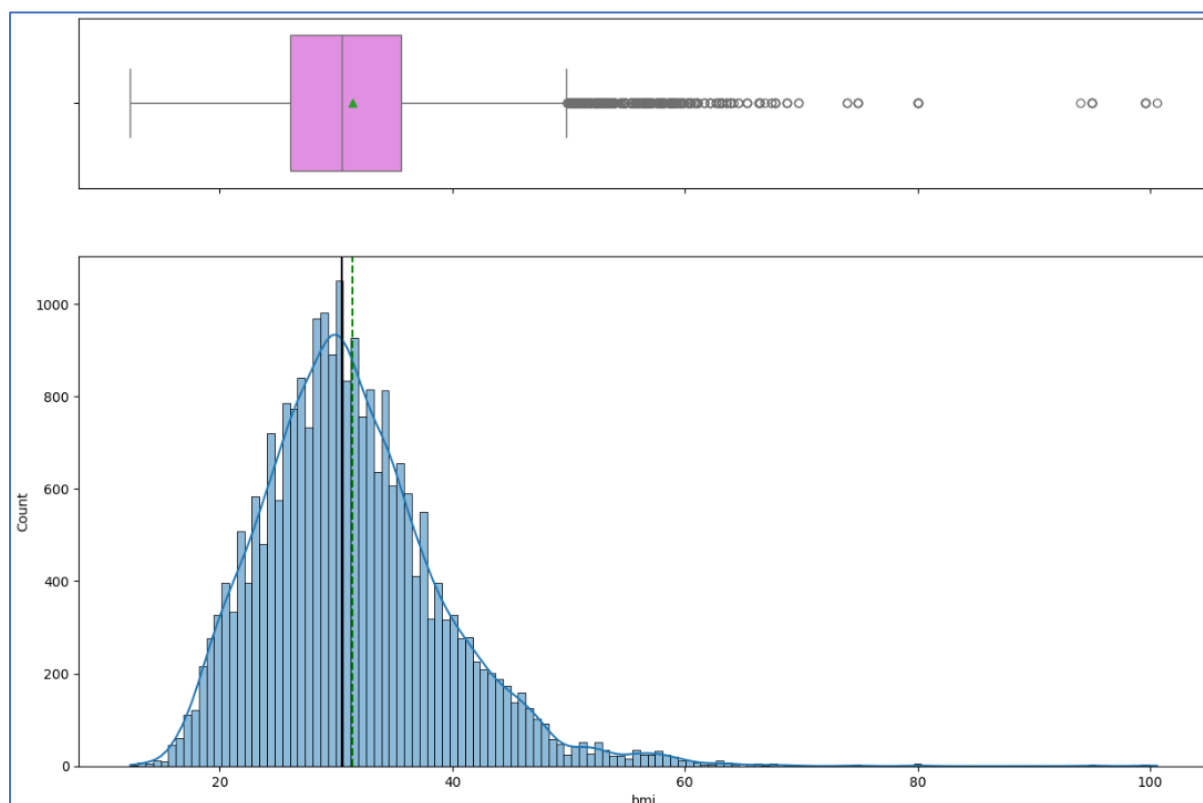


Fig 7. bmi

9.1.8 weight

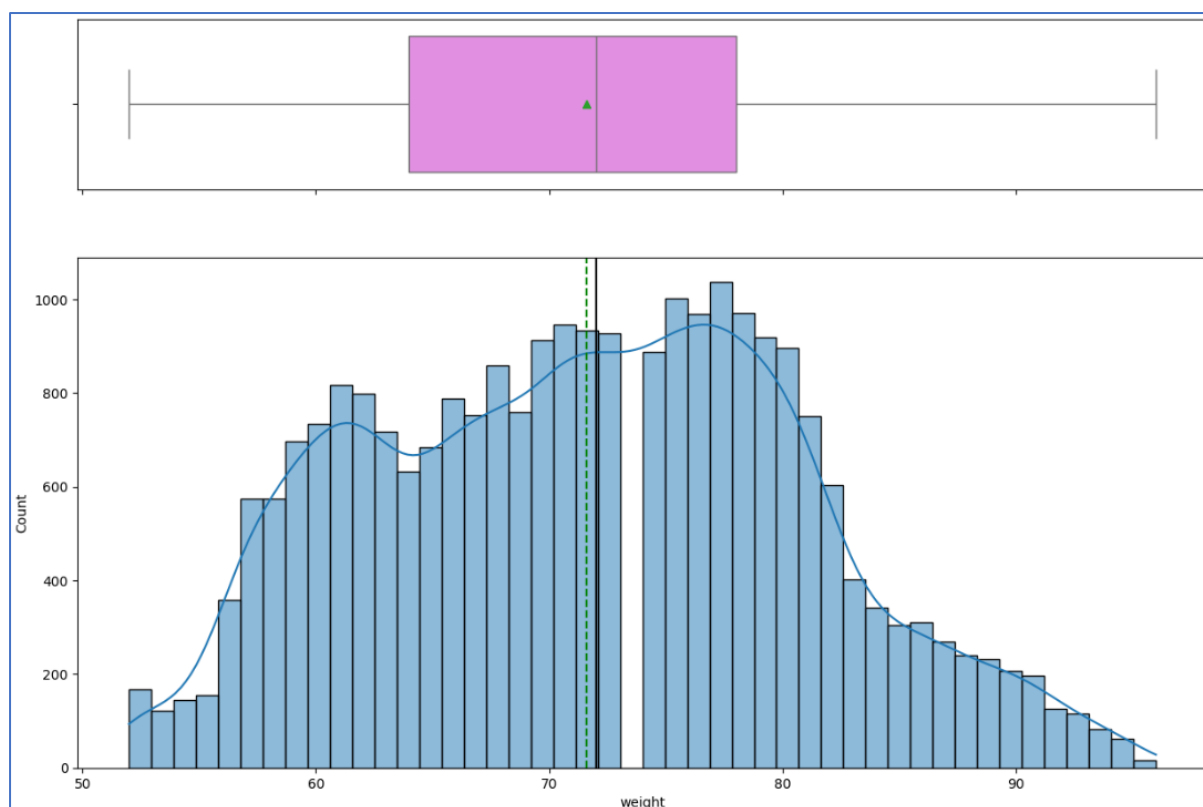


Fig 8. weight

9.1.9 weight_change_in_last_one_year

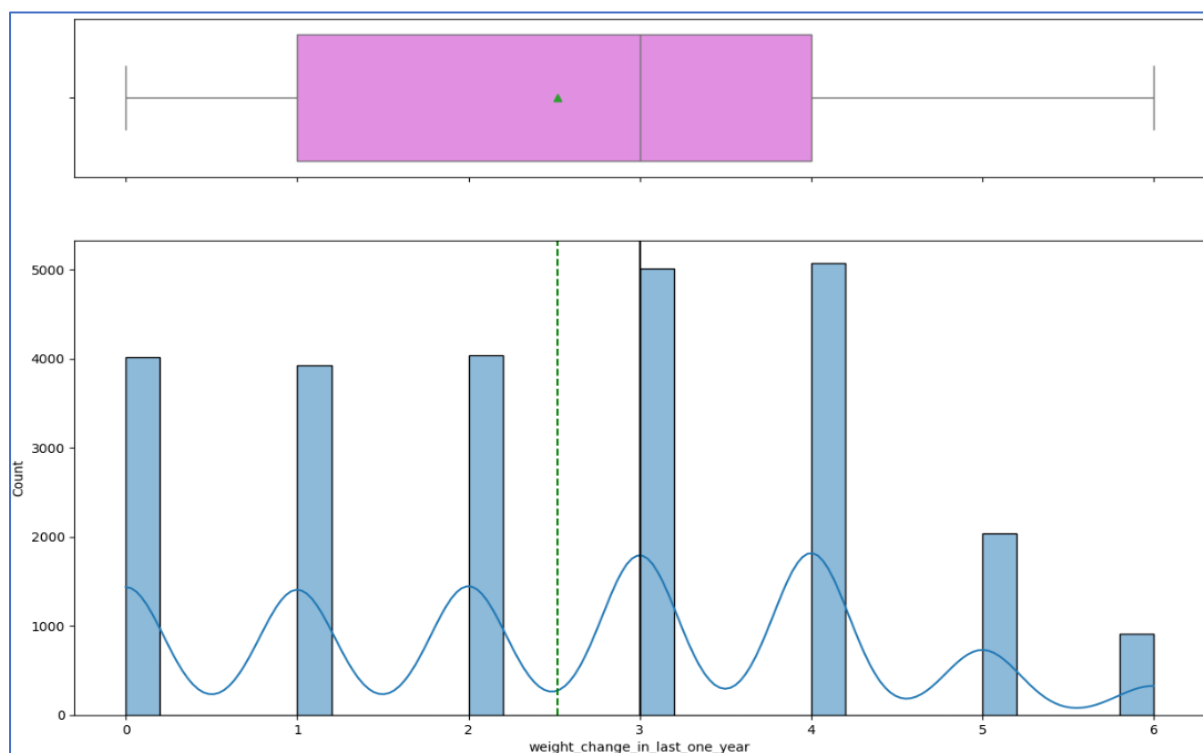


Fig 9. weight_change_in_last_one_year

9.1.10 fat_percentage

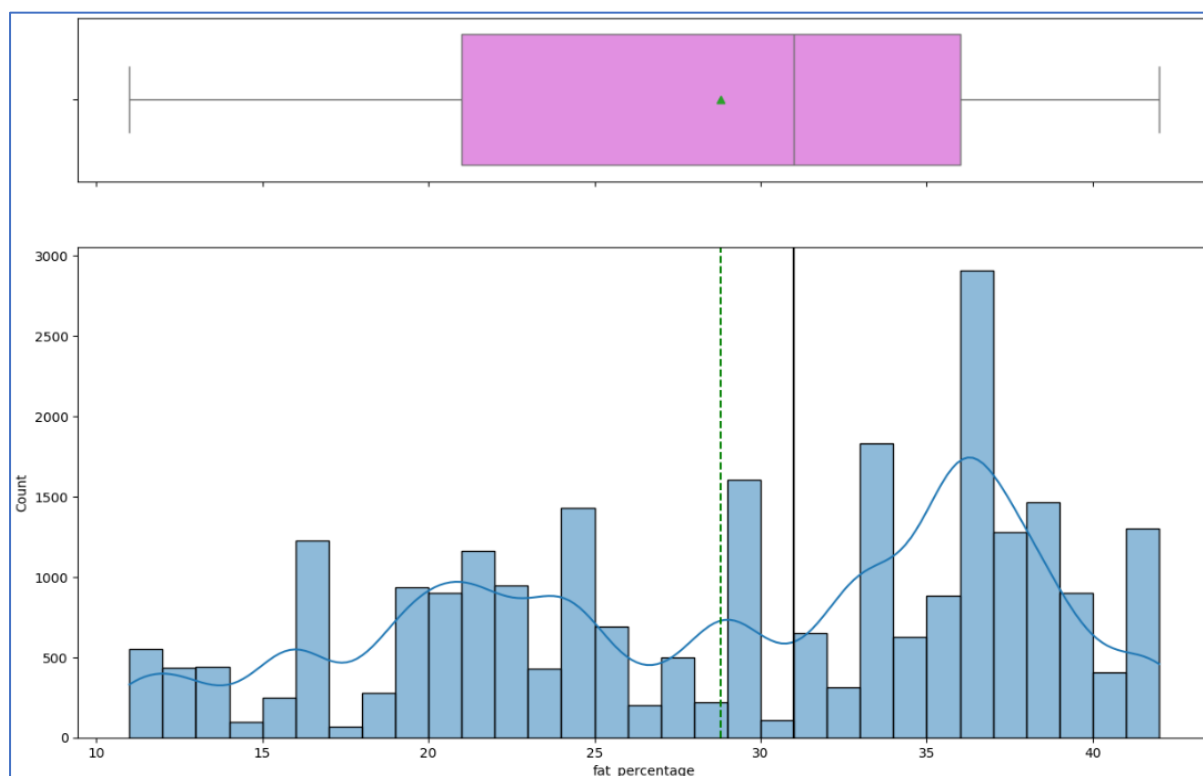


Fig 10. fat_percentage

9.1.11 insurance_cost

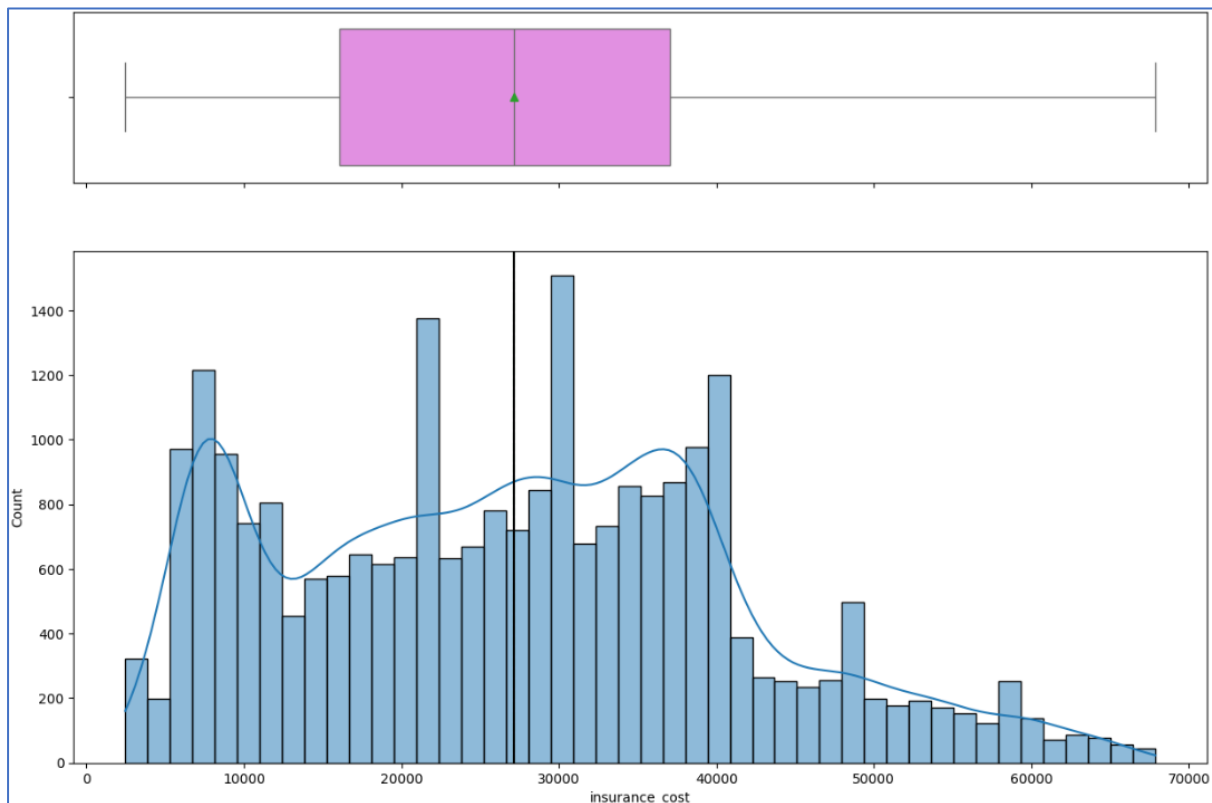


Fig 11. insurance_cost

9.1.12 Year_last_admitted

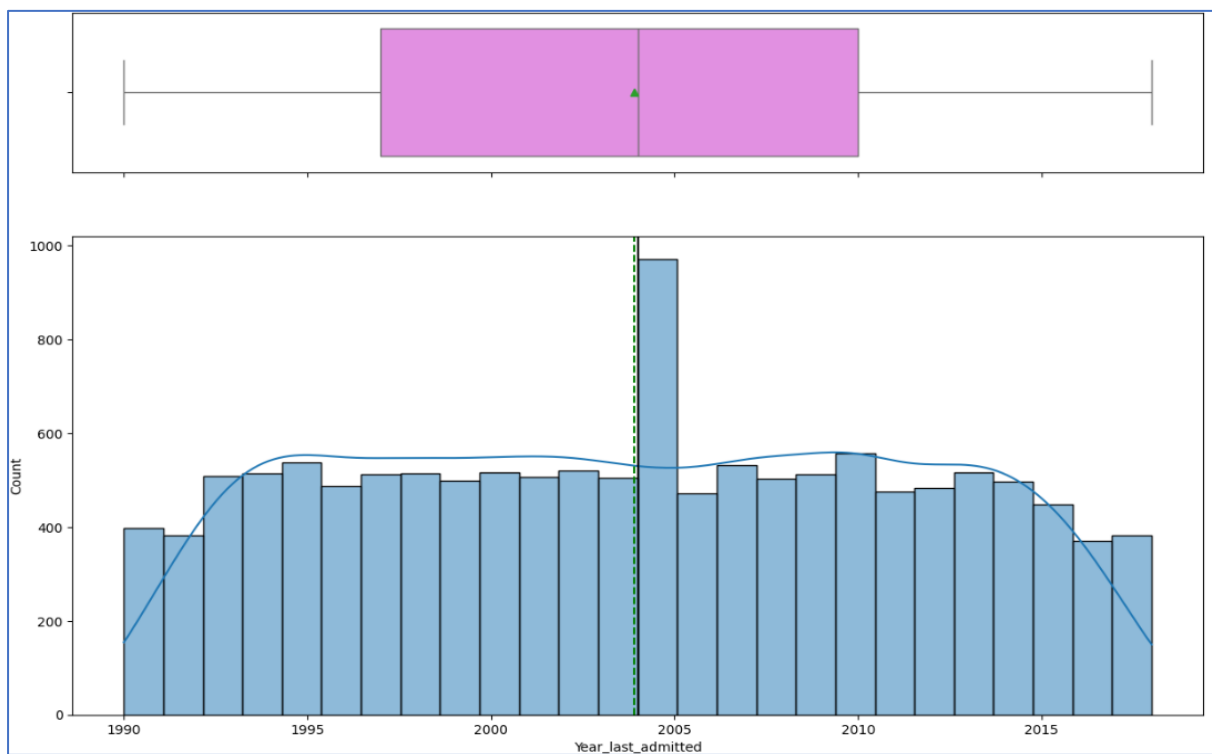


Fig 12. Year_last_admitted

9.1.13 adventure_sports

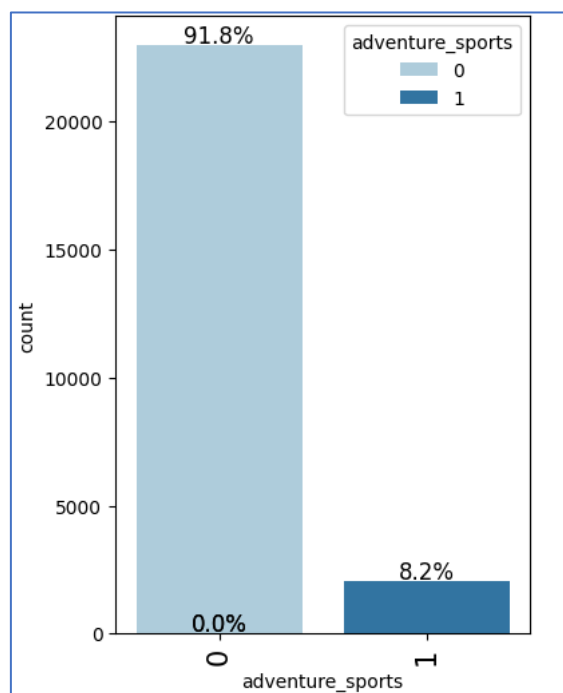


Fig 13. adventure_sports

9.1.14 Occupation

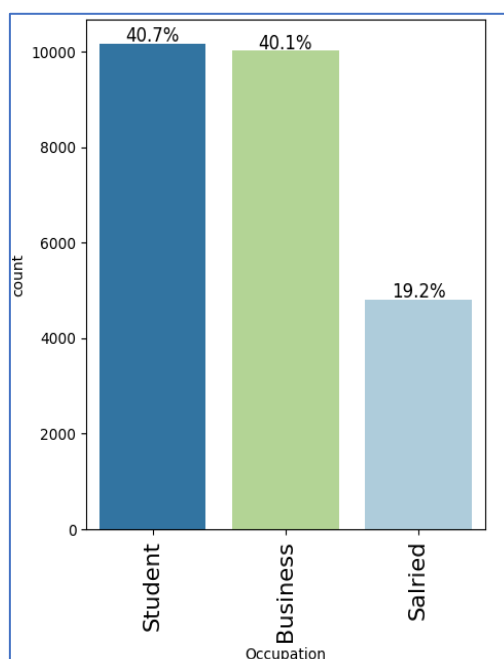


Fig 14. Occupation

9.1.15 cholesterol_level

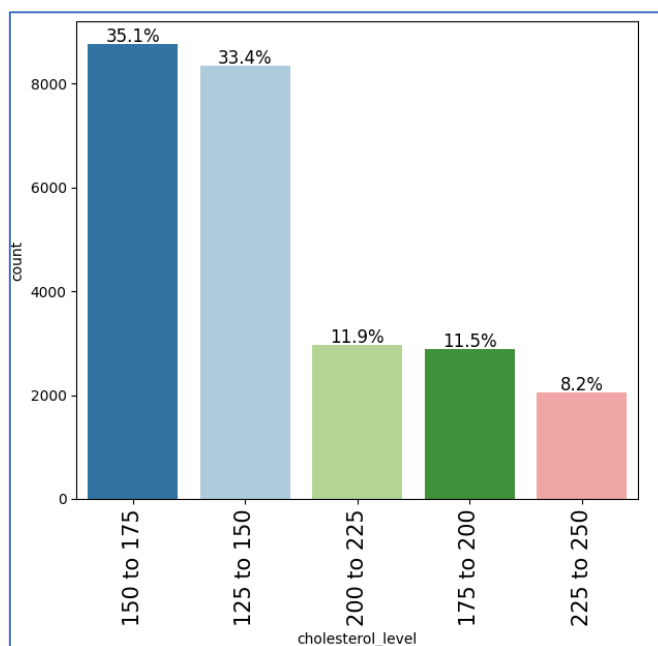


Fig 15. cholesterol_level

9.1.16 Gender

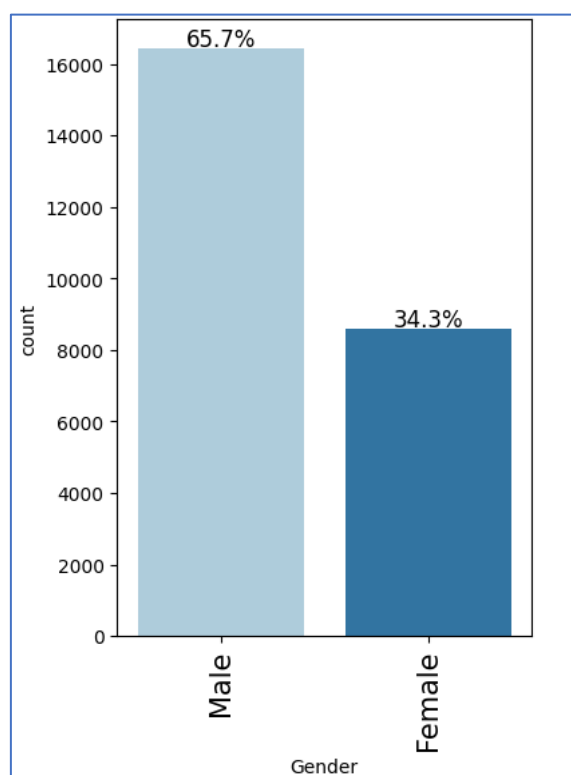


Fig 16. Gender

9.1.17 smoking_status

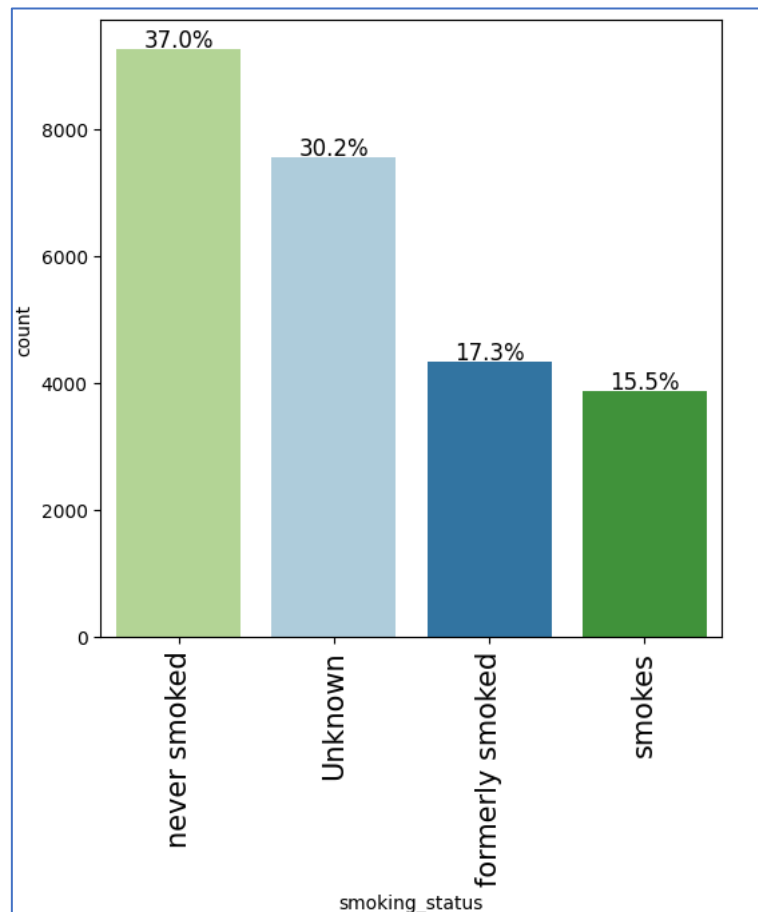


Fig 17. smoking_status

9.1.18 Location

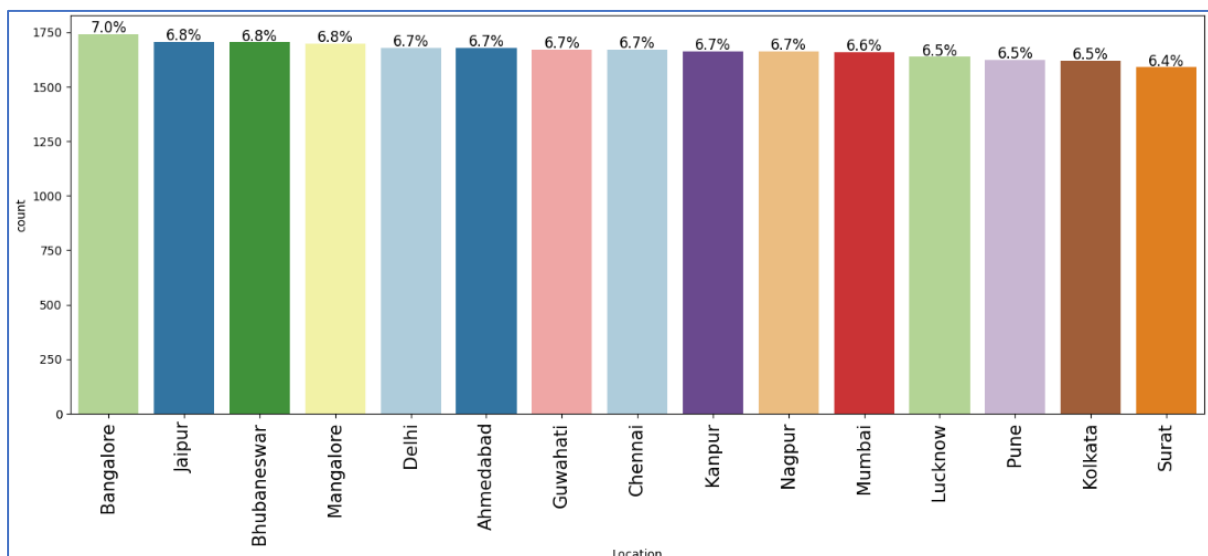


Fig 18. Location

9.1.19 covered_by_any_other_company

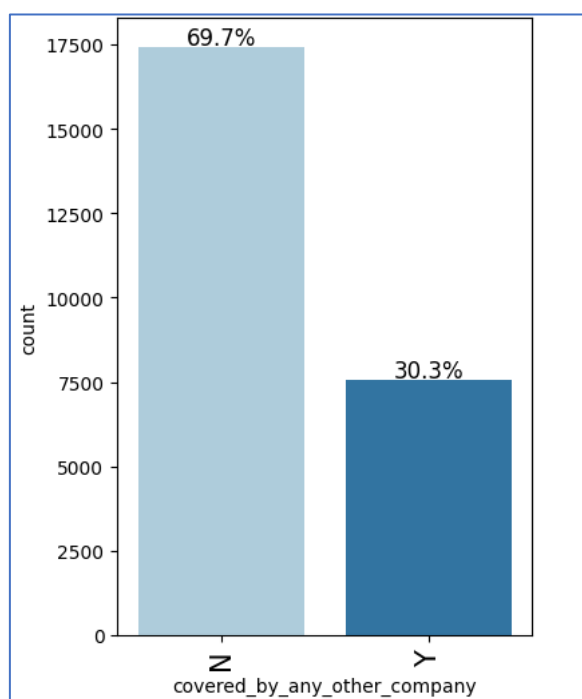


Fig 19. covered_by_any_other_company

9.1.20 Alcohol

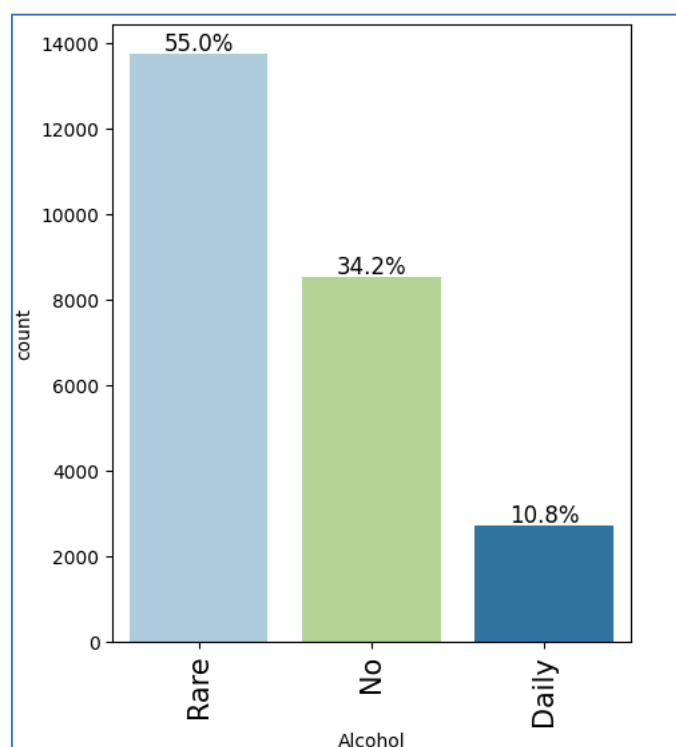


Fig 20. Alcohol

9.1.21 Exercise

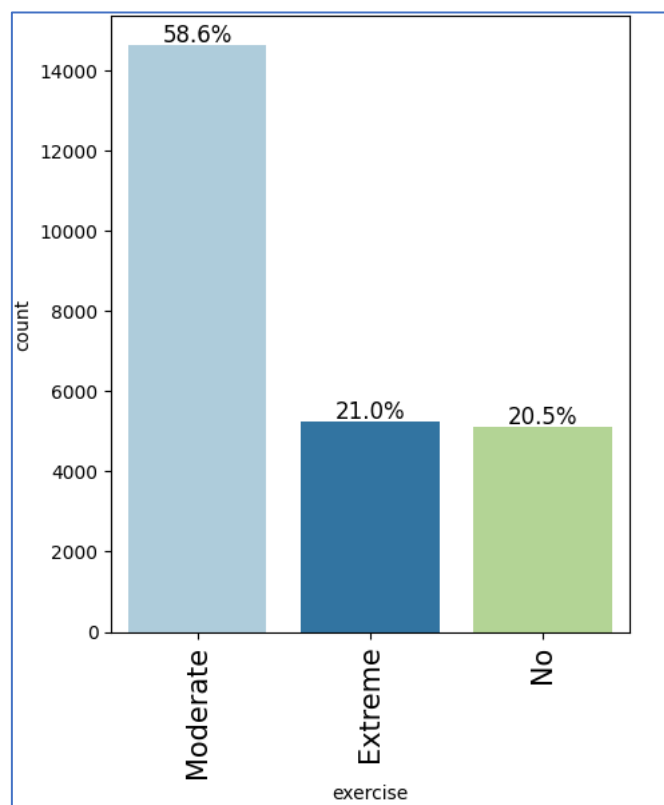


Fig 21. Exercise

9.1.22 heart_decs_history

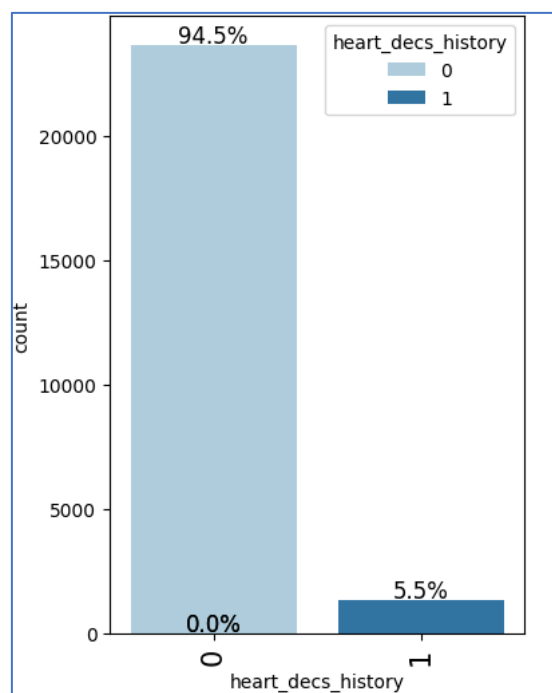


Fig 22. heart_decs_history

9.1.23 other_major_decs_history

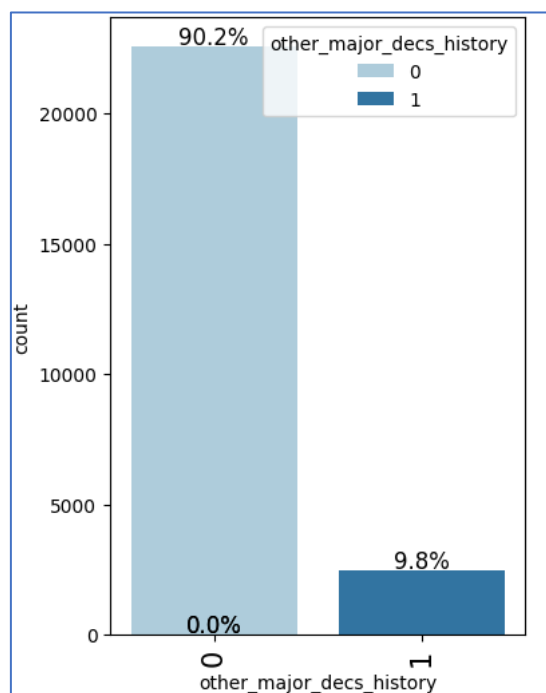


Fig 23. other_major_decs_history

9.2 Bivariate Analysis

9.2.1 weight vs insurance_cost

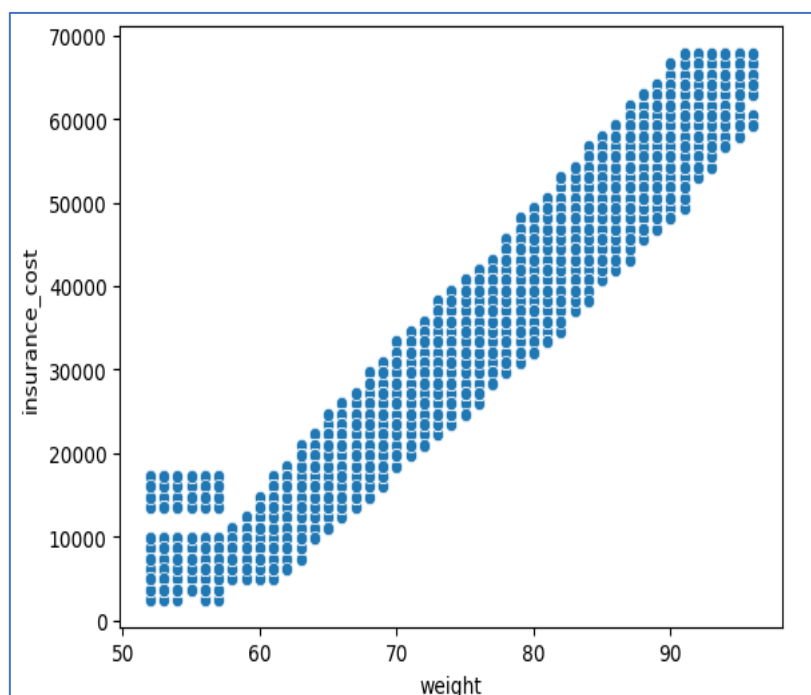


Fig 24. weight vs insurance_cost

9.2.2 Occupation vs insurance_cost

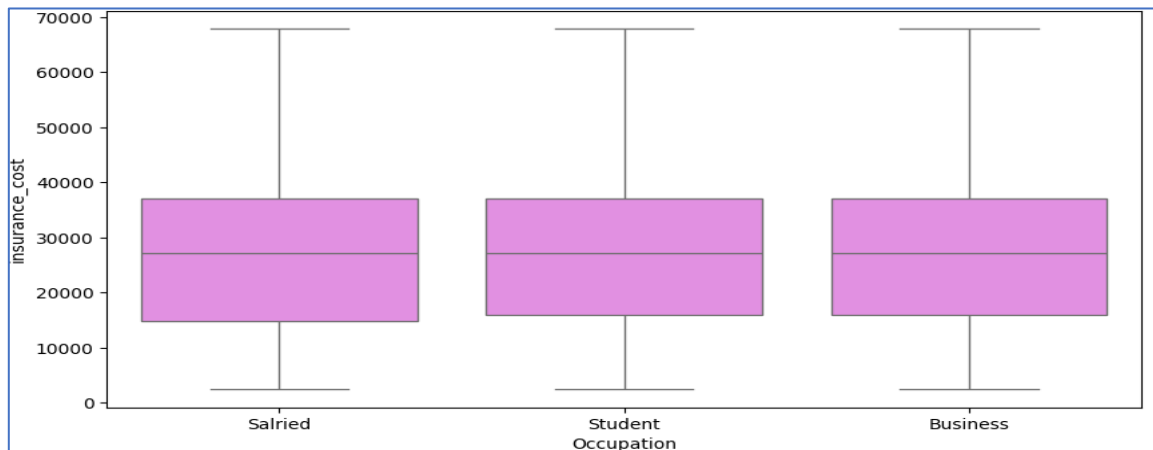


Fig 25. Occupation vs insurance_cost

9.2.3 Gender vs insurance_cost

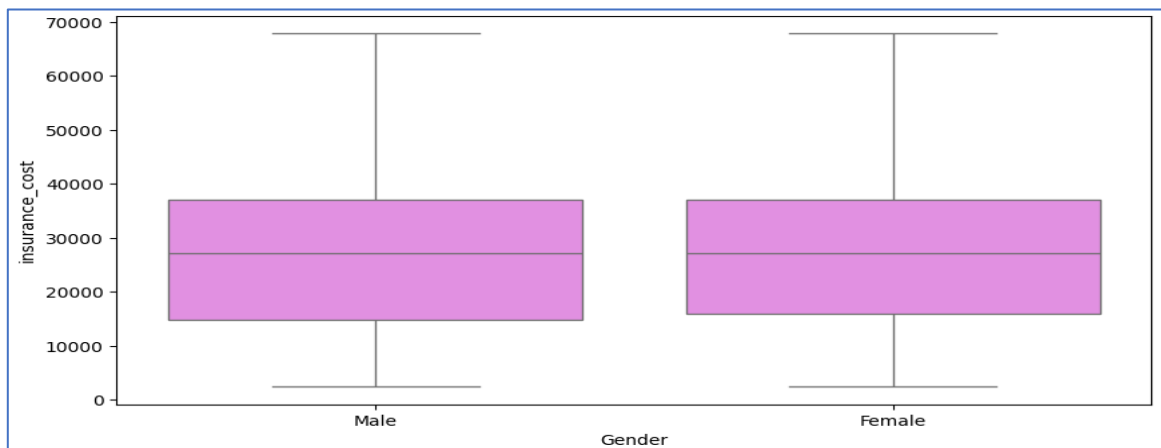


Fig 26. Gender vs insurance_cost

9.2.4 smoking_status vs insurance_cost

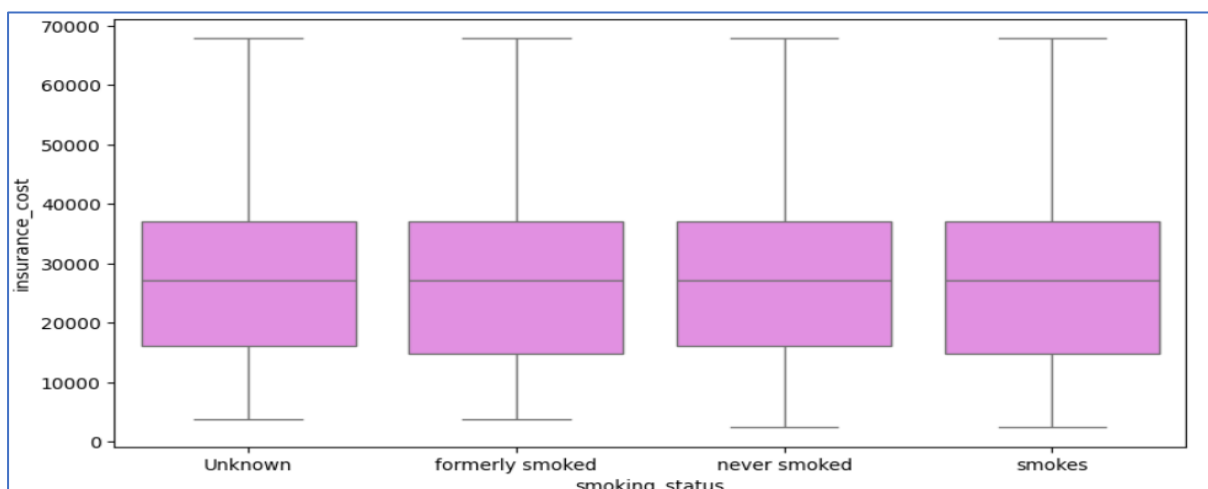


Fig 27. smoking_status vs insurance_cost

9.2.5 Location vs insurance_cost

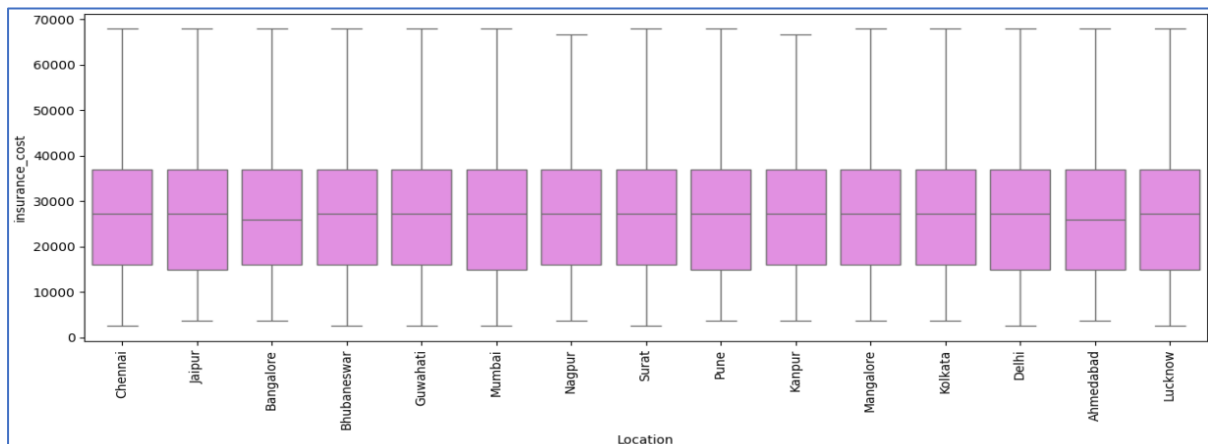


Fig 28. Location vs insurance_cost

9.2.6 covered_by_any_other_company vs insurance_cost

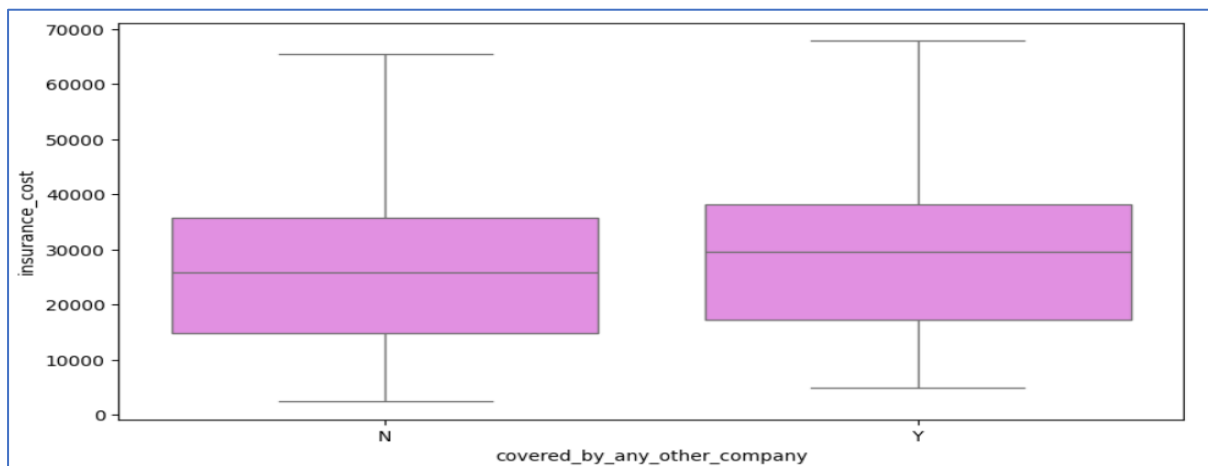


Fig 29. covered_by_any_other_company vs insurance_cost

9.2.7 Alcohol vs insurance_cost



Fig 30. Alcohol vs insurance_cost

9.2.8 exercise vs insurance_cost

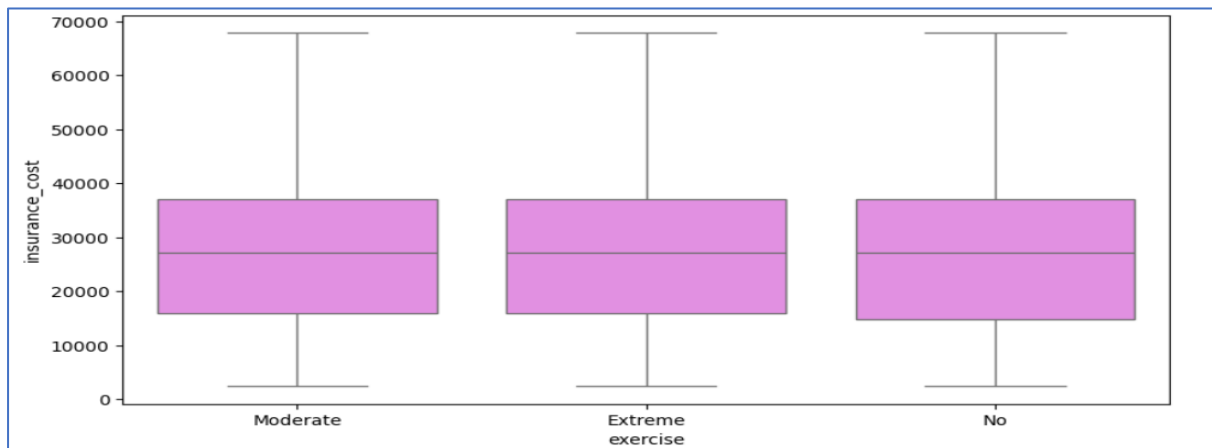


Fig 31. exercise vs insurance_cost

9.2.9 adventure_sports vs insurance_cost

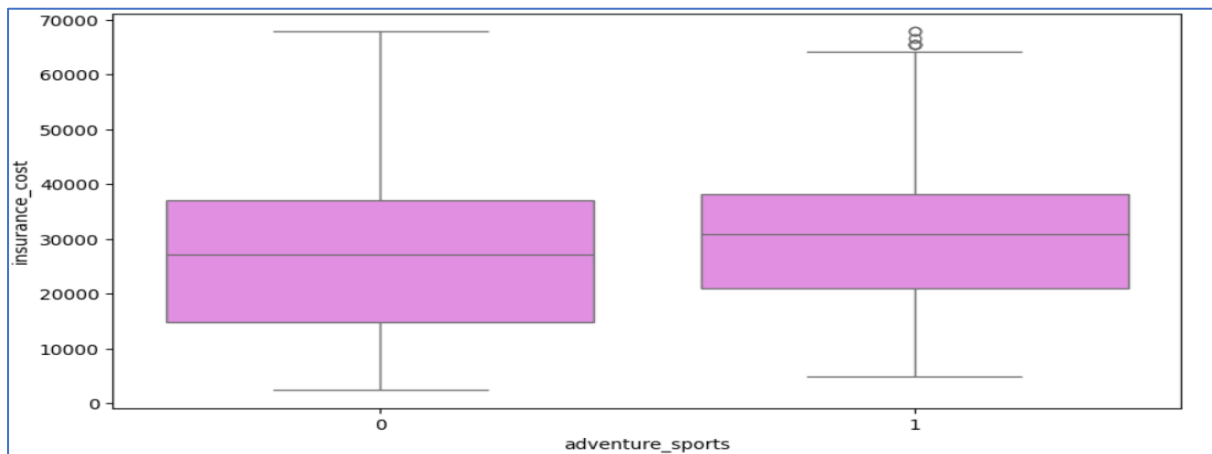


Fig 32. adventure_sports vs insurance_cost

9.2.10 cholesterol_level vs insurance_cost

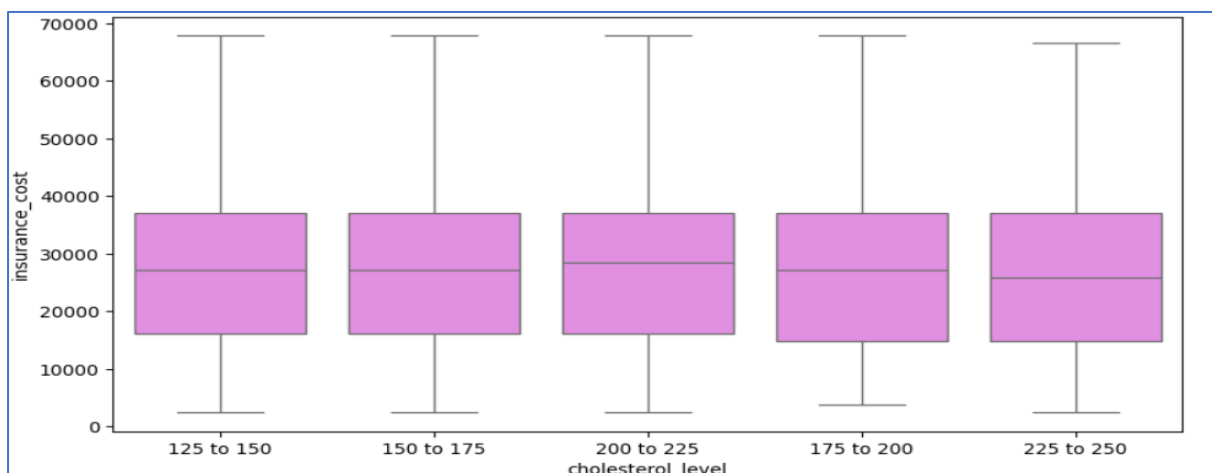


Fig 33. cholesterol_level vs insurance_cost

9.2.11 heart_decs_history vs insurance_cost

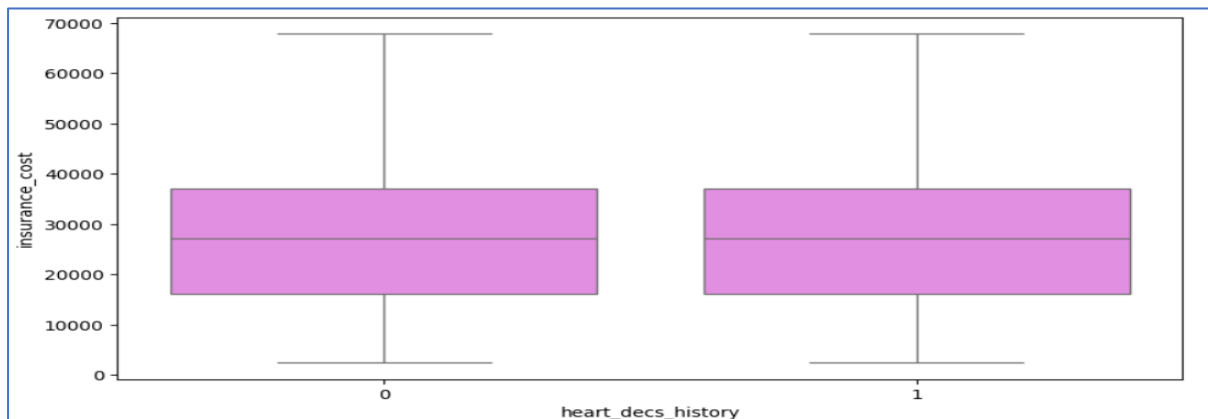


Fig 34. heart_decs_history vs insurance_cost

9.2.12 other_major_decs_history vs insurance_cost

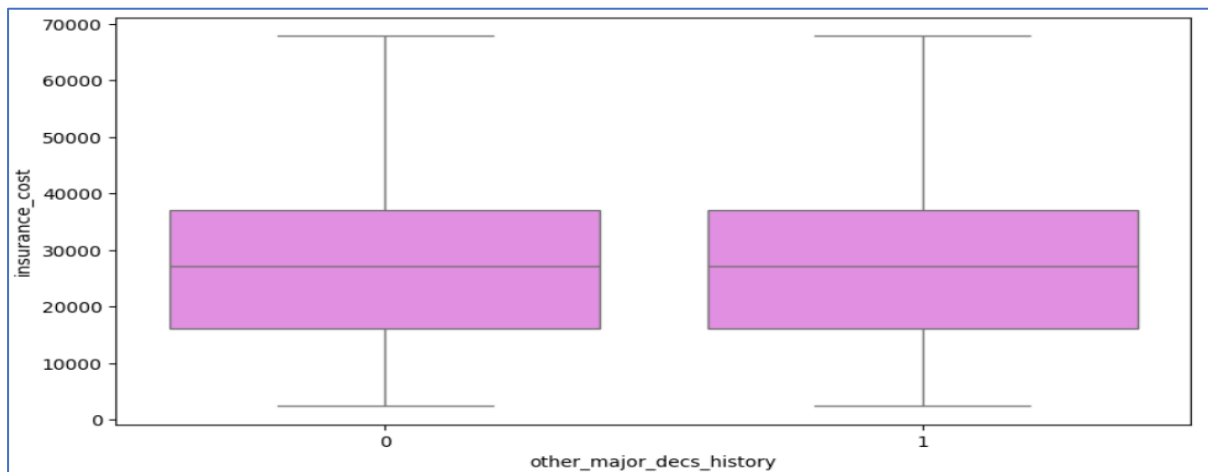


Fig 35. other_major_decs_history vs insurance_cost

9.2.13 Year_last_admitted vs insurance_cost

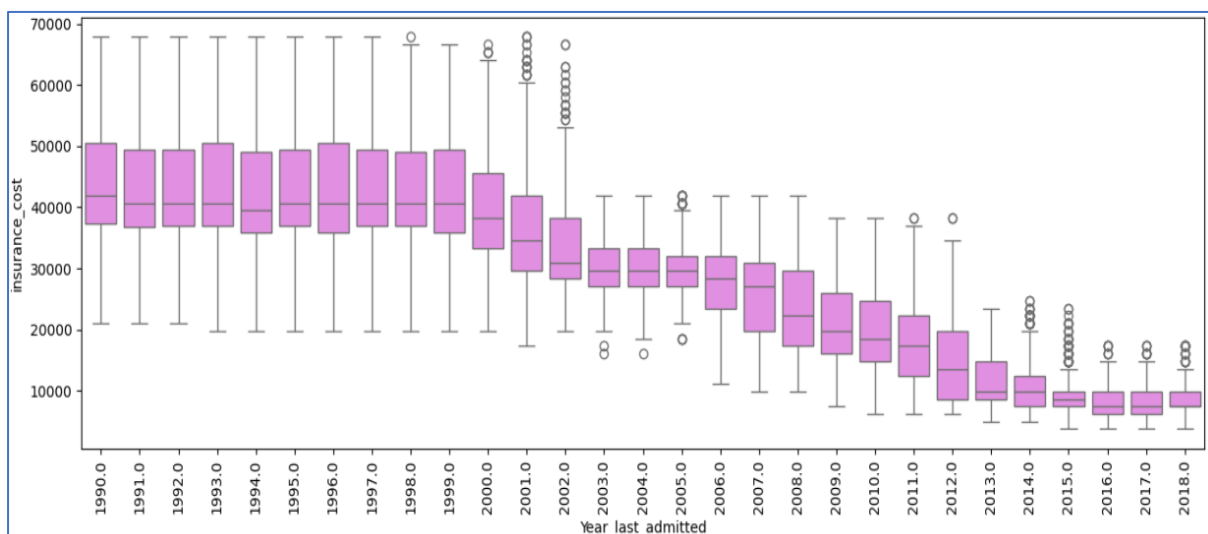


Fig 36. Year_last_admitted vs insurance_cost

9.3 Multivariate Analysis

9.3.1 Weight vs insurance_cost (covered_by_any_other_company)

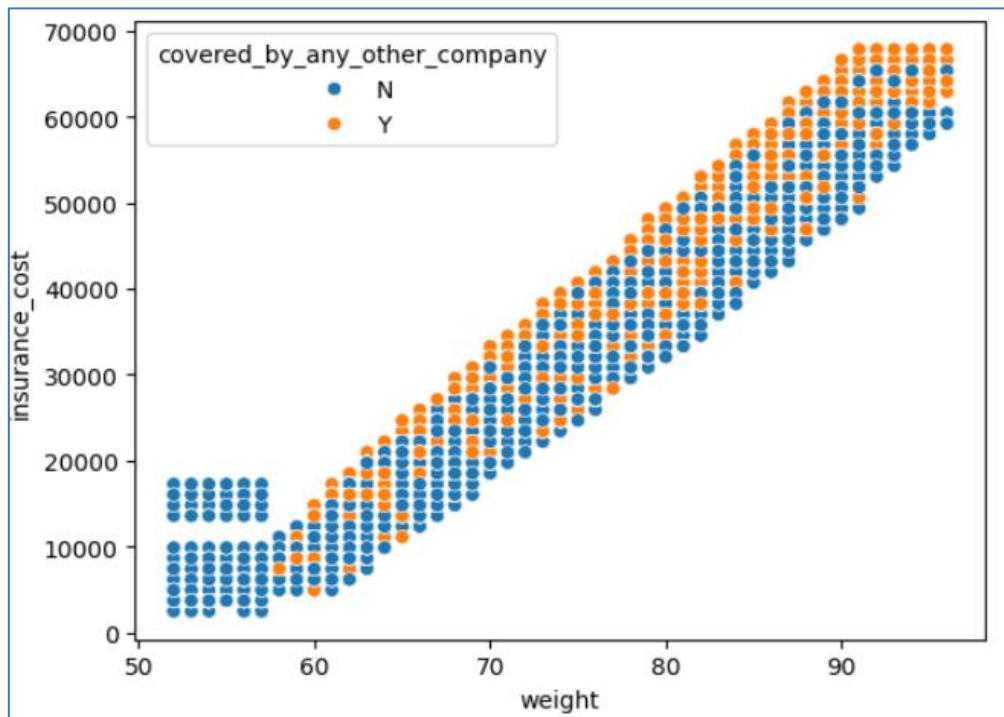


Fig 37. weight vs insurance_cost (covered_by_any_other_company)

9.3.2 weight vs insurance_cost (adventure_sports)

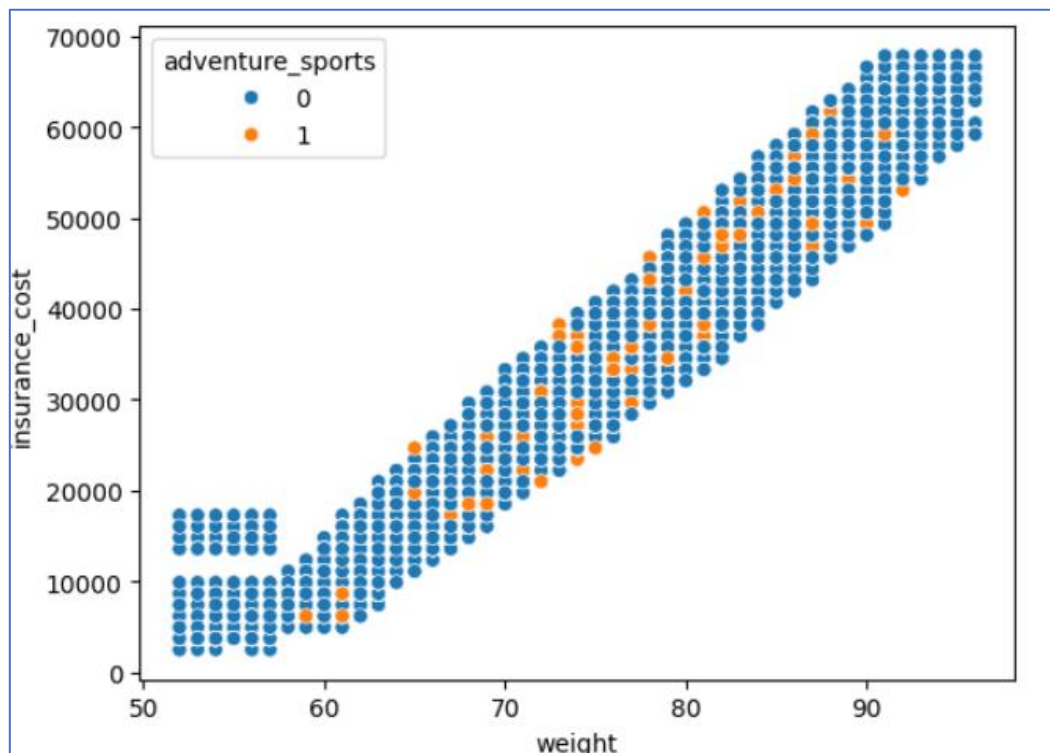


Fig 38. weight vs insurance_cost (adventure_sports)

9.3.3 weight vs insurance_cost (Year_last_admitted)

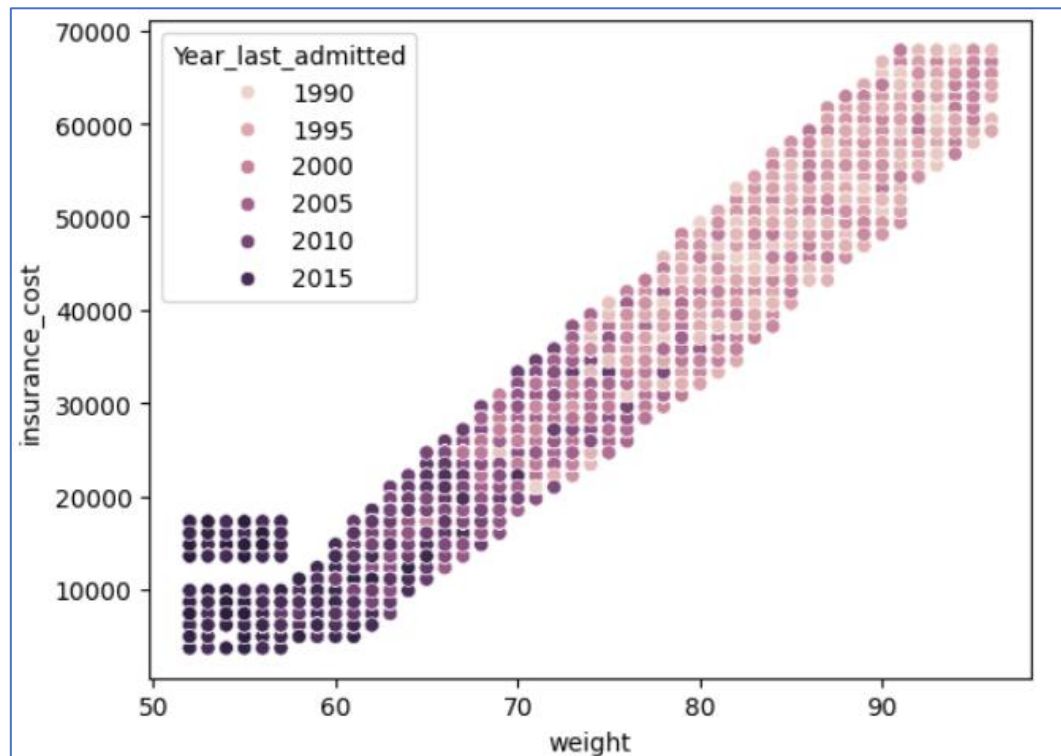


Fig 39. weight vs insurance_cost (Year_last_admitted)

9.4 Bivariate Analysis: Effect of variables on heart_decs_history

9.4.1 Gender vs heart_decs_history

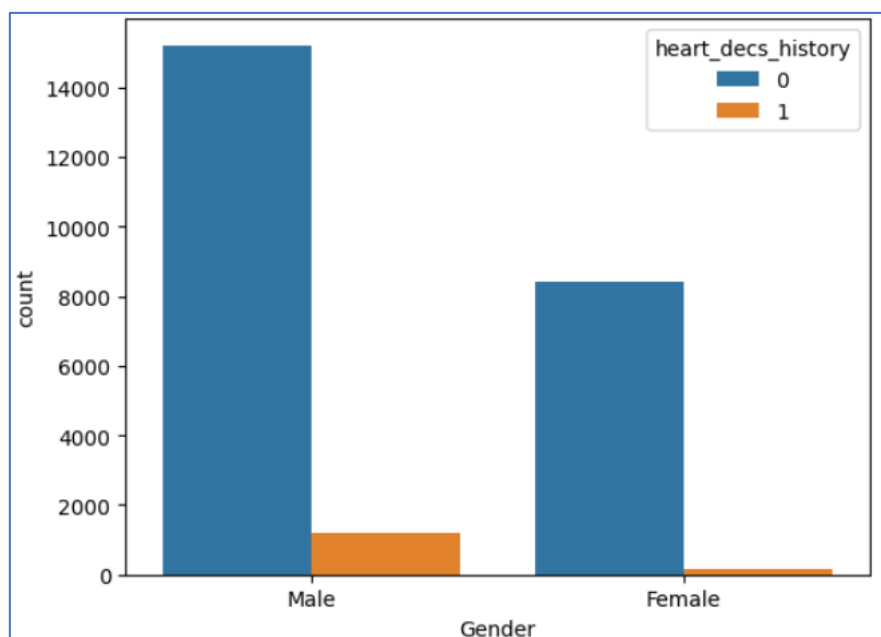


Fig 40. Gender vs heart_decs_history

9.4.2 age vs heart_decs_history

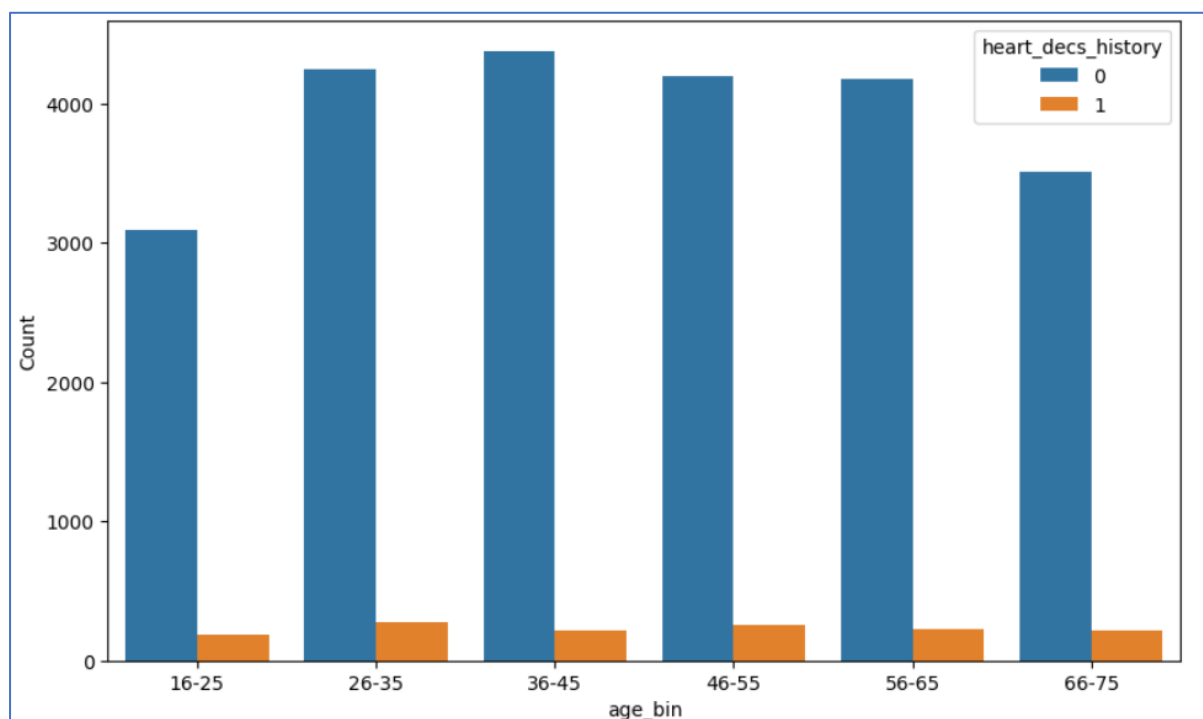


Fig 41. age vs heart_decs_history

9.4.3 daily_avg_steps vs heart_decs_history

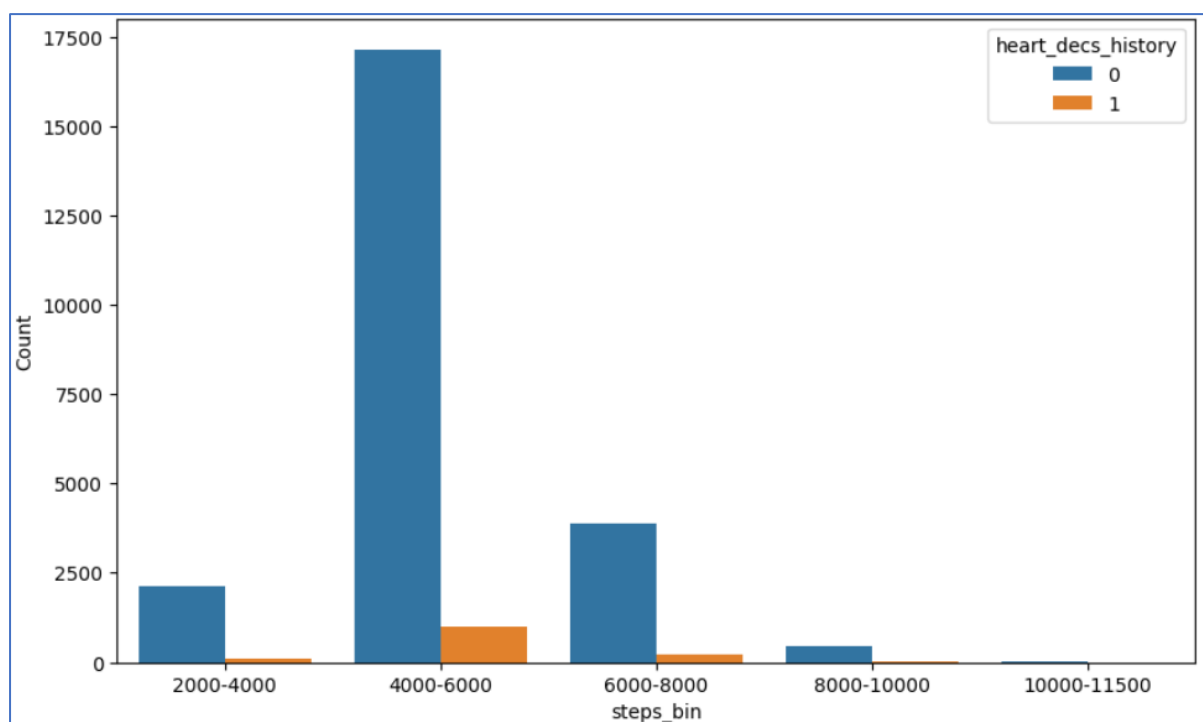


Fig 42. daily_avg_steps vs heart_decs_history

9.4.4 exercise vs heart_decs_history

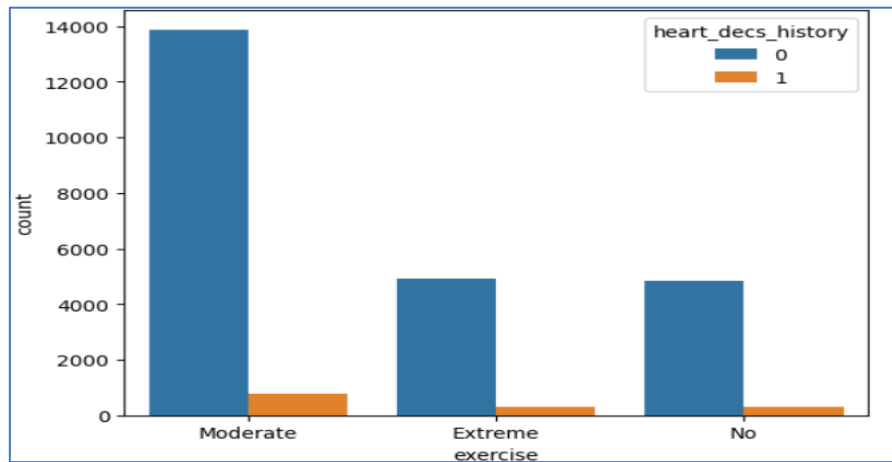


Fig 43. exercise vs heart_decs_history

9.4.5 cholesterol_level vs heart_decs_history

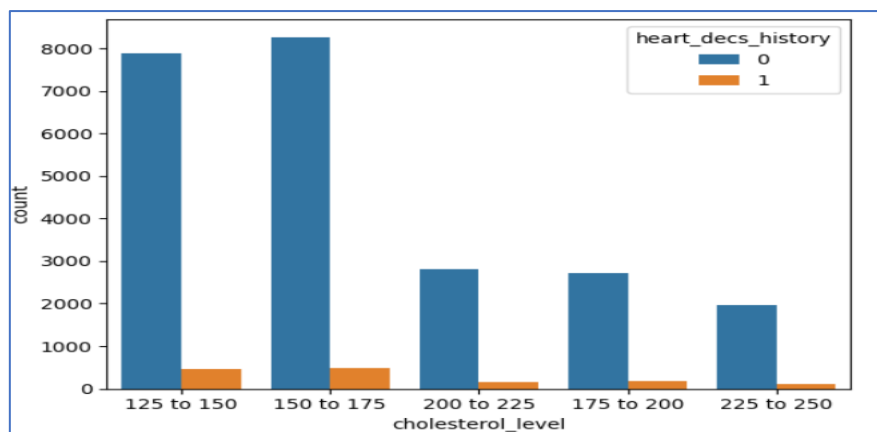


Fig 44. cholesterol_level vs heart_decs_history

9.4.6 avg_glucose_level vs heart_decs_history

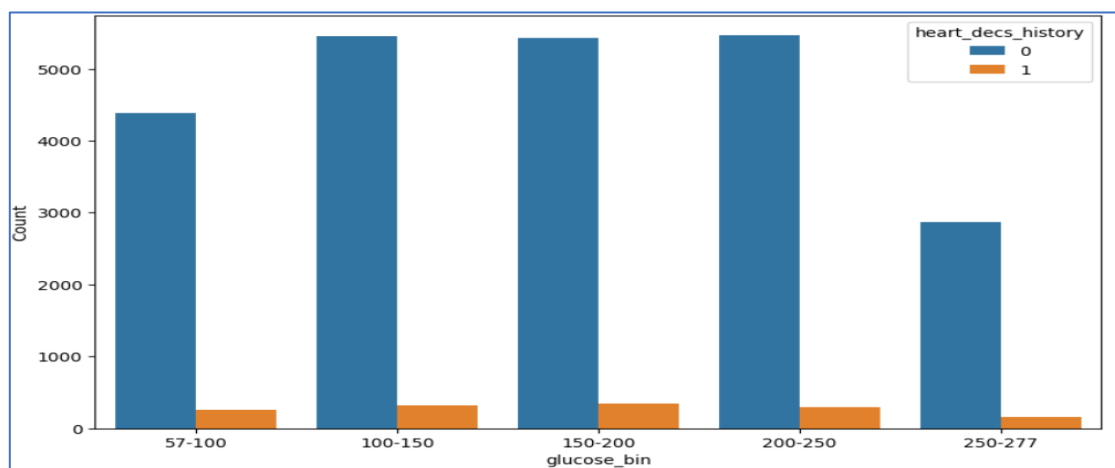


Fig 45. avg_glucose_level vs heart_decs_history

9.4.7 bmi vs heart_decs_history

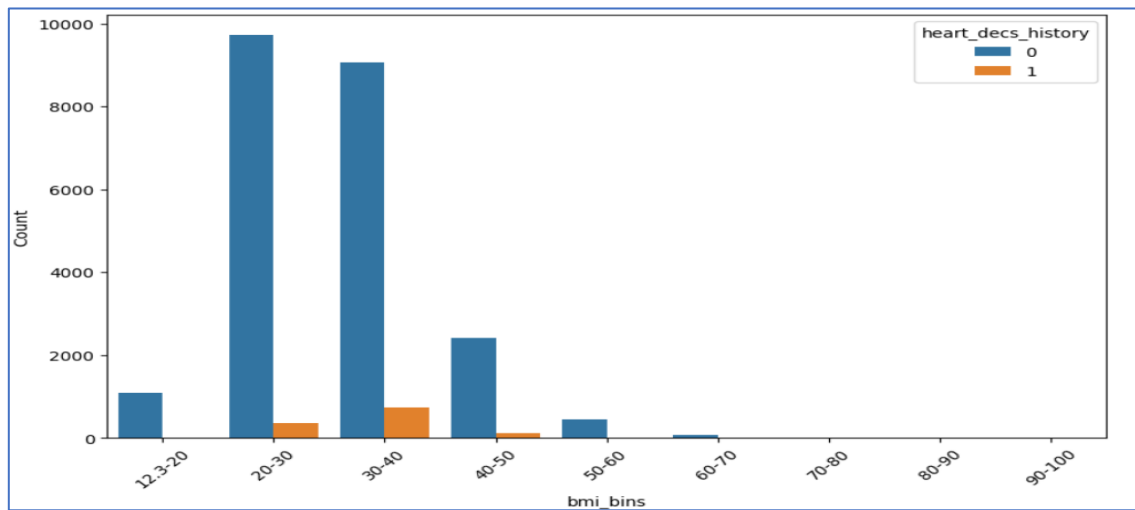


Fig 46. bmi vs heart_decs_history

9.4.8 smoking_status vs heart_decs_history

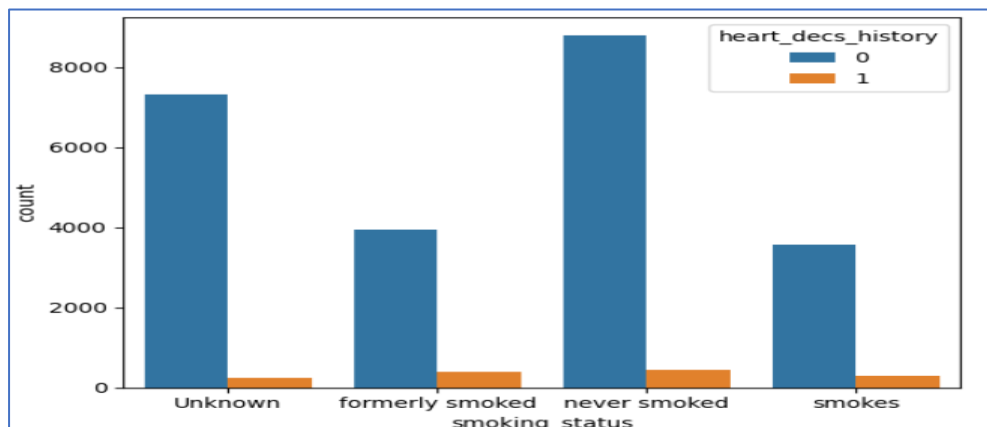


Fig 47. smoking_status vs heart_decs_history

9.4.9 Alcohol vs heart_decs_history

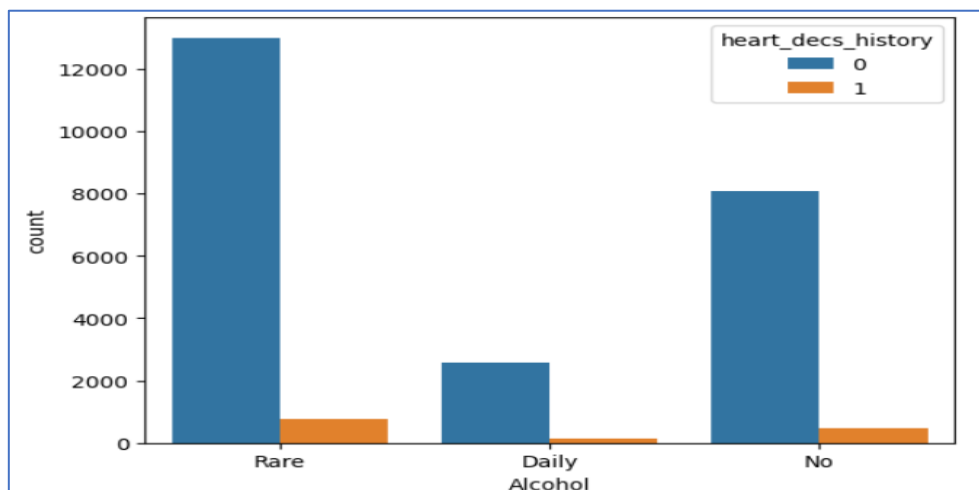


Fig 48. Alcohol vs heart_decs_history

9.4.10 weight vs heart_decs_history

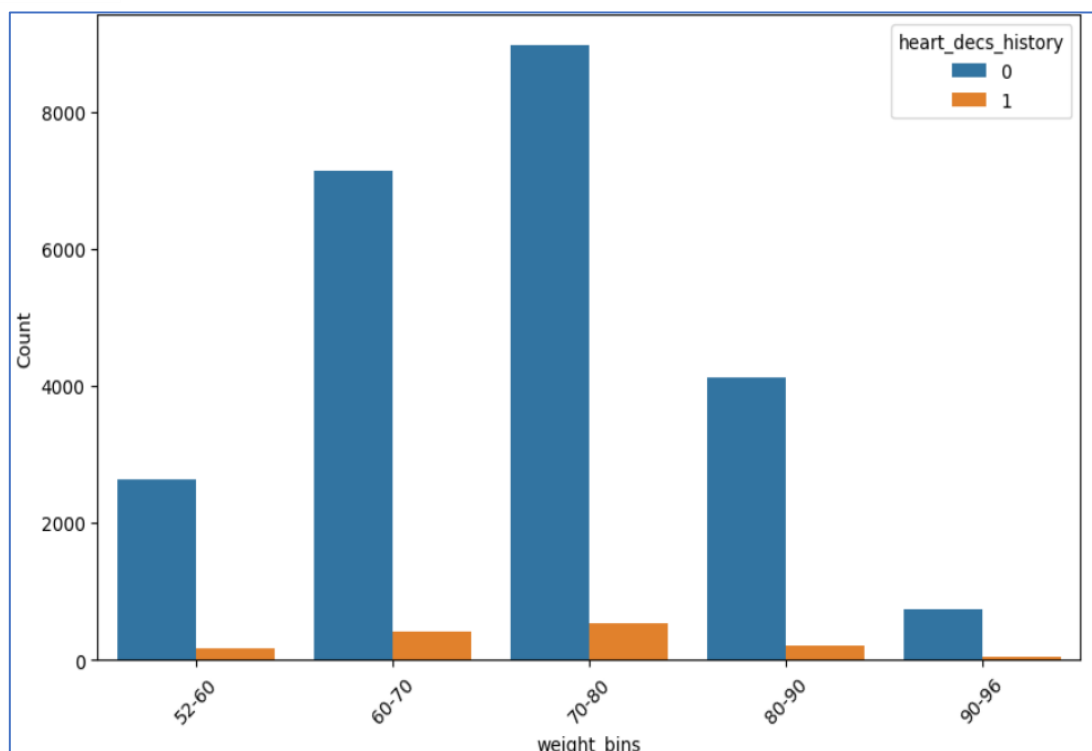


Fig 49. weight vs heart_decs_history

9.4.11 fat_percentage vs heart_decs_history

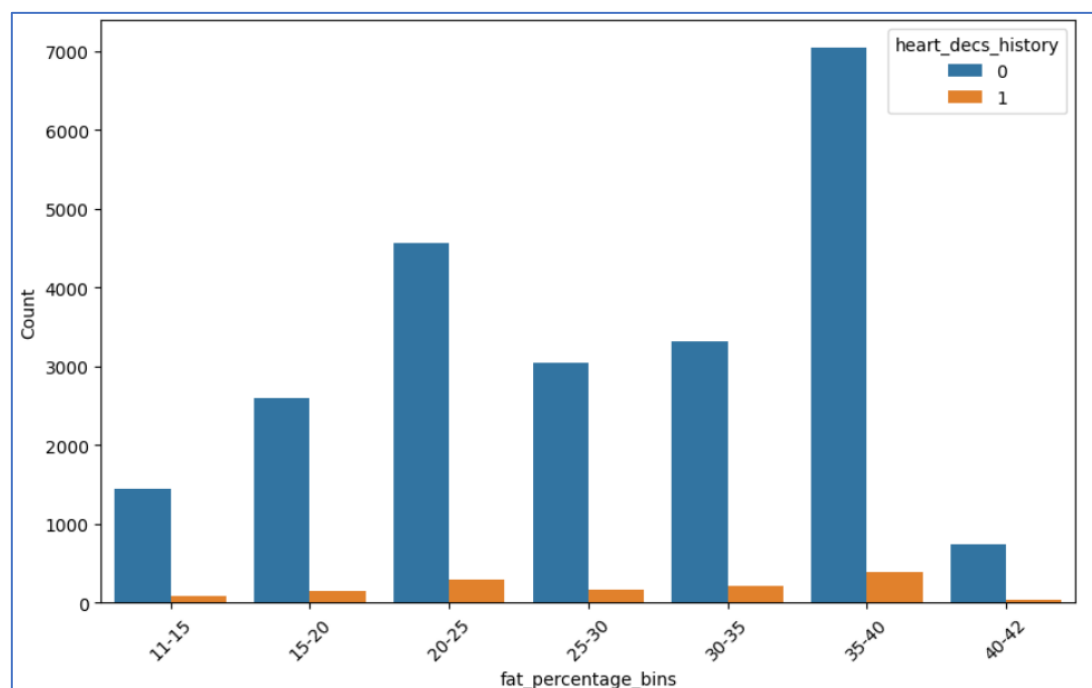


Fig 50. fat_percentage vs heart_decs_history

9.5 Bivariate Analysis: Effect of variables on other_major_decs_history

9.5.1 Gender vs other_major_decs_history

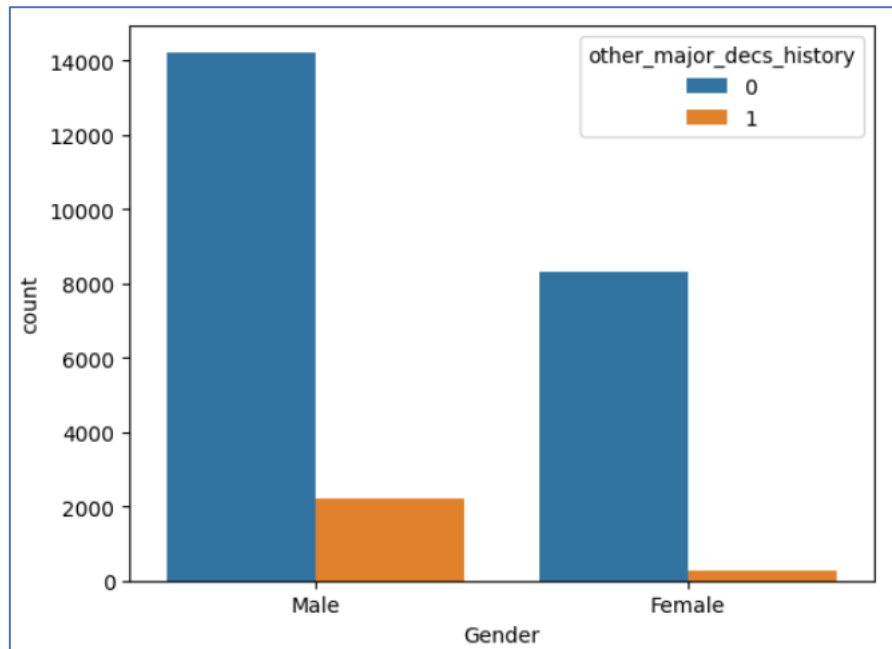


Fig 51. Gender vs other_major_decs_history

9.5.2 age vs other_major_decs_history

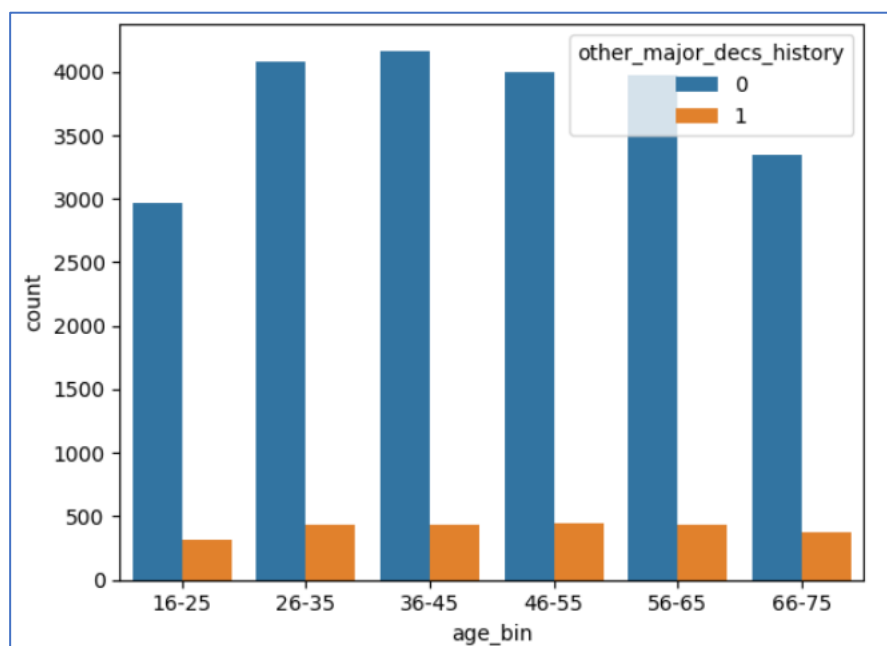


Fig 52. age vs other_major_decs_history

9.5.3 daily_avg_steps vs other_major_decs_history

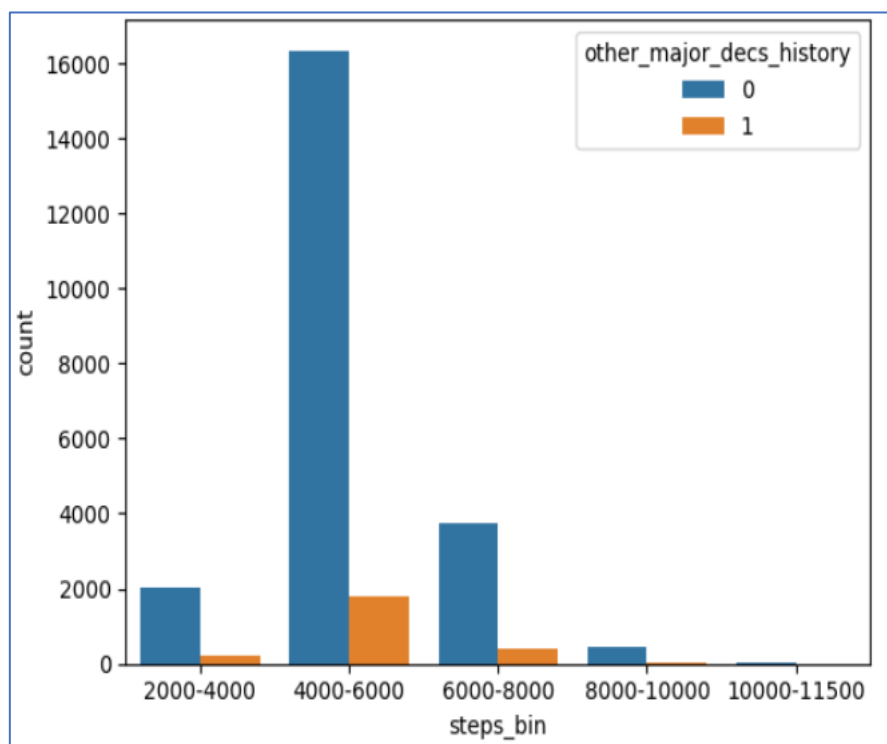


Fig 53. daily_avg_steps vs other_major_decs_history

9.5.4 exercise vs other_major_decs_history

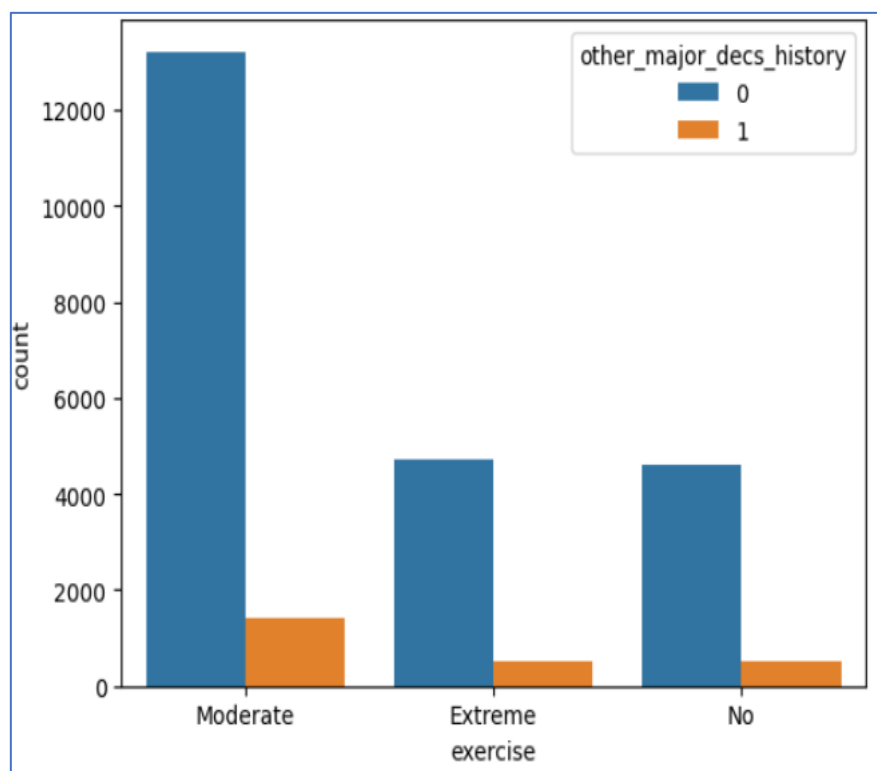


Fig 54. exercise vs other_major_decs_history

9.5.5 cholesterol_level vs other_major_decs_history

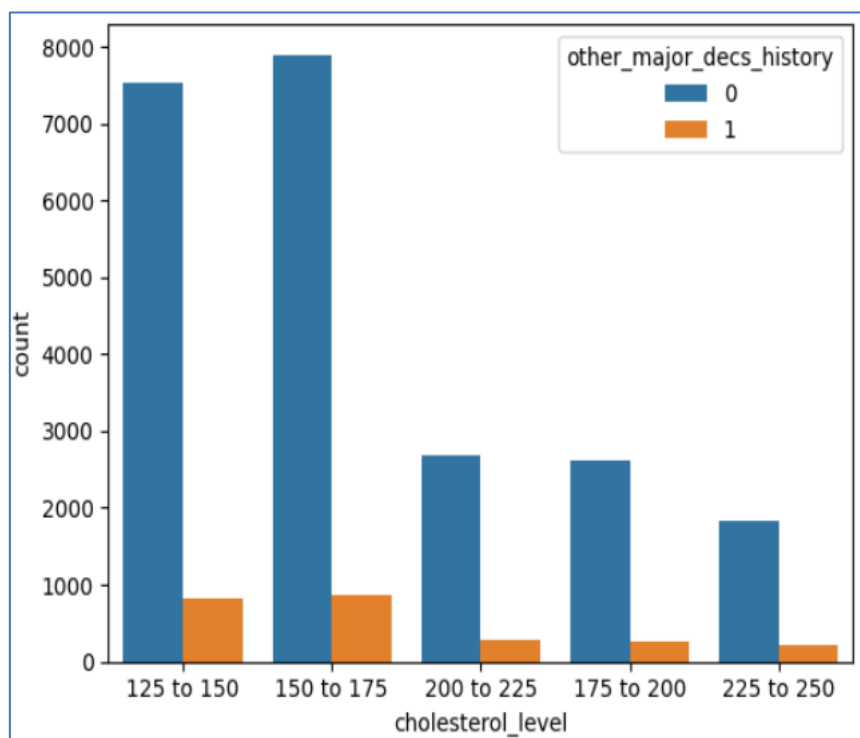


Fig 55. cholesterol_level vs other_major_decs_history

9.5.6 avg_glucose_level vs other_major_decs_history

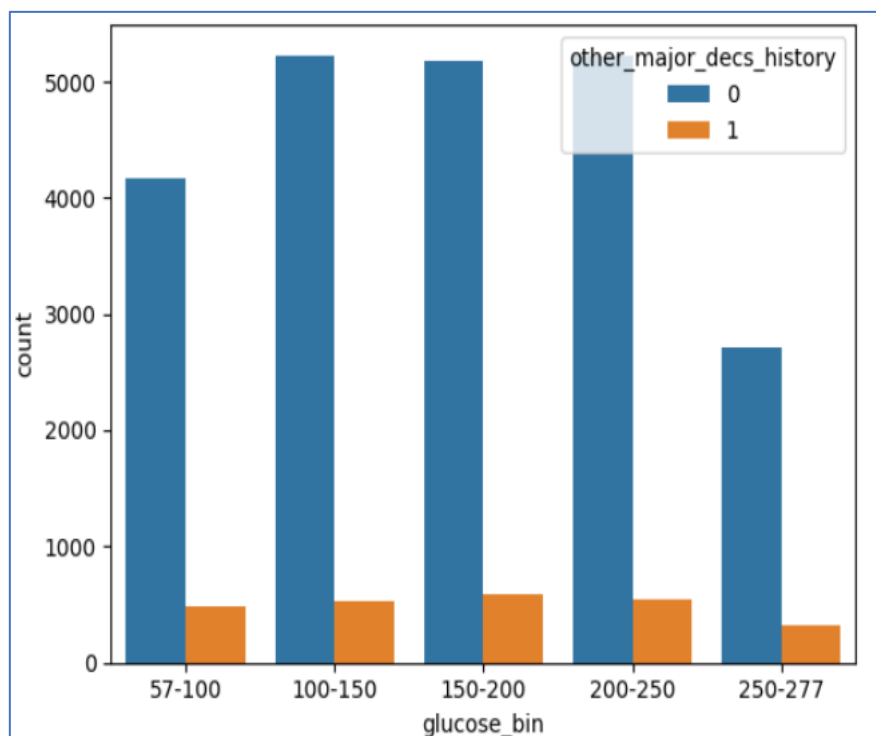


Fig 56. avg_glucose_level vs other_major_decs_history

9.5.7 bmi vs other_major_decs_history

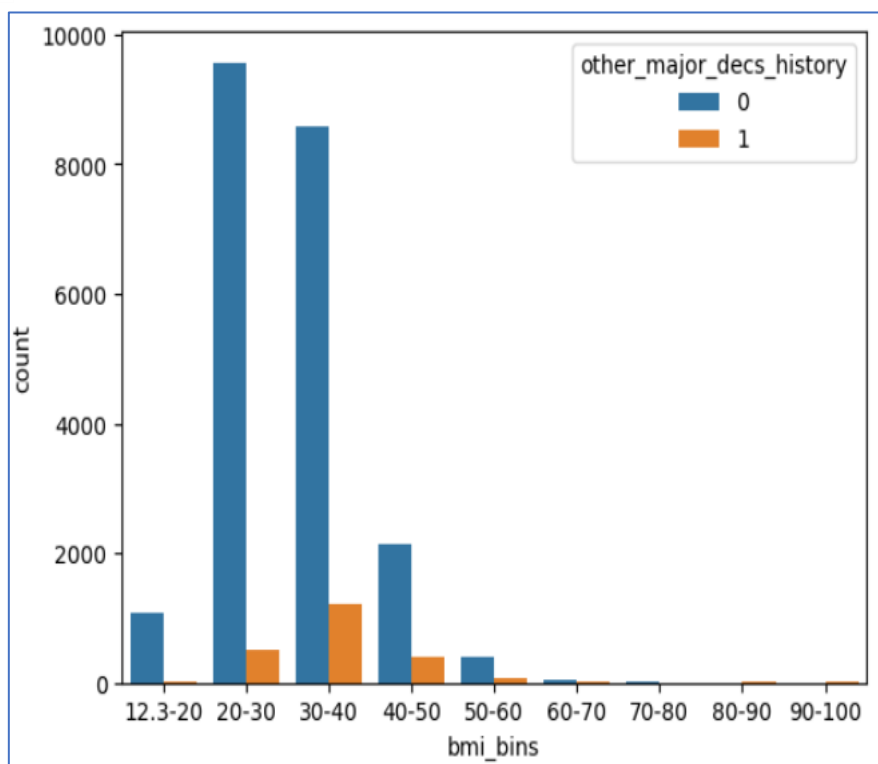


Fig 57. bmi vs other_major_decs_history

9.5.8 smoking_status vs other_major_decs_history

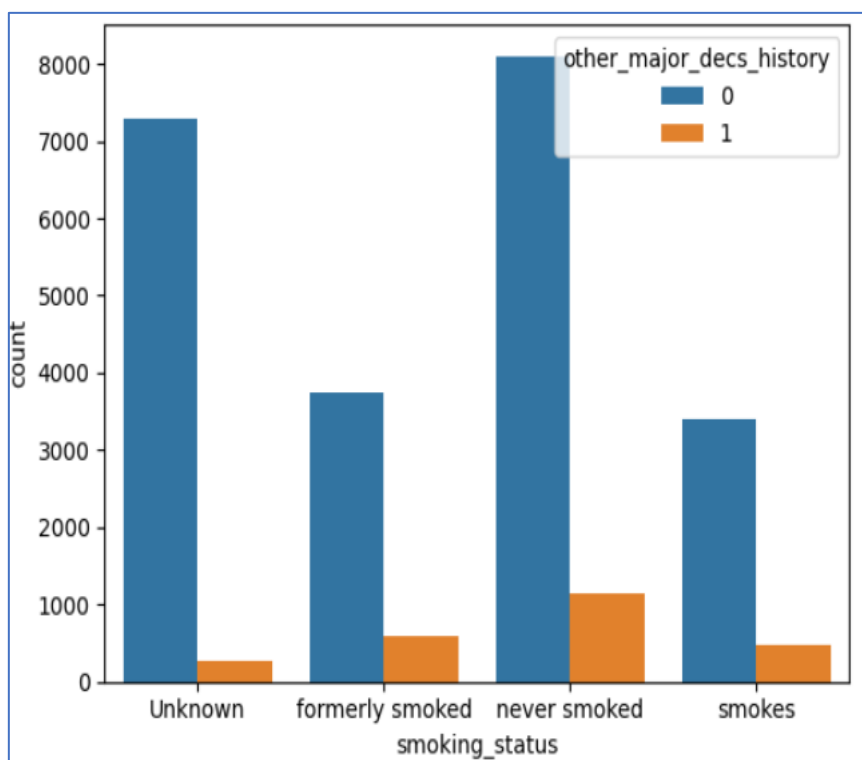


Fig 58. smoking_status vs other_major_decs_history

9.5.9 Alcohol vs other_major_decs_history

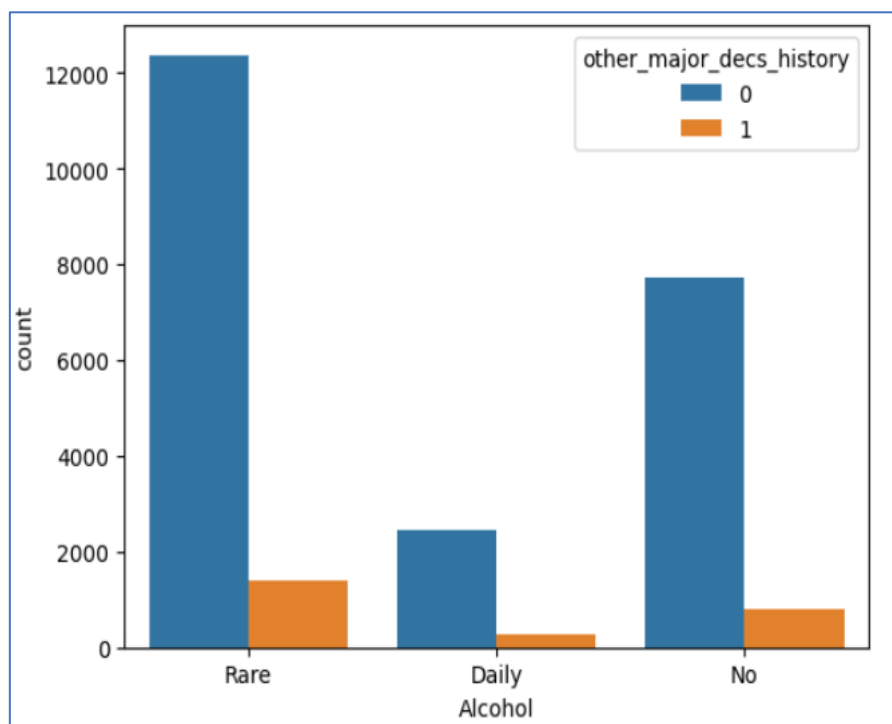


Fig 59. Alcohol vs other_major_decs_history

9.5.10 weight vs other_major_decs_history

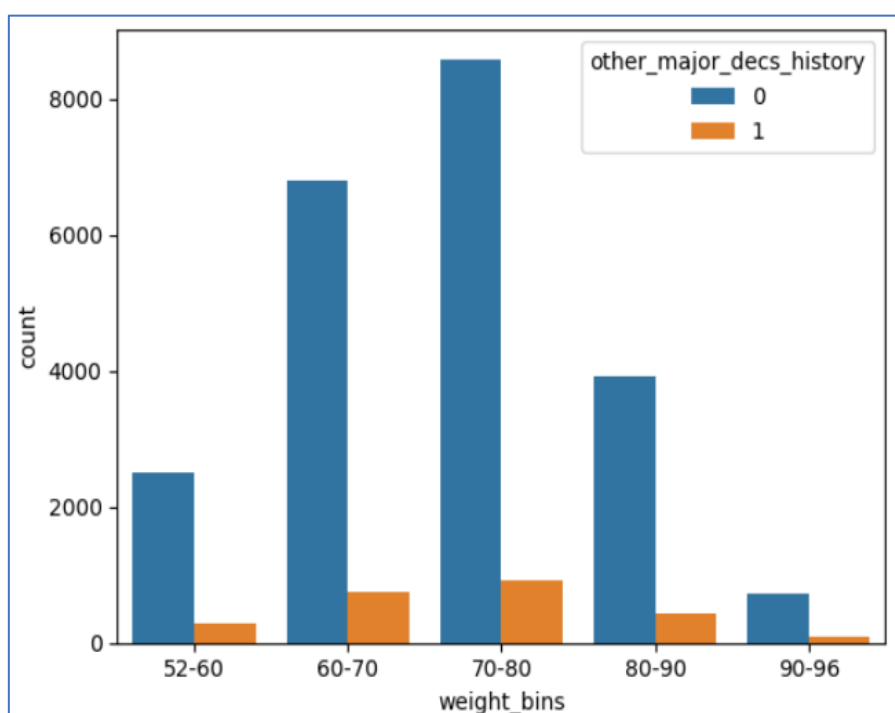


Fig 60. weight vs other_major_decs_history

9.5.11 fat_percentage vs other_major_decs_history

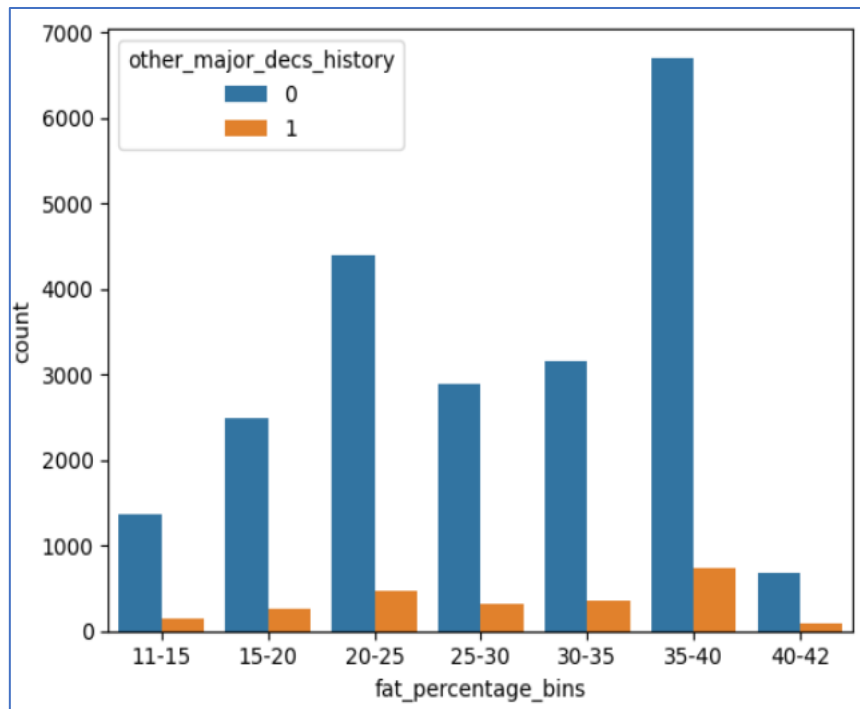


Fig 61. fat_percentage vs other_major_decs_history

9.6 Bivariate Analysis: Other Predictor Variables

9.6.1 covered_by_any_other_company vs heart_decs_history

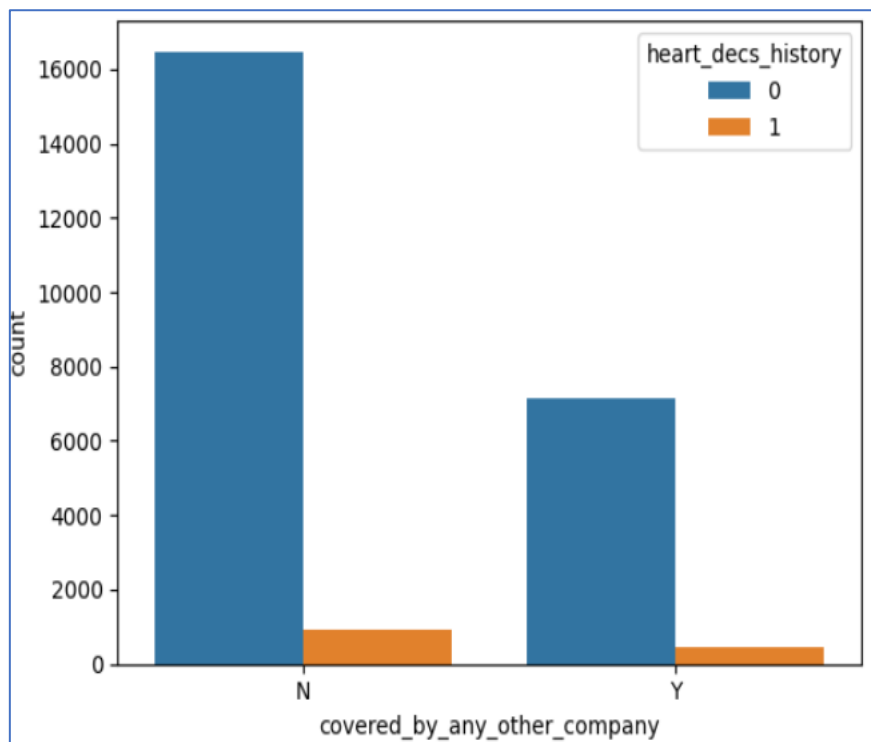


Fig 62. covered_by_any_other_company vs heart_decs_history

9.6.2 covered_by_any_other_company vs other_major_decs_history

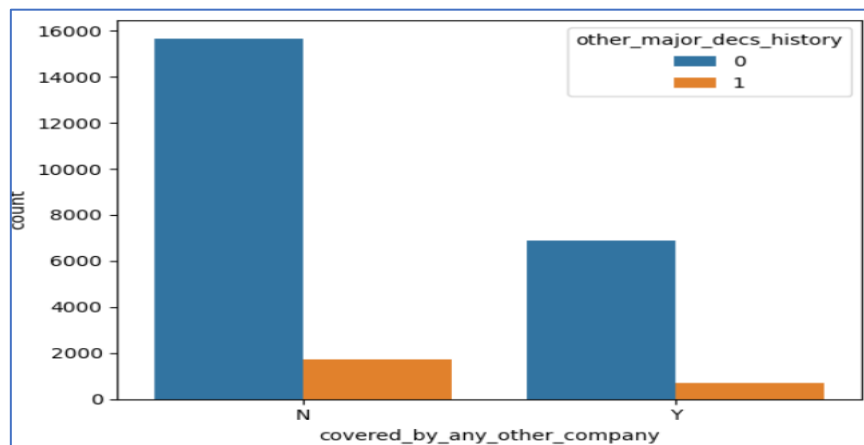


Fig 63. covered_by_any_other_company vs other_major_decs_history

9.6.3 Alcohol vs exercise

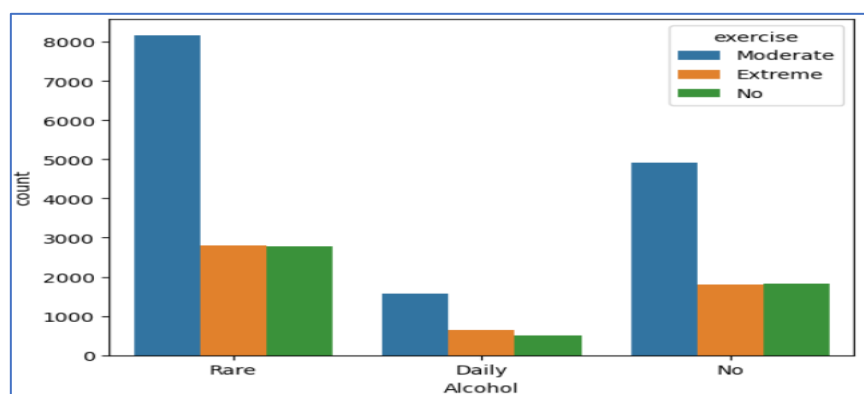


Fig 64. Alcohol vs exercise

9.6.4 Gender vs exercise

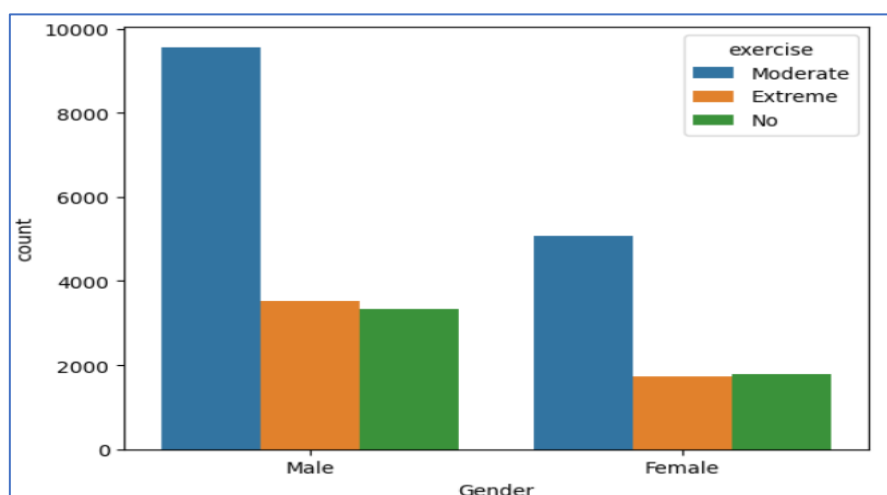


Fig 65. Gender vs exercise

9.6.5 adventure_sports vs heart_decs_history

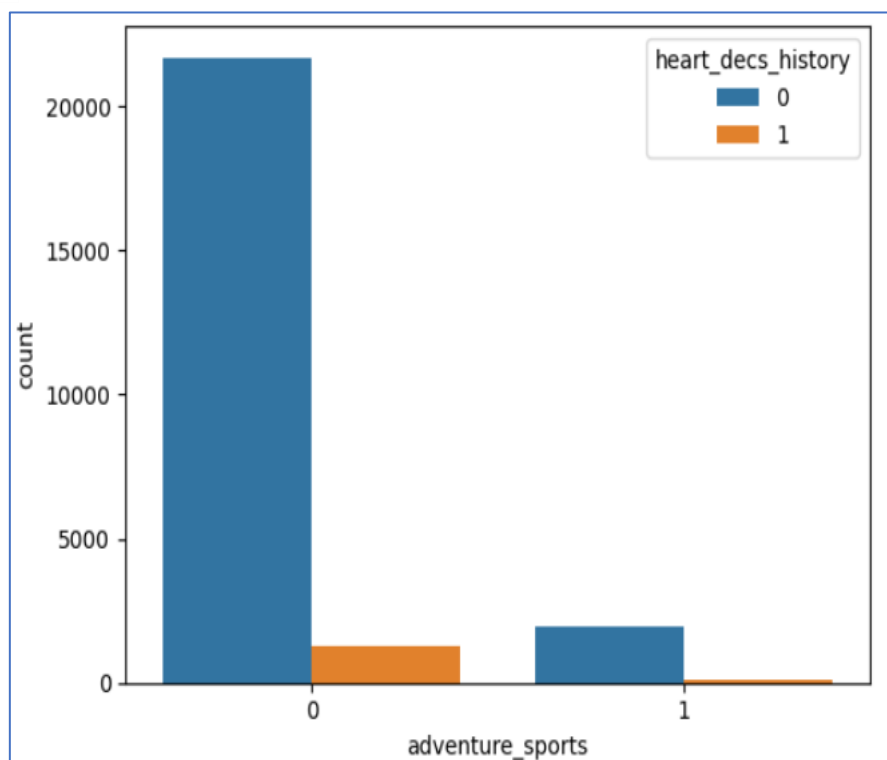


Fig 66. adventure_sports vs heart_decs_history

9.6.6 adventure_sports vs other_major_decs_history

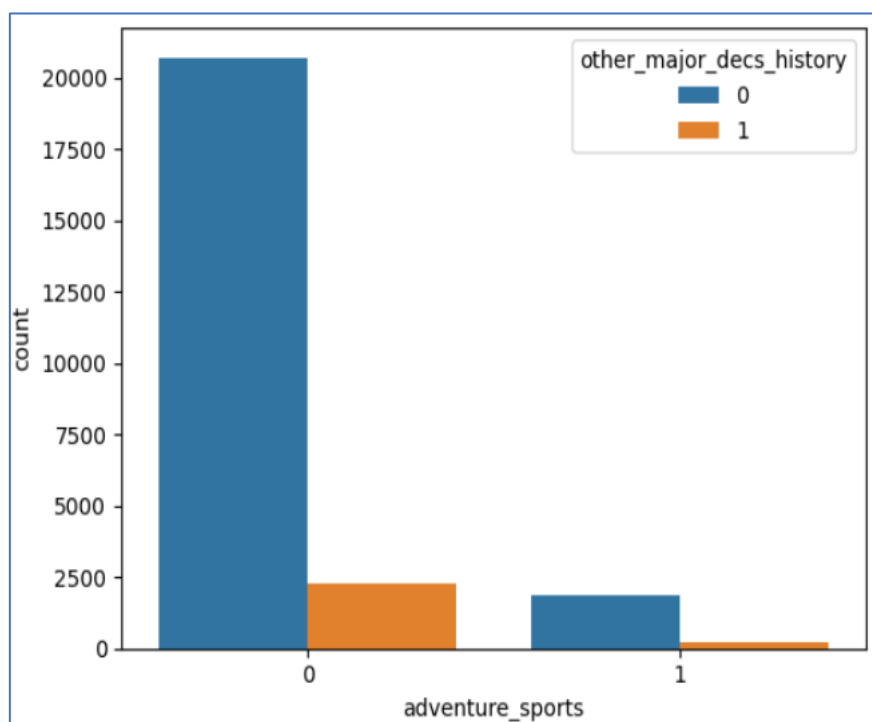


Fig 67. adventure_sports vs other_major_decs_history

9.7 K-means Clustering

9.7.1 Elbow Plot

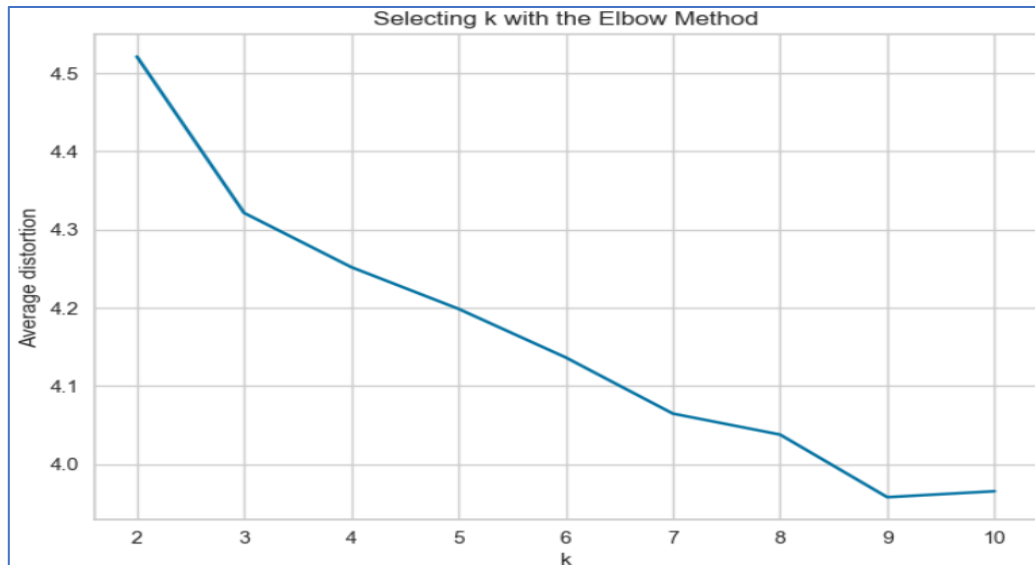


Fig 68. Elbow Plot

9.7.2 Silhouette Score Plot

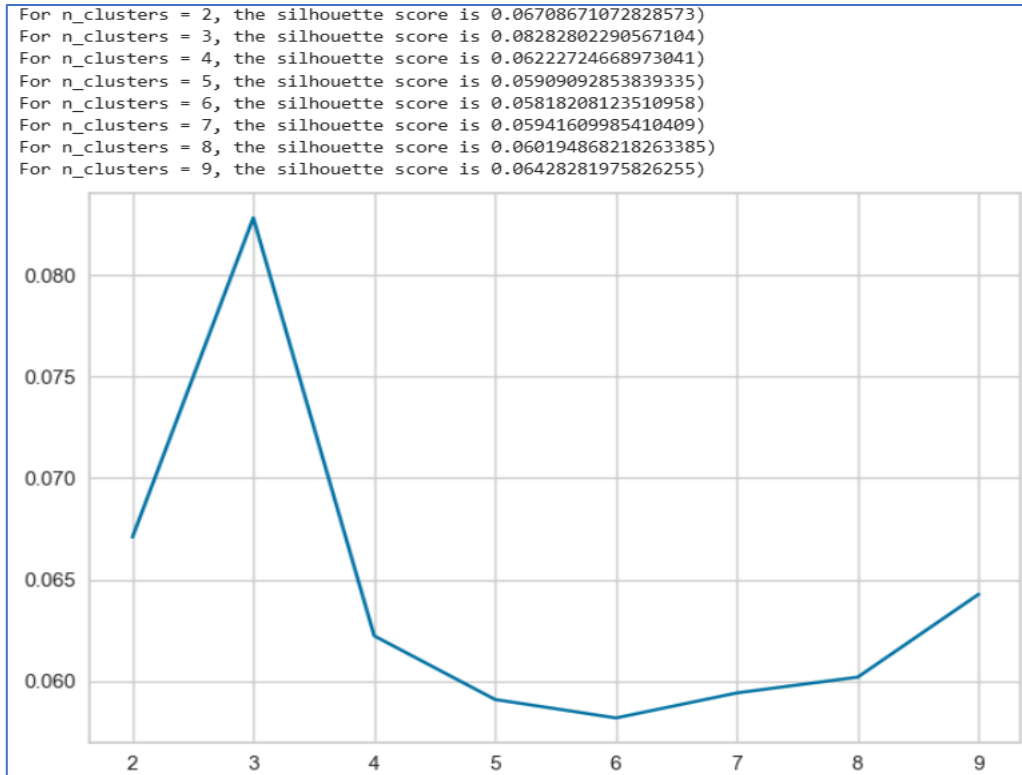


Fig 69. Silhouette Score Plot

9.7.3 Silhouette Score Plot (k=3)

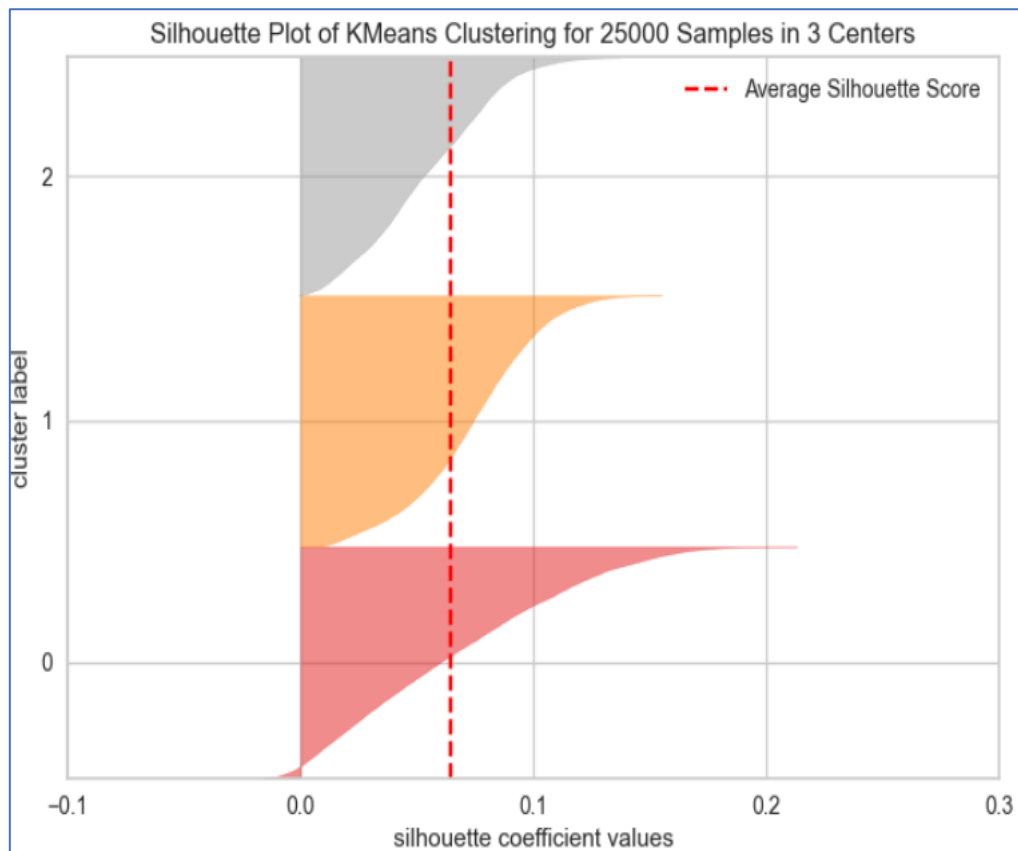


Fig 70. Silhouette Plot (k=3)

9.7.4 Cluster Profile using mean

	years_of_insurance_with_us	regular_checkup_last_year	adventure_sports	Occupation	visited_doctor_last_1_year	cholesterol_level	daily_avg_steps
KM_segments							
0	3.563474	1.070610	0.055679	1.010393	3.075806	1.273530	5221.439495
1	4.720384	0.436547	0.111271	1.002302	3.129592	1.253621	5212.335635
2	4.003263	0.738989	0.084829	0.999592	3.136623	1.273246	5203.557504

age	heart_decs_history	other_major_decs_history	Gender	avg_glucose_level	bmi	smoking_status	weight	covered_by_any_other_company
44.863978	0.050153	0.000000	0.637136	168.500866	30.966990	1.357172	65.340510	0.229811
44.931319	0.042494	0.000192	0.623981	166.346667	30.703456	1.315300	78.931127	0.391655
45.131729	0.128467	1.000000	0.894372	167.761011	34.584932	1.738581	71.485318	0.290783

exercise	weight_change_in_last_one_year	fat_percentage	insurance_cost	Years_since_last_admitted	count_in_each_segments
0.994968	3.052792	28.879238	17355.803349	3.529654	12123
0.994916	1.893621	28.731607	38562.943885	18.867530	10425
0.991843	2.528140	28.824225	27023.694127	10.419250	2452

Table 1. K-means: Cluster profiling using mean